PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

AND THE

STRUCTURAL OUTLINE

FOR

B.A. & B.Sc. GENERAL THIRD YEAR EXAMINATION, 2015

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CONTENTS

Sr. No.	*Particulars*	*Page No.*
1.	Structural Outline	(i)
2.	English (Compulsory) (for B.A. Candidates only)	1
3.	Panjabi (Compulsory) (for B.A. Candidates only)	3
4.	History and Culture of Punjab (for B.A. Candidates only)	5
5.	English (Elective)	7
6.	Hindi (Elective)	10
7.	Sanskrit (Elective)	13
8.	Panjabi (Elective)	16
9.	Urdu (Elective)	18
10.	Persian (Elective)	21
11.	Arabic (Elective)	23
12.	French (Elective)	24
13.	German (Elective)	27
14.	Russian (Elective)	29
15.	Bengali (Elective)	32
16.	Tamil (Elective)	34
17.	Kannada (Elective)	35
18.	Malayalam (Elective)	36
19.	Telugu (Elective)	37
20.	Physical Education	38
21.	Education	48
22.	Adult Education	52
23.	Music (Vocal)	55
24.	Music (Instrumental)	58
25.	Tabla (Instrumental Music)	61
26.	Indian Classical Dance	64
27.	Fine Arts	67
28.	History of Art	72
29.	Ancient Indian History, Culture & Archaeology	76
30.	Defence and Strategic Studies	79
31.	History	84
32.	Political Science	88
33.	Economics	92
34.	Sociology	96
35.	Public Administration	101
36.	Philosophy	105
37.	Psychology	109
38.	Geography	112
39.	Gandhian Studies	121
40.	Journalism & Mass Communication	125
41.	Police Administration	130
42.	Women's Studies	139
43.	Human Rights and Duties	142
44.	Computer Science	147
45	Environment Conversvation	152
46.	Home Science	156
47.	Agriculture (Elective)	163
48.	Mathematics	166
49.	Statistics	172
50.	Applied Statistics	176
51.	Physics	179
52.	Chemistry	187
53.	Botany	199
54.	Zoology	208
55.	Electronics	224
56.	Bio-Chemistry	229
57.	Microbiology	234

STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL AND HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION FOR FIRST YEAR 2013 EXAMINATION (ALREADY HELD); SECOND YEAR 2014 (ALREADY HELD); THIRD YEAR 2015 EXAMINATION

The new B.A./B.Sc. (General & Honours) course (at + stage) in the 10+2+3 system of education has been structured keeping in view certain basic guidelines as under :--

1. *(i)* The scheme should provide a broad-based, interrelated and comprehensive programme of study, built upon the + 2 stage, which ultimately may act as a sound foundation for postgraduate education, participation in competitive examinations, and joining professional courses like those in Law, Education, etc.

   (ii) It is so structured and organised that it is in conformity with the needs of the country in the present context. Apart from providing good general education, it should have an applied component as part of the course work so that it creates awareness of the practical problems facing the society and generates employability of the youth in diverse spheres of activity.

   (iii) It is so designed and formulated that it provides flexibility and relative freedom of choice of courses to the students according to their special interests and capabilities.

   (iv) It is based on intensive teacher–student interaction.

2. The B.A./B.Sc. courses have been so structured and revised that they are comparable both in content and quality.

3. To cope with the fast increasing knowledge in each discipline, it has been considered desirable to cover each elective subject in two papers.

The two-paper scheme will enable the teachers to teach the subject in detail and examine the students accordingly. Effort has to be made to improve the level and adequacy of content of each course in such a manner that the total scheme provides distinct upgradation of the existing curricula for the first degree. In the absence of such an upgradation, the increase in the total educational period from 14 to 15 years for the B.A./B.Sc. will be unjustifiable.

Keeping in view the above objectives and guidelines, the following structural outline has been introduced for the B.A./B.Sc. (General) degree course :

Admission to B.A./B.Sc. (General and Honours) course for the second and third year (for the students who have passed First Year under this scheme) for the session 2012-2013 (2013 examination) shall be as hereinafter mentioned.

Note : A person who has passed B.A./B.Sc. 1st year or 2nd year examination from other Universities in India may be allowed to migrate to this University subject to the condition that he shall have to clear the deficient subject but the total number of credits required to be earned shall remain the same.

B.A. (General)

The B.A. (General) programme of study shall consist of **24** credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2013 Examination (already held)

COMPULSORY

(a)	Punjabi – Two Papers	1 Credit	These papers constitute one subject in each year.
	OR		
	*History & Culture of Punjab –One Paper	1 Credit	
(b)	English	1 Credit	

ELECTIVE

+ Any three elective subjects of two credits each including languages and Elective vocational subject (To be studied in selected colleges)	6 Credits
Total : 8 Credits	

****Environment Education and Road Safety :** 70 marks

2nd Year 2014 Examination (already held)

Same as in the 1st year

3rd Year 2015 Examination

Same as in the 1st year and 2nd year examination.

Note :

1. In subjects having practicals, the theory papers and practicals together will be of 2 credit value. The candidate will have to pass in theory and practical/s separately.

2. Each paper of one credit shall be allocated 3 hours of teaching per week during each academic year. However, for subjects having practicals, three hours of teaching will be allocated to each theory paper and two hours for a practical per week or as per requirement of the subject concerned.

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* For B.A./B.Sc. History and Culture of Punjab is allowed as an option in lieu of Punjabi (compulsory) to those students who had already offered earlier in the First Year of 2013 examination.

** This is a compulsory qualifying paper, which the students have to study in the B.A./B.Sc. 1st year. The students are required to qualify this paper either in the 1st year/2nd year/3rd year of the course. The examination will be conducted by the University.

3. B.A. (General) degree holders shall be eligible for admission to the Master's course in any of the elective subjects studied by them during all the three years of the programme of study, earning six credits in each, provided they fulfil the eligibility conditions.

4. A student would offer any science subject, including Mathematics, only if he has passed that subject in the qualifying examination or qualifies in the subject as a deficient/additional subject from the concerned Board/University/Council in the Supplementary Examination subsequent to the admission.

Provided further that a student can offer :

(a) Statistics only if he takes up Mathematics.

(b) Applied Statistics only if he takes up other subject(s) excluding Mathematics.

B.Sc. (General)

The B.Sc. (General) programme of study shall consist of **20** credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2013 Examination (already held)

Punjabi/History & Culture of Punjab	1 Credit
+ Three Elective Subjects of 2 credits each including Elective Vocational subject (To be studied in selected colleges)	6 Credits
Total :	**7 Credits**

** Environment Education and Road Safety : 70 marks

2nd Year 2014 Examination (already held)

English	1 Credit
+ Three Elective Subjects of 2 credits each including Elective Vocational subject (To be studied in selected colleges) (the same as in the 1st year)	6 Credits
Total :	**7 Credits**

3rd Year 2015 Examination

Three Elective Subjects of 2 credits each (the same as in the 1st and 2nd year)	6 Credits
Total :	**20 Credits**

** This is a compulsory qualifying paper, which the students have to study in the B.A./B.Sc. 1st year. The students are required to qualify this paper either in the 1st year/2nd year/3rd year of the course. The examination will be conducted by the University.

Note : 1. The two credits allocated to an elective subject, to be studied in an academic year as spelled out above, shall be covered in two theory papers and a practical or practicals as per requirements of the subject concerned.

2. In the instructional process, it is desireable to strike a balance in the teaching and learning strategies. A close interaction between the teacher and the student is necessary to achieve the above objectives.

3. Each elective subject of 2 credits for the B.Sc. shall be allocated five hours of teaching and four hours of practical work per week during the course of the academic year or as per requirements of the subject concerned.

4. B.Sc. (General) degree holders shall be eligible for admission to the M.Sc. course in any of the three elective subjects which they have studied during all the three years of the degree course, earning 6 credits in each, provided the candidate fulfils the eligibility conditions.

B.A./B.Sc. Honours

In addition to the main objectives listed in the preamble to the B.A./B.Sc. (General) degree course, the Honours Course seeks to provide course work of advanced nature in one of the elective subjects already being studied from 1st year onwards by a student. Both in content and quality, it would be of a level as gives adequate specialized knowledge in a subject even to make it a culminating point in the programme of study if a student so chooses. Or else, it should provide strong, additional foundation for pursuing the subject at the M.A./M.Sc. level.

Keeping in view the objectives stated in the preamble to the B.A./B.Sc. (General) degree course and that enunciated above, the structural outline for the Honours degree course shall be as under :

B.A. (Honours)

The B.A. Honours Course shall carry 28 credits (each credit carrying the equivalence of 100 marks). Apart from the structure of the programme of study prescribed for the B.A. (General) degree course, credit value of the subjects/papers and hours of teaching allocated to them as already stated in the scheme, the Honours Course shall include four papers of one credit each of an advanced nature in one of the elective subjects a student shall study in all the three years of the degree course.

A student may offer Honours in Second Year in any one of the elective subjects, to be studied by him in all the three years of the course.

Provided that he has obtained at least 50 % marks in the subject of Honours in the First Year of the B.A. (General) Course.

For the B.A. (Honours) Course, the distribution of work during the three years will be as under :--

1st Year : The same as in B.A. (General). (8 credits)

2nd Year : The same as in 2nd year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honour degree.
(8+2 = 10 credits)

3rd Year : The same as in 3rd year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.
(8+2 = 10 credits)
Total Credits : 28

A candidate eligible to appear as a private candidate in B.A. examination may also offer Honours papers in any of the elective subjects offered for B.A. (General) in all three years provided he fulfils the requirement of the Regulation.

B.Sc. (Honours)

The B.Sc. (Honours) programme of study shall consist of 24 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

A student may offer Honours in any one of the elective subjects to be studied by him in all the three years of the course provided he has obtained at least 50% marks in the subject concerned in the first year examination of the B.Sc. (General) course.

Of the 24 credits, each student shall offer courses in each year as under :--

1st Year : The same as for First Year of B.Sc. (General). (7 credits)

2nd Year : The same as for 2nd Year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honours degree.
(7+2 = 9 credits)

3rd Year : The same as for 3rd year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.
(6+2 = 8 credits)
Total Credits : 24

Both in B.A. Honours and B.Sc. Honours Courses, each paper of one credit shall be allocated 3 hours of teaching, or as per requirements of the subject concerned.

Note : Instruction through audio and/or video cassettes may form a part of Languages course.

GUIDELINES REGARDING CONTINUOUS ASSESSEMENT OF REGULAR STUDENTS OF B.A./B.SC./B.COM./B.C.A. COURSES

IMPORTANT NOTE

(i) In order to incorporate an element of continuous assessment of students, the Colleges will conduct two mandatory House Tests in theory papers – one in the month of September/October and the other in December/January every year.

(ii) (a) For September Test, there will be only one paper of one hour's duration in each subject, and for December Test, there will be paper/s on the pattern of annual examination conducted by the University.

There will be a Special Test for those students who could not fulfil the conditions of eligibility. It will not be held to provide an opportunity to all students to improve their earlier score. Those students who are exempted by the Principal of the College from appearing in the House Test/s in September and/or December/January will also be allowed to appear in the Special Test; this Test will determine their eligibility for admission to the examination as well as their score for Internal Assessment.

(b) With a view to meet the grievance of students, if any, on account of scores obtained by them, the answer-books will be shown to them. Difference of opinion on the issue, if any, will be sorted out with the help of respective Heads of departments as well as the Principal of the College.

(iii) Whereas the September House Test will carry weightage of 40 per cent, the December House Test will have weightage of 60 per cent in each subject/paper. The total weightage for both the Tests taken together shall be10 per cent of the total marks in each theory subject/paper. The weightage of 10 per cent marks shall be added to each paper of B.A./B.Sc./B.Com./B.C.A. I, II and III Year which will, henceforth, carry weightage of maximum marks allotted to each paper. A candidate will have to pass in theory and practical/s separately.

(iv) The record of marks secured by the students in the two House Tests will be sent by the respective Colleges so as to reach the office of Controller of Examinations latest by 15th March, failing which the result of the students shall be shown as 'RLA' and the entire responsibility for this would lie with the Principal/s of the College/s.

(v) The Colleges will continue to forward the internal assessment of the students for Practicals. Projects and similar other activities, wherever applicable, to the Controller of Examinations, as usual, so as to reach his office latest by 15th March.

SPECIAL NOTE :

(i) Each theory question paper will be set out of the marks allotted to each theory paper and 10% marks of the maximum marks of each paper will be internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

(iii) It will not be mandatory for the students to separately pass in the internal assessment.

PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING IN VARIOUS SUBJECTS FOR B.A. (GENERAL) AND B.Sc. (GENERAL), THIRD YEAR EXAMINATION, 2015.

ENGLISH (Compulsory)

(For B.A. Candidates Only)
B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015
Outlines of Tests, Syllabi and Courses of Reading

Objective: The main aim of teaching English (Compulsory) to B.A. III students is to equip them to use language in an efficient and effective manner. The idea is to empower an average student in such a way that learning English becomes a pleasurable endeavor, and ceases to be a nightmare, which is what it is, right now. It is with this aim in mind that new text books have been chosen for all the three years of B.A. English (Compulsory) courses. The focus of the new text books is two-fold: to teach finer nuances of language through an integrated approach; and to teach composition and grammar through activity-based, not rule-based approach. For this reason, we have consciously chosen text books that de-emphasize literature and re-set our focus on the study of language. Given the way English language proficiency levels have touched a new low in our context, it was felt that such a corrective measure was not only much needed, but was almost inescapable.

Note:

(i) There will be **one paper** in all the three years of B.A. English (Compulsory).
(ii) The paper shall be divided into two sections, i.e. **Section A & B.** The distribution of marks in each section shall be indicated separately against the questions.
(iii) Section A shall deal with the text and Section B, with composition and grammar (again based largely upon the text).
(iv) The questions should be set strictly in accordance with the pattern of question paper outlined in the syllabus.
(v) For the successful implementation of this syllabus, it is necessary that we reflect objectives of this course, first in our teaching practices, and then in designing question paper/s and evaluating answer scripts of the students. The examination shall be held at the end of the term and / or year, as is recommended by the university, from time to time.

Max. Marks : 100
Theory : 90 Marks
Int. Ass. : 10 Marks
Time : 03 Hours

Text Prescribed

Insights: A Course in English Literature and Language, Ed., K. Elango, Orient Blackswan Pvt. Ltd.,Hyderabad, 2013. (Revised Edition).

Unit II Chapter 'Emotional Intelligence' deleted.

Unit VII Chapter 'Nano Technology' deleted.

Section - A (Poetry & Prose)

Q. 1. The examiner will set *fourteen* short questions from *Poetry & Prose* Sections of the prescribed text, out of which a student shall be required to attempt only *ten,* selecting, at least, five from each section (to be answered each in not more than 60-80 words). These questions may be drawn from each of the seven units given in the text.

(20 Marks)

Q. 2. The examiner shall set *eight* questions (on the pattern of questions for **Critical Analysis** suggested in the prescribed text) again from *Poetry & Prose* Sections of the text, out of which a student is expected to attempt only *five,* selecting, at least, two from each sections (to be answered in not more than 150-180 words). The questions should be chosen in such a manner that all the seven units given in the text are covered.

(25 Marks)

Section-B (Composition & Grammar)

Q. 3. Write an essay (in not more than 500 words), choosing *one* topic out of the given *three.* A wide range of topics should be given and in this respect ideas may be drawn from the prescribed text.

(15 Marks)

Q. 4. A question on Précis writing shall be set, without any internal choice. For this purpose, the passage chosen should be simple, lucid and coherent and must not exceed 240-250 words.

(10 Marks)

Q. 5. (a) Antonyms-***five*** out of ***eight.*** (5 Marks)

(b) Correct the sentences- ***five*** out of ***eight.*** (5 Marks)

(c) Idioms and Phrases to be used in sentences. ***Five*** out of ***eight.*** (5 Marks)

(d) One word substitution- ***five*** out of ***eight***. (5 Marks)

ਪੰਜਾਬੀ (ਲਾਜ਼ਮੀ)

ਸਿਰਫ ਬੀ. ਏ. ਦੇ ਵਿਦਿਆਰਥੀਆਂ ਲਈ

(ਬੀ. ਏ. (ਜਨਰਲ) ਭਾਗ ਤੀਜਾ 2015 ਦੇ ਇਮਤਿਹਾਨ ਲਈ)

ਪੇਪਰ : ਏ

ਕੁੱਲ ਅੰਕ: 50
ਲਿਖਤੀ: 45
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 5
ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1. ਮੱਧਕਾਲ ਦੀਆਂ ਕਵਿਤਾਵਾਂ ਦਾ ਅਧਿਐਨ — 10 ਅੰਕ
2. ਨਾਵਲ ਦਾ ਅਧਿਐਨ — 10 ਅੰਕ
3. ਨਿਬੰਧਾਂ ਦਾ ਅਧਿਐਨ — 10 ਅੰਕ
4. ਉਪਰੋਕਤ ਪੁਸਤਕਾਂ ਤੇ ਆਧਾਰਤ ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ — 15 ਅੰਕ

ਕੋਰਸ

1. **ਗੁਰਮਤਿ ਅਤੇ ਸੂਫ਼ੀ ਕਾਵਿ**, (ਸੰਪਾ.) ਪ੍ਰੋ. ਪਰਮਜੀਤ ਸਿੰਘ ਸਿੱਧੂ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।
   ਜ਼ਰੂਰੀ ਨੋਟ: –
   i. ਵਜੀਦ ਦੇ ਸਲੋਕ, ਪੰਨਾ: 64, 65, 66
   ii. ਸੁਲਤਾਨ ਬਾਹੂ, ਪੰਨਾ: 71, 72, 73

   ਅਤੇ

   ਖ਼ਵਾਜ਼ਾ ਗੁਲਾਮ ਫਰੀਦ ਦੀਆਂ ਸਾਰੀਆਂ ਕਵਿਤਾਵਾਂ ਸਿਲੇਬੱਸ ਤੋਂ ਬਾਹਰ ਹਨ।
2. **ਤਫ਼ਤੀਸ਼**, ਮਿੱਤਰ ਸੈਨ ਮੀਤ, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ, ਲੁਧਿਆਣਾ
3. **ਚੋਣਵੇਂ ਲੇਖ**, ਪ੍ਰਿੰਸੀਪਲ ਤੇਜਾ ਸਿੰਘ, ਪੰਜਾਬੀ ਸਾਹਿਤ ਪ੍ਰਕਾਸ਼ਨ, ਅੰਮ੍ਰਿਤਸਰ।

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1. ਕਾਵਿ-ਸੰਗ੍ਰਹਿ ਵਿਚੋਂ ਪ੍ਰਸੰਗ ਸਹਿਤ ਵਿਆਖਿਆ (ਤਿੰਨ ਵਿਚੋਂ ਇਕ) — 5 ਅੰਕ
2. ਕਵਿਤਾ ਦਾ ਸਾਰ ਜਾਂ ਕੇਂਦਰੀ ਭਾਵ (ਤਿੰਨ ਵਿਚੋਂ ਇਕ) — 5 ਅੰਕ
3. ਨਾਵਲ ਦਾ ਵਿਸ਼ਾ/ਸਾਰ/ਪਾਤਰ ਚਿਤਰਨ ਬਾਰੇ ਪ੍ਰਸ਼ਨ (ਦੋ ਵਿਚੋਂ ਇਕ) — 10 ਅੰਕ
4. 'ਚੋਣਵੇਂ ਲੇਖ' ਵਿਚੋਂ ਕਿਸੇ ਇਕ ਕਾਂਡ ਦਾ ਸਾਰ (ਤਿੰਨਾਂ ਵਿਚੋਂ ਇਕ) — 10 ਅੰਕ
5. ਗੁਰਮਤਿ ਅਤੇ ਸੂਫ਼ੀ ਕਾਵਿ, ਨਾਵਲ ਅਤੇ ਨਿਬੰਧਾਂ ਦੀ ਪੁਸਤਕ ਦੇ ਆਧਾਰ ਤੇ ਲਘੂ-ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ : (ਉੱਤਰ ਇਕ ਤੋਂ ਤਿੰਨ ਸਤਰਾਂ ਤੱਕ)
   (ੳ) ਕਾਵਿ-ਸੰਗ੍ਰਹਿ ਵਿਚਲੀਆਂ ਕਵਿਤਾਵਾਂ ਨਾਲ ਸੰਬੰਧਤ ਲਘੂ-ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠ ਵਿਚੋਂ ਪੰਜ) — 5 x 1=5 ਅੰਕ
   (ਅ) ਨਾਵਲ ਵਿਚਲੇ ਕਾਂਡਾਂ/ ਘਟਨਾਵਾਂ ਅਤੇ ਤੱਥਾਂ ਨਾਲ ਸੰਬੰਧਤ ਲਘੂ-ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠ ਵਿਚੋਂ ਪੰਜ) — 5 x 1=5 ਅੰਕ
   (ਏ) 'ਚੋਣਵੇਂ ਲੇਖ' ਪੁਸਤਕ ਵਿਚੋਂ ਲਘੂ-ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠ ਵਿਚੋਂ ਪੰਜ) — 5 x 1=5 ਅੰਕ

ਪੇਪਰ : ਬੀ

ਕੁੱਲ ਅੰਕ: 5●
ਲਿਖਤੀ: 45
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 5
ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1.	ਲੇਖਕਾਂ ਦਾ ਜੀਵਨ, ਰਚਨਾ ਅਤੇ ਯੋਗਦਾਨ	5 ਅੰਕ
2.	ਪੈਰ੍ਹਾ ਰਚਨਾ	1● ਅੰਕ
3.	ਗੁਰਮੁਖੀ ਲਿੱਪੀ	1● ਅੰਕ
4.	ਅਖਬਾਰਾਂ ਲਈ ਪ੍ਰੈੱਸ ਨੋਟ	8 ਅੰਕ
5.	ਸ਼ਬਦਾਂ ਦੇ ਅਰਥ	4 ਅੰਕ
6.	ਵਿਆਕਰਣ	8 ਅੰਕ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1.	ਕਾਵਿ-ਸੰਗ੍ਰਹਿ ਵਿਚੋਂ ਨਿਰਧਾਰਤ ਕਵੀਆਂ ਦੇ ਜੀਵਨ, ਰਚਨਾ ਅਤੇ ਯੋਗਦਾਨ ਬਾਰੇ ਪ੍ਰਸ਼ਨ (ਤਿੰਨਾਂ 'ਚੋਂ ਇਕ)	5 ਅੰਕ
2.	ਲਗਭਗ 25● ਸ਼ਬਦਾਂ ਵਿਚ ਪੈਰ੍ਹਾ ਰਚਨਾ ਕਰਨੀ (ਚਾਰ ਵਿਚੋਂ ਇਕ)	1● ਅੰਕ
3.	ਗੁਰਮੁਖੀ ਲਿੱਪੀ ਦੀ ਮੁੱਢਲੀ ਜਾਣਕਾਰੀ ਨਾਲ ਸੰਬੰਧਤ ਪ੍ਰਸ਼ਨ (ਦੋ ਵਿਚੋਂ ਇਕ) ਗੁਰਮੁਖੀ ਲਿੱਪੀ ਦਾ ਜਨਮ ਅਤੇ ਵਿਕਾਸ, ਨਾਮਕਰਣ, ਗੁਰਮੁਖੀ ਲਿੱਪੀ ਦੀ ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਲਈ ਅਨੁਕੂਲਤਾ ਅਤੇ ਪ੍ਰਮੁੱਖ ਵਿਸ਼ੇਸ਼ਤਾਵਾਂ	1● ਅੰਕ
4.	ਅਖ਼ਬਾਰਾਂ ਲਈ ਪ੍ਰੈੱਸ ਨੋਟ ਤਿਆਰ ਕਰਨੇ ਕਾਲਜ ਦੀਆਂ ਸਾਹਿਤਕ, ਸਭਿਆਚਾਰਕ, ਖੇਡਾਂ ਅਤੇ ਅਕਾਦਮਿਕ ਖੇਤਰ ਨਾਲ ਸੰਬੰਧਤ ਸਰਗਰਮੀਆਂ ਬਾਰੇ (ਲਗਭਗ 2●● ਸ਼ਬਦਾਂ ਵਿਚ) (ਸਿਰਫ ਇਕ ਪ੍ਰੈੱਸ ਨੋਟ ਹੀ ਤਿਆਰ ਕਰਨ ਲਈ ਕਿਹਾ ਜਾਵੇ)	8 ਅੰਕ
5.	ਕਾਵਿ-ਸੰਗ੍ਰਹਿ ਵਿਚੋਂ ਅੱਠ ਸ਼ਬਦਾਂ ਦੇ ਅਰਥ ਦੱਸੋ (ਦਸ ਵਿਚੋਂ ਅੱਠ)	4 ਅੰਕ
6.	(ੳ) ਬਹੁ-ਅਰਥਕ ਸ਼ਬਦਾਂ ਦੀ ਵਾਕਾਂ ਵਿਚ ਵਰਤੋਂ (ਛੇ ਵਿਚੋਂ ਚਾਰ)	4 ਅੰਕ
	(ਅ) ਸ਼ਬਦ-ਜੁੱਟਾਂ ਦੇ ਅਰਥਾਂ ਦਾ ਵਾਕਾਂ ਦੁਆਰਾ ਅੰਤਰ ਸਪਸ਼ਟ ਕਰਨਾ (ਛੇ ਵਿਚੋਂ ਚਾਰ)	4 ਅੰਕ

ਸਹਾਇਕ ਪੁਸਤਕਾਂ

1. ਪੰਜਾਬੀ ਸੰਚਾਰ ਯੋਗਤਾ ਅਭਿਆਸ, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ
2. ਕਾਲਜ ਪੰਜਾਬੀ ਵਿਆਕਰਣ, ਡਾ. ਹਰਕੀਰਤ ਸਿੰਘ ਤੇ ਗਿਆਨੀ ਲਾਲ ਸਿੰਘ, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।

ਨੋਟ : ਹਫਤੇ ਦੇ 6+3 = 9 ਪੀਰੀਅਡ

HISTORY AND CULTURE OF PUNJAB

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

INSTRUCTIONS FOR THE PAPER SETTER AND CANDIDATES :

1. *The syllabus has been divided into four units.*
   There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short answer type questions, spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any **9** short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

2. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper setter must put note (2) in the question paper.

HISTORY AND CULTURE OF PUNJAB 1849-1966 A.D.

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 hours

Objectives : To introduce the students to the history of Punjab and culture in modern times.

Pedagogy : Lectures, library work and discussions.

Unit-I

1. Main features of the early British administration.
2. British policy towards the army, agriculture, industry, trade and commerce.
3. Spread of modern education at primary, secondary and higher levels.

Unit-II

4. The foundation, programme and impact of the socio-religious reform movements : Arya Samaj, Singh Sabhas, Ahmadiyas.
5. Contribution to the growth of political consciousness by the Namdhari movement, uprising of 1907 and Ghadar movement.

6. Contribution to the freedom struggle with special reference to the Gurdwara Reform Movement, Jallianwala Bagh, Bhagat Singh, Non-cooperation and Civil Disobedience.

Unit-III

7. Significant developments leading to independence and partition.
8. Rebuilding of social and economic life after partition.
9. The main stages in the movement for the Punjabi Speaking State; Reorganisation Act of 1966.

Unit-IV

10. New trends in Social life : Gender discrimination; emigration from Punjab.
11. New trends in Economic life : Modernisation of agriculture; land reforms.
12. Development of Punjabi literature with special reference to prose, poetry and drama.

Suggested Readings :

1. Fauja Singh (ed.) : *History and Culture of the Punjab, Part II,* Publication Bureau, Punjabi University, Patiala, 1987.

N.B. : The required detail and depth would conform to the treatment of the subject in the above survey. It would also form the basis for one to two sentence answer questions.

2. Fauja Singh : *Freedom Struggle in the Punjab,* Punjabi University, Patiala, 1974.
3. J.S. Grewal : *The Sikhs of the Punjab*, The New Cambridge History of India, Orient Longman, 1990, Foundation Books, New Delhi, 1994.
4. Khushwant Singh : *A History of the Sikhs*, Vol. II, 1839-1988, Oxford University Press, Delhi, 1991.
5. Satya, M. Rai : *Heroic Tradition in the Punjab,* 1900-1947, Punjabi University, Patiala, 1978.
6. P.N. Chopra, B.N. Puri and M.N. Das : *A Social, Cultural & Economic History of India,* Vol. III, Macmillan, Delhi, 1974.
7. K.C. Yadav and S.M. Fogat : *Haryana Aitihasik Simhavalokan* (Hindi), Haryana Sahitya Akademy, Chandigarh, 1991.
8. R.S. Johar, and J.S Khanna : *Studies in Punjab Economy,* GNDU, Amritsar, 1983.
9. Monica Das Gupta, T.N. Krishna & Lincon C. Chen : *Women's Health in India, Risk and Vulnerability*, Oxford, OUP, 1998.

ENGLISH (Elective)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015
Outlines of Tests, Syllabi and Courses of Reading

Objective:

The main aim of teaching English (Elective) to B.A. III students is to enable them to approach a wide variety of literary texts and genres with critically sensitive and analytical understanding. The idea is to introduce the students to the basic concepts of literature and also empower them to read, analyze and write about a poem, prose essay or drama in an independent manner. It is with this modest aim in mind that the new text books have been selected for B.A. English Third Year(Elective) courses. The focus of the new text books is two-fold: to teach finer nuances of literature and language through an integrated approach; and to help and motivate the students to develop basic tools of analyzing a variety of literary texts/genres.

Note:

(i) There will be two papers in all the three years of B.A. English (Elective) courses. Rather than divide the papers vertically, there will be horizontal division in terms of texts, composition and grammar. As two books are prescribed for each course, these two books shall be distributed across two papers and shall not be included in the same paper. So, **Modern Indian Literature : Poems and Short Stories** shall figure in **Paper A** and **R.K. Narayan's *Waiting for the Mahatma*** shall be included in **Paper B.** questions on composition and grammar shall, however, be included in Paper B. This is being done to help the students, who shall now find both the papers proportionately divided and so easily manageable.

(ii) Each theory paper shall be of 90 marks, as 10 marks in each paper shall be reserved for Internal Assessment. Each paper shall be further sub-divided into Sections, I & II. The mode of distribution of marks shall, however, vary from paper to paper. There shall be six questions in each paper. All the questions will be compulsory. Though internal choice may be offered in some of the questions, there will be no external choice.

(iii) Each paper shall include minimum 15 literary terms/concepts appropriate to the particular literary genre included in that paper.

(iv) The questions should be set strictly in accordance with the pattern of question paper outlined in the syllabus.

(v) For the successful implementation of this syllabus, it is necessary that we reflect objectives of this course, first in our teaching practices, and then in designing question paper/s and evaluating answer scripts of our students. The examination shall be held at the end of the term and/or year, as is recommended by the university, from time to time.

Texts Prescribed

1. **Modern Indian Literature : Poems and Short Stories,** Edited by the Department of English**,** University of Delhi, OUP, 2007 New Delhi.
2. **R.K. Narayan, *Waiting for the Mahatma,*** Hyderabad: Orient Paperbacks, 2007.

Testing Pattern:

Paper A

Max. Marks : 100
Theory : 90Marks
Internal Assessment: 10 Marks
Time :Three Hours

Section I

{Literary Terms/Concepts: *Indian Literatures, Indian Languages, Colonialism, Post-colonial, Multicultural Society, Orientalism, Hybridity, Ideology, Gender, Race, Class, Caste, Nation, Importance of Translation in India, Methods of Translation}*

Q. 1 It shall be on literary terms/concepts. *Eight* terms shall be given in all, and the students will be required to do *five.*

(15 Marks)

Q. 2 The examiner will set seven short questions (each to be answered in 60-80 words), based on **Modern Indian Literature : Poems and Short Stories** out of which the students shall be required to attempt only *five.*

(15 Marks)

Q. 3. There will be three long questions, out of which two long questions are to be answered each in about 180-200 words. These questions shall be based on *Modern India Literature : Poems and Short Stories* .

(15 Marks)

Section - II

Q. 4. Note-making (A passage of about 1000 words is to be given for this purpose)

(15 Marks)

Q. 5. An unseen Passage for Comprehension (about 1000 words, again) with ten multiple choice questions is to be given.

(15 Marks)

Q. 6. Applied Grammar:

(a) 5 Pairs of words to be used in sentences (altar/alter, compliment/complement etc.).

(1×5=5 Marks)

(b) First find one word for many and then use it in a sentence.

(5 Marks)

(c) First change the form of nouns/verbs/adjectives and then make sentences. (1×5=5 Marks)

Paper B

Max. Marks : 100
Theory : 90Marks
Internal Assessment: 10 Marks
Time :Three Hours

Section I

{Literary Concepts/Terms: *Fiction, Non-fiction, Narrative, Bildungroman, Picaresque Novel, Novel of Sensibility, Historical Romance, Gothic Novel, Realistic Novel, Psychological Novel,Political Novel, Regional Novel,Point of View, Omniscient Author, First Person Narrator, Stream of Consciousness, Protagonist/Antagonist, Plot/Story, Character, Structure}*

Q. 1. Literary terms/concepts, *five* out of *eight* (15 Marks)

Q. 2. Short questions based on the prescribed novel, *ten* out of *twelve* (each in 60-80 words) (15 Marks)

Q. 3. Long questions based on the novel, *three* out of *five* dealing with the incidents, theme(s), character(s), symbols etc., (each in 180-200 words)

(15 Marks)

Section - II

Q. 4. Essay on any one (out of the given three) topic of international importance (in about 700 words)

(20 marks)

Q. 5. Report-writing (in about 300 words) on an incident/situation, conference/seminar, problem/state of education/poverty/unemployment or similar issues

(10 Marks)

Q. 6. Translation from Hindi/Punjabi into English (Passage of about 400 Words)

Or

Paragraph on any *one* out of the *two* given topics (For Foreign Students only)

(15 Marks)

हिन्दी (ऐच्छिक)

बी. ए. (सामान्य) तृतीय वर्ष परीक्षा, 2015

पत्र - एक **पूर्णांक : 90 + 10 = 100**

समय : 3 घण्टे

1. **कुरुक्षेत्र :** अंक : 22

रामधारी सिंह 'दिनकर' प्रकाशक - राजपाल एण्ड संज, नई दिल्ली ।

(क) 6 - 6 अंकों की दो संदर्भ सहित व्याख्याएँ करनी होंगी। कुल चार काव्यांश दिए जायेंगे।

(ख) 10 अंकों का एक आलोचनात्मक प्रश्न करना होगा। कुल दो प्रश्न पूछे जायेंगे।

2. **गद्य फुलवारी,** सम्पादक डॉ॰ शहाबुदीन शेख, प्रकाशक - राजपाल एण्ड संज, नई दिल्ली। अंक: 16
केवल निम्नलिखित पाठ निर्धारित हैं -

'आँसुओं की होली' (प्रेमचंद), 'अकेली' (मन्नु भंडारी), 'चीफ की दावत' (भीष्म साहनी), 'सुभान खाँ' (रामवृक्ष बेनीपुरी), 'भाभी' (महादेवी वर्मा), 'सदाचार का ताबीज' (हरिशंकर परसाई), 'महात्मा गाँधी' (रामकुमार वर्मा), 'मैं धोबी हूँ' (शिवपूजन सहाय), 'गप - शप' (नामवर सिंह), 'जमनोत्री की यात्रा' (विष्णु प्रभाकर)।

3. इस खंड में तीन - तीन अंकों के चार लघु - उत्तरापेक्षी प्रश्न करनें होंगे। कुल आठ प्रश्न पूछे जाएंगे। प्रत्येक उत्तर की शब्द - सीमा 50 होगी। ये प्रश्न इस पत्र के पूर्वोक्त दो खंडों 'कुरुक्षेत्र' एवं 'गद्य फुलवारी' पर आधाारित होंगे। अंक : 12

4. **हिन्दी साहित्य का इतिहास :** अंक : 20
केवल निम्नलिखित गद्य - विधाओं का उद्‌भव और विकास : उपन्यास, कहानी, नाटक, निबन्ध, आत्मकथा, जीवनी, संस्मरण, रेखाचित्र ।
इन गद्य विधाओं में से 10 - 10 अंकों के कम से कम चार प्रश्न पूछे जायेंगे, छात्रों को केवल दो करने होंगे।

5. **वस्तुनिष्ठ प्रश्न :** अंक : 20

नोट : उपर्युक्त तीन खंडों की सामग्री के आधाार पर 2 - 2 अंकों के दस वस्तुनिष्ठ प्रश्नों के उत्तर पूछे जायेंगे। कोई विकल्प नहीं होगा।

6. **आंतरिक मूल्यांकन :** अंक : 10

पत्र - दो **पूर्णांक : 90 +10 = 100**

समय : 3 घण्टे

1. **निबन्ध लेखन** (केवल साहित्यिक और सामाजिक विषयों पर) अंक : 16

कुल आठ निबन्धों में से किसी एक निबन्ध पर लिखने के लिए कहा जाएगा।

2. **समीक्षा सिद्धान्त :** अंक : 24

(क) काव्य की परिभाषा तथा भेद, महाकाव्य, खंडकाव्य, गीतिकाव्य की परिभाषा तथा विशेषताएँ।

(ख) गद्य विधाएँ - निबन्ध, संस्मरण, जीवनी तथा आत्मकथा के स्वरूप और तत्त्वों का समान्य परिचय।

उपर्युक्त खंड (क) और (ख) से संबंधित 12 - 12 अंकों के चार प्रश्न पूछे जायेंगे। इनमें से केवल दो के उत्तर देने होंगे।

3. **छन्द – अंलकार परिचय :** अंक : 30

क) केवल निम्नलिखित अलंकार निर्धारित हैं :-

अनुप्रास, यमक, श्लेष, वक्रोक्ति, उपमा, रूपक, अतिषयोक्ति, विरोधाभास, उत्प्रेक्षा, प्रतीप।

ख) केवल निम्नलिखित छंद निर्धारित हैं-

दोहा, सोरठा, चौपाई, रोला, कुण्डलियाँ, सवैया, द्रुतविलम्बित, हरिगीतिका, उपेन्द्रव्रजा, इन्द्रवज्रा।

खंड (क) और (ख) में से 5 - 5 अलंकारों और छन्दों के केवल लक्षण और उदाहरण पूछे जायेंगे, जिनमें से 3 - 3 के उत्तर देने होंगे।

4. **हिन्दी भाषा और उसकी लिपि :** अंक : 20

(क) तार/निमन्त्राण पत्र, विज्ञप्ति/विज्ञापन का रूप तैयार करना।

(ख) देवनागरी लिपि : विकास, गुणदोष, सुधाार के उपाय।

इनमें से 10 - 10 अंकों के कुल चार प्रश्न पूछे जाएंगे, जिनमें से छात्रों को दो के उत्तर देने होंगे।

5. **आतंरिक मूल्यांकन** अंक : 10

सहायक पुस्तकें :

1. हिन्दी का आत्मकथा-साहित्य, विश्वबंधु 'व्यथित', राधा पब्लिकेशन, नई दिल्ली।

2. काव्य के तत्त्व : आचार्य देवेन्द्रनाथ शर्मा, लोकभारती प्रकाशन, इलाहाबाद-1

संस्कृत (इलैक्टिव)

बी.ए.(जनरल) तृतीय वर्ष परीक्षा, 2015

PAPER-A: **उपनिषद्, रामायण एवं शब्दावली** **पूर्णांक: 90+10=100 अंक**

(आन्तरिक परीक्षा- 10, लिखित परीक्षा- 90) **समय – 3 घण्टे**

टिप्पणी- प्रश्नपत्र का माध्यम हिन्दी होगा । उत्तरों का माध्यम संस्कृत, हिन्दी, पंजाबी या अंग्रेजी में से कोई एक भाषा होगी ।

उद्देश्य-

- वैदिक काव्य के माध्यम से वैदिक भाषा के स्वरूप तथा जन्मभूमि के प्रति अनुराग की भावना भरना।
- वैदिक दर्शन के सामान्य सिद्धान्त -ईश्वर, त्याग, कर्म, विद्या, अविद्या, सम्भूति, असम्भूति आदि औपनिषदिक विषयों का अध्ययन कराना।
- आदि काव्य वाल्मीकि रामायण तथा लौकिक काव्य के सौन्दर्य का दिग्दर्शन कराना ।
- इसके अतिरिक्त व्यावहारिक शब्दावली द्वारा विद्यार्थियों की संस्कृत के प्रति रुचि जागृत करना।

(क) **अथर्ववेद**(काण्ड-12, प्रथमसूक्त/भूमिसूक्त के प्रथम बीस मन्त्र) - अनुवाद मात्र **1 × 10 = 10 अंक**

(ख) **ईशोपनिषद्** (मन्त्र/मन्त्रांश का अनुवाद एवं व्याख्या) **1 × 10 = 10 अंक**

(ग) **वाल्मीकिरामायण**(सुन्दरकाण्ड, सर्ग-15, प्रथम तीस श्लोक, गीता प्रैस गोरखपुर) श्लोक व्याख्या **2× 10 = 20अंक**

(घ) **रघुवंश (प्रथम सर्ग) –कालिदास** **30अंक**

(i) श्लोको का सप्रंसग अनुवाद व व्याख्या 2× 10= 20 अंक

(ii) सूक्ति-सप्रंसग व्याख्या(तीन में से एक) 1× 10= 10 अंक

(ङ) **व्यावहारिक संस्कृत शब्दावली** **20 अंक**

(i) वाद्य, खेल, सम्बन्धसूचक एवं प्रशासनिक शब्दसूची 5× 2= 10 अंक

1. ढिंढोरा	- डिण्डिम:	2. ढोल	- पटह:
3. ढोलक	- ढोलक:	4. तवला	- मुरज:
5. तानपुरा	- तानपुरा:	6. तुरही (शहनाई)	- तूर्यम्
7. नगाड़ा	- दुन्दुभि:	8. पियानो	- तन्त्रीवाद्यम्
9. बांसुरी	- मुरली	10. मंजीरा	- मञ्जीरम्
11. सारंगी(वायलिन)	- सारंगी	12. सितार	- वीणा
13. हारमोनियम	- मनोहारिवाद्यम्	14. गेंद	- कन्दुक:
15. टेनिस का खेल	- टेनिस-क्रीडा	16. नेट	- जालम्
17. फुटबाल	- पादकन्दुक:	18. बैड मिंटन	- पत्रिक्रीडा
19. मैच	- क्रीडाप्रतियोगिता	20. रेफरी	- निर्णायक:
21. रैकेट	- काष्ठपरिष्कर:	22. बालीबाल	- क्षेपकन्दुक:

23. हाकी का खेल	– यष्टिक्रीडा	24. पिता/माता	– जनकः, पिता/जननी, माता
25. चाचा, चाची	– पितृव्यः, पितृव्या।	26. पति, पत्नी	– पतिः, पत्नी(भार्या, जाया)
27. नाना, नानी	– मातामहः, मातामही	28. छोटा भाई, बड़ा भाई	– अनुजः, अग्रजः
29. दादा, दादी	– पितामहः, पितामही	30. पुत्र, पुत्री	– पुत्रः/आत्मजः, पुत्री/आत्मजा।
31. जीजा	– भगिनीपतिः	32. जंवाई(दामाद)	– जामाता
33. देवर, देवरानी	– देवरः, याता	34. मामा, मामी	– मातुलः, मातुली
35. साला, साली	– श्यालः(स्यालः), श्याली(स्याली)	36. सास, ससुर	– श्वश्रूः, श्वशुरः
37. समधी(कुड़म), समधिन(कुड़मनी)	– सम्बन्धी, सम्बन्धिनी	38. पोता, पोती	– पौत्रः, पौत्री
39. भतीजा,भतीजी	– भ्रातृपुत्रः/ भ्रातृव्यः, भ्रातृपुत्री/ भ्रातृव्या	40. भानजा,भानजी	– भागिनेयः, भागिनेयी
41. चचेरा भाई, चचेरी बहिन	– पितृव्यपुत्रः, पितृव्यपुत्री	42. भाभी	– भ्रातृजाया
43. Administration	– प्रशासनम्	44. Appeal	– निवेदनम्
45. Approval	– अनुमोदनम	46. Ban	– प्रतिबन्धः
47. Cabinet	- मन्त्रिमण्डलम्	48. Corporation	- निगमः
49. Director	- निदेशकः	50. Employment	- आजीविका, वृत्तिः
51. Initials	- आद्यक्षरम्	52. Judgement	- निर्णयः
53. Gazetted Officer	- राजपत्रिताधिकारी	54. Notification	- अधिसूचना
55. Procedure	- कार्यविधिः	56. Receipt	- प्राप्तिः
57. Senior	- वरिष्ठः	58. Subordinate	- अधीनस्थः
59. Transfer	- स्थानान्तरणम्	60. Vacancy	- रिक्तिः

(ii) पंजाबी में प्रयुक्त संस्कृत के तत्सम तथा तद्भव शब्दों की सूची 5× 2= 10 अंक

संस्कृत शब्द	पंजाबी शब्द	संस्कृत शब्द	पंजाबी शब्द
1. चिरम्- (विलम्ब,देरी, दीर्घकाल)	चिर	11. पदम् (पैर)	पउं
2. वेला (समय, अवसर)	बेला	12. नयनम् (नेत्र)	नैणूं
3. शाकः (हरि सब्जी)	साग	13. पुच्छम् (पूंछ)	पूंछ
4. कर्करम् (ढेला, कंकर)	कंकर	14. बाल्यम् (बचपन)	बालम
5. चतुष्कम् (चौंक)	चौंक, चौंका	15. बीजम् (बीज)	बीं, बीउं
6. द्वयम् (दो, जोड़ा)	दोंह	16. भयम् (डर)	भैं, भउं, भंऊ
7. धावनम् (धोना)	धावण	17. माषम् (उड़द)	मांह, महां
8 .नखम् (नाखून)	नउंह, नहुं, नौंह	18. मनोघटितम् (स्वकल्पित)	मनघड़ंत
9. निषिद्धम् (बुरा, अपवित्र)	नखिंध	19. मुखम् (मुंह)	मूंह, मूं
10. नीलम्(एक रत्न)	नीलम, लीलम	20. युगलम् (जोड़ा)	जुगलां

21. रटितम् (रटा हुआ)	रटंत	31. तदा (तब)	तद
22. लोहितम्(खून)	लहूं	32. यदा (जब)	जद
23. वसितम्, वसितिः(वस्ती)	वसूं, वसों	33. कृतः (किया)	कीत्ता
24. हृदयम् (हृदय, दिल)	हिउं, हियां	34. चन्द्रः (चांद)	चन्द
25. श्वश्रूः(सास)	सस्सू	35. त्वम् (तुम)	तूं
26. वधूः (बहू)	वहू	36. निम्बः (नीम)	निम्म
27. अद्य (आज)	अज्ज	37. पिप्पलः (पीपल)	पिप्पल
28. अर्धः (आधा)	अद्धा	38. बलदः (बैल)	बलद, बौलद
29. उच्चैः(ऊँचा)	उच्चा	39. हस्तः (हाथ)	हत्थ
30. कदा (कब)	कद	40. सर्वः (सब)	सब्भ

नोट- 1. पत्र का अध्ययनसमय छह पीरियड प्रतिसप्ताह होगा ।
2. सभी प्रश्नों में शतप्रतिशत विकल्प आवश्यक हैं ।

PAPER-B: व्याकरण, साहित्य एवं अनुवाद

पूर्णांकः 90+10=100 अंक

(आन्तरिक परीक्षा- 10, लिखित परीक्षा- 90) **समयः 3 घण्टे**

टिप्पणी- प्रश्न पत्र का माध्यम हिन्दी होगा । उत्तरों का माध्यम संस्कृत, हिन्दी, पञ्जांबी या अंग्रेजी में से कोई एक भाषा होगी ।

उद्देश्य-

- सन्धि समास, अलंकारों द्वारा छात्र की व्याकरण पृष्ठभूमि तथा अलंकार परिज्ञान को सुदृढ. करना।
- वैदिक तथा लौकिक संस्कृत साहित्य के इतिहास का परिज्ञान कराना।
- संस्कृत निबन्ध लेखन में प्रवीण करना।

(क) विसर्ग सन्धि 5X 2 = 10 अंक

(ख) समास - (अव्ययीभाव, बहुव्रीहि) 5X 2 = 10 अंक

(ग) अलंकार (लक्षण, उदाहरण व स्पष्टीकरण) 2 X 10 = 20अंक

(उपमा, उत्प्रेक्षा, रूपक, विभावना, विशेषोक्ति, अनुप्रास, यमक, अर्थान्तरन्यास, श्लेष व दृष्टान्त-तीन में से दो)

(घ) वैदिक इतिहास (लघु प्रश्न/टिप्पणी) 2X 5 = 10 अंक

(i) वेदों का सामान्य परिचय

(ii) वेदांग साहित्य का सामान्य परिचय

(ङ) लौकिक संस्कृत साहित्य का इतिहास(लघु प्रश्न) 2X 7½ = 15 अंक

(भास, कालिदास, भवभूति, बाणभट्ट, माघ, भर्तृहरि व दण्डी की कृतियों का परिचय)

(च) णिच् तथा सन् प्रत्यय 5X 2 = 10 अंक

(i) णिच् प्रत्यय केवल भू, पठ्, हस्, गम्, कृ, अधि+इङ्, खाद्, चल्, नश् व नम् धातु से लट् प्रथम पुरुष, एकवचन ही प्रष्टव्य हैं।

(ii) सन् प्रत्यय केवल स्वप्, घ्रा, पठ्, कृ, गम्, मृ, मुच्, दा, तृ तथा भू धातु से लट् प्रथम पुरुष, एकवचन ही प्रष्टव्य हैं।

(छ) संस्कृत निबन्ध (तीन में से एक विषय पर दस पंक्तियों में) 1X 15 = 15 अंक

(विषय- संस्कृतभाषाया महत्त्वम्, मम प्रियः कविः, मम प्रियं पुस्तकम्, दीपावली, सत्संगतिः, पर्यावरणसुरक्षा)

नोट- 1. पत्र का अध्ययनसमय छह पीरियड प्रतिसप्ताह होगा ।
2. सभी प्रश्नों में शतप्रतिशत विकल्प आवश्यक हैं ।

ਪੰਜਾਬੀ (ਇਲੈਕਟਿਵ)

ਬੀ. ਏ. (ਜਨਰਲ) ਭਾਗ ਤੀਜਾ 2015 ਦੇ ਇਮਤਿਹਾਨ ਲਈ

ਪੇਪਰ : ਏ

ਕੁੱਲ ਅੰਕ: 100
ਲਿਖਤੀ: 90
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 10
ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1.	ਆਧੁਨਿਕ ਪੰਜਾਬੀ ਕਵਿਤਾ ਦਾ ਅਧਿਐਨ	40 ਅੰਕ
2.	ਪੰਜਾਬੀ ਨਾਟਕ ਦਾ ਅਧਿਐਨ	25 ਅੰਕ
3.	ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਬਾਰੇ ਨਿਬੰਧਾਂ ਦਾ ਅਧਿਐਨ	25 ਅੰਕ

ਕੋਰਸ

1. **ਨਕਸ਼ ਨੁਹਾਰ**, (ਸੰਪਾ.) ਡਾ. ਜਸਵਿੰਦਰ ਸਿੰਘ, ਆਧੁਨਿਕ ਪੰਜਾਬੀ ਕਵਿਤਾ (1901 ਈ. ਤੋਂ 2000 ਈ.), ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ
2. **ਕਾਮਾਗਾਟਾ ਮਾਰੂ**, ਹਰਚਰਨ ਸਿੰਘ, ਕਸਤੂਰੀ ਲਾਲ ਐਂਡ ਸੰਨਜ਼, ਅੰਮ੍ਰਿਤਸਰ
3. **ਪੰਜ ਆਬ**, (ਸੰਪਾ.) ਪ੍ਰੋ. ਭੁਪਿੰਦਰ ਸਿੰਘ ਖਹਿਰਾ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1.	(ੳ)	ਕਾਵਿ ਪੁਸਤਕ ਵਿਚੋਂ ਪ੍ਰਸੰਗ ਸਹਿਤ ਵਿਆਖਿਆ (ਚਾਰ ਵਿਚੋਂ ਦੋ)	2x10= 20 ਅੰਕ
	(ਅ)	ਤਿੰਨ ਕਵਿਤਾਵਾਂ ਵਿਚੋਂ ਕਿਸੇ ਇਕ ਦਾ ਸਾਰ	10 ਅੰਕ
	(ੲ)	ਤਿੰਨ ਕਵਿਤਾਵਾਂ ਵਿਚੋਂ ਕਿਸੇ ਇਕ ਦਾ ਕੇਂਦਰੀ ਭਾਵ	10 ਅੰਕ
2.	(ੳ)	ਨਾਟਕ ਦੀ ਪੁਸਤਕ ਵਿਚੋਂ ਸੰਖੇਪ/ ਲਘੂ ਪ੍ਰਸ਼ਨ (*ਅੱਠ ਵਿਚੋਂ ਪੰਜ ਕਰਨੇ ਹਨ*)	5x3 =15 ਅੰਕ
	(ਅ)	ਨਾਟਕ ਦਾ ਵਿਸ਼ਾ/ ਪਲਾਟ/ ਪਾਤਰ ਚਿਤਰਣ/ ਰੰਗਮੰਚ ਅਤੇ ਸਾਹਿਤਕ ਆਲੋਚਨਾ (*ਦੋ ਪ੍ਰਸ਼ਨਾਂ ਵਿਚੋਂ ਇਕ ਕਰਨਾ ਹੈ*)	10 ਅੰਕ
3.	(ੳ)	ਨਿਬੰਧਾਂ ਦੀ ਪੁਸਤਕ ਵਿਚੋਂ ਸੰਖੇਪ/ਲਘੂ ਪ੍ਰਸ਼ਨ (ਅੱਠ ਪ੍ਰਸ਼ਨਾਂ ਵਿਚੋਂ ਪੰਜ ਕਰਨੇ ਹਨ)	5x3 =15 ਅੰਕ
	(ਅ)	ਨਿਬੰਧ ਵਿਸ਼ਾ-ਵਸਤੂ/ ਸਾਹਿਤਕ ਆਲੋਚਨਾ (ਦੋ ਪ੍ਰਸ਼ਨਾਂ ਵਿਚੋਂ ਇਕ ਕਰਨਾ ਹੈ)	10 ਅੰਕ

ਪੇਪਰ : ਬੀ

ਕੁੱਲ ਅੰਕ: 1●●
ਲਿਖਤੀ: 9●
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 1●
ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ (19●1 ਈ. ਤੋਂ 2●●● ਈ. ਤੱਕ) 35 ਅੰਕ
2. ਸਾਹਿਤਕ ਵਾਦ 15 ਅੰਕ
3. ਸਾਹਿਤ ਦੇ ਰੂਪ 2● ਅੰਕ
4. ਲੋਕਧਾਰਾ ਅਤੇ ਪੰਜਾਬੀ ਲੋਕਧਾਰਾ 2● ਅੰਕ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1. **ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ** (19●1 ਈ. ਤੋਂ 2●●● ਈ.)
   (ੳ) ਕਵਿਤਾ, ਨਾਵਲ, ਨਾਟਕ, ਕਹਾਣੀ, ਵਾਰਤਕ, ਜੀਵਨੀ, ਸਵੈ-ਜੀਵਨੀ ਨਾਲ ਸੰਬੰਧਿਤ ਇਤਿਹਾਸਮੂਲਕ ਤੇ ਪ੍ਰਵਿਰਤੀਮੂਲਕ ਪ੍ਰਸ਼ਨ (ਦੋ ਵਿਚੋਂ ਇਕ) 2● ਅੰਕ
   ਨੋਟ: ਪੇਪਰ ਸੈਟਰ ਵੱਲੋਂ ਇਕ ਪ੍ਰਸ਼ਨ ਇਤਿਹਾਸਮੂਲਕ ਅਤੇ ਇਕ ਪ੍ਰਵਿਰਤੀਮੂਲਕ ਪੁੱਛਿਆ ਜਾਵੇ।
   (ਅ) 19●1 ਤੋਂ ਹੁਣ ਤੱਕ ਦੇ ਪੰਜਾਬੀ ਸਾਹਿਤ ਦੇ ਇਤਿਹਾਸ ਵਿਚੋਂ ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ 15 x 1=15 ਅੰਕ
   (ਉੱਤਰ ਦੋ-ਤਿੰਨ ਸਤਰਾਂ ਵਿਚ ਹੋਵੇ) (ਵੀਹ ਵਿਚੋਂ ਪੰਦਰਾਂ ਪ੍ਰਸ਼ਨ ਕਰਨੇ ਹਨ)

2. **ਸਾਹਿਤਕ ਵਾਦ** 15 ਅੰਕ
   ਸਨਾਤਨਵਾਦ, ਰੋਮਾਂਸਵਾਦ, ਯਥਾਰਥਵਾਦ, ਰਹੱਸਵਾਦ, ਪ੍ਰਗਤੀਵਾਦ (ਦੋ ਵਿਚੋਂ ਇਕ)

3. **ਸਾਹਿਤ ਦੇ ਰੂਪ** : ਨਜ਼ਮ, ਗੀਤ, ਗ਼ਜ਼ਲ, ਨਾਵਲ, ਕਹਾਣੀ, ਨਾਟਕ, ਸਫ਼ਰਨਾਮਾ, ਜੀਵਨੀ, ਰੇਖਾ-ਚਿੱਤਰ, ਸਵੈ-ਜੀਵਨੀ (ਚਾਰ ਵਿਚੋਂ ਦੋ ਸਵਾਲ ਕਰੋ) 1●+1● =20 ਅੰਕ

4. **ਲੋਕਧਾਰਾ ਅਤੇ ਪੰਜਾਬੀ ਲੋਕਧਾਰਾ** 1●+1●=2● ਅੰਕ
   (ੳ) ਲੋਕਧਾਰਾ ਦੀ ਪਰਿਭਾਸ਼ਾ ਅਤੇ ਵਿਸ਼ੇਸ਼ਤਾਵਾਂ
   (ਅ) ਜਨਮ, ਵਿਆਹ ਤੇ ਮੌਤ ਨਾਲ ਸੰਬੰਧਤ ਲੋਕ ਵਿਸ਼ਵਾਸ
   (ਏ) ਲੋਕ ਸਾਹਿਤ ਦੀ ਪਰਿਭਾਸ਼ਾ ਅਤੇ ਸਰੂਪ
   (ਸ) ਲੋਕਗੀਤ ਦੀ ਪਰਿਭਾਸ਼ਾ ਅਤੇ ਸਰੂਪ
   (ਤਿੰਨ ਵਿਚੋਂ ਦੋ ਸਵਾਲ ਕਰੋ)

<u>ਸਹਾਇਕ ਪੁਸਤਕਾਂ:</u>

1. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦੀ ਉਤਪਤੀ ਤੇ ਵਿਕਾਸ, ਡਾ. ਪਰਮਿੰਦਰ ਸਿੰਘ ਤੇ ਕਿਰਪਾਲ ਸਿੰਘ ਕਸੇਲ, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ, ਲੁਧਿਆਣਾ।
2. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ (ਆਧੁਨਿਕ ਕਾਲ), ਭਾਸ਼ਾ ਵਿਭਾਗ ਪੰਜਾਬ, ਪਟਿਆਲਾ

ਨੋਟ : ਸਮੁੱਚੇ ਪਾਠਕ੍ਰਮ ਲਈ ਹਫਤੇ ਵਿੱਚ 6+6=12 ਪੀਰੀਅਡ।

URDU (Elective)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Outlines of tests, syllabi and courses of reading

Paper-A 100 marks

Time : 3 hours

1. Explanation of Prose Passage 30 marks

2. Explanation of Verses

   I. Ghazal 20 marks

   II. Nazm 10 marks

3. Summary of a lesson or a poem 10 marks

4. Note on a Prose writer or a poet (which is prescribed in your course) 20 marks

5. Question on the basis of the following forms of prose and poetry:-

   1. Ghazal 2. Nazm 3. Azad Nazm 4. Paband Nazm

   5. Qasida 6. Marsiya 7. Masnavi

   1. Afsana 2. Novel 3. Drama 4. Inshaiya 10 marks

Book Prescribed

Shaoor-e-Adab, Maktaba Jamia, New Delhi-25.

I. All the lessons contained in the prose portion are prescribed except the following:-

1. Lucknow Ka Chehlam
2. Yeh London Hai
3. Natak

II. The complete portion of poetry is prescribed in the course.

Paper-B 100 marks

Time : 3 hours

Short Story, Composition and Literary History of Urdu Literature :-

1. *Study of Short Stories* : Summary and Central Idea 20 marks
2. *Composition* : Essay Writing 20 marks
3. *History of Urdu Literature* 60 marks

1. Urdu Ki Ibtida Aur Irtiqa
2. Urdu Ka Ahd-a-Zarreen, with special reference to the following :-

   (i) Ibrahim Zauq (ii) Asad-ulla-Khan Ghalib (iii) Momin Khan Momin

   (iv) Bahadur Shah Zafar.
3. Urdu Shairi Ke Naye Rujhanat with special reference to the following poets : -

   (i) Mohd. Husain Azad (ii) Altaf Hussain Hali (iii) Brij Narain Chakbast

   (iv) Dr. Iqbal.
4. Taraqqi Pasand Tehrik aur Urdu Shairi with special reference to the following poets :-

   (i) Asrar-ul-Haq Majaz (ii) Faiz Ahmed Faiz (iii) Ali Sardar Jafri

   (iv) Moin Alhsan Jazbi
5. Urdu Nasr Ka Irtiqa Aur Fort William College.
6. Urdu Nasr Ka Ahd-e-Zarreen with special reference to the following :-

   (i) Sir Syyed Ahmad Khan (ii) Maulana Shibli Nomani (iii) Altaf Hussain Hali
7. Modern Urdu Fiction :

   General information about the contribution of :

   (i) Krishan Chander (ii) Rajinder Singh Bedi (iii) Qurratulain Haider

   (iv) Ismet Chughtai & Kanhiya Lal Kapoor.

Book Prescribed

Urdu Afsane, Inshaiye aur Drame, Compiled by Mohd. Qasim Siddiqui, Educational Book House, Aligarh.

All the stories Inshaiye Aur Drame except following two :-

1. Nai Aur Purani Tehzib ke Takkar
2. Rustam-o-Suhrab.

Books Recommended

1. Mukhtsar Tarikh Adab-e-Urdu by Sayyed Ejaz Husain.
2. Tarikh Adab Urdu by Ram Babu Saxena.
3. Urdu Adab Ki Tarikh by Azeem-ul-Haq Junaidi.

PERSIAN (Elective)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

There will be two paper setter/examiner.

Paper-A : PROSE	Marks :	100
	Time :	3 hours
1. Translation of text pieces into English, Hindi, Urdu, Panjabi or Persian		25 marks
2. Explanation of text pieces into English , Hindi, Urdu, Panjabi or Persian		25 marks
3. Summary or central idea of the text prescribed as in Dastanha-ye-Kutah		20 marks
4. Simple direct questions on the life and works of the authors		20 marks
5. Short notes.		10 marks
	Total :	100 marks

Book prescribed

Nisabe Jadide Farsi (Published By Jayyad Press Ballimaran Delhi)

Only following portions from Prose Section :

(i) Intekhab-e-Sarzamin-e-Hind-Ali Asghar Hikmat, p. 33-60.

(ii) Dastanha-ye-Kutah.:-

(a) Azaan-e-Maghrib by Saeed Nafisi.

(b) Khukushi By Mohd. Hijazi.

Paper-B : POETRY Marks : 100

Time : 3 hours

1.	Translation of text pieces into English, Urdu, Hindi, Punjabi or Persian	25 marks
2.	Explanation of text pieces into English, Hindi, Panjabi, Urdu or Persian	25 marks
3.	Central idea of the poem	20 marks
4.	Simple direct questions on the life and works of the poets as prescribed in the text.	20 marks
5.	Short notes.	10 marks
	Total :	100 marks

Book Prescribed

Nisab-e-Jadeed-e-Farsi (Published by Jayyad Press Ballimaran, Delhi). Only the following portions of poetry section :

Az Gzalliyat-e-Hafiz :

Agar Aan Turke Shirazi Be Dast Aarad Dile Maara.

Saaqi Benur Badah Bar Afrooz Jam-e-Maa.

Doosh deedam Keh Malayek Dare Maiykhaneh Zadand.

Gazalliyat-e-Khdsrow :

Jan Ze Tan Burdi-o-Dar Jaani Hunooz.

Madeh Pandam ke Man Dar Seeneh Saudayee Digar Daram.

Masanaviy-e-Maulana Rum :

Deed Musa Yek Shabani Rah Be Rah.

Wahi Aamad Suy-e-Musa Az Khusa.

Qasayed-e-Urfi :

Dar Wasf-e-Kashmir.

ARABIC (Elective)

(*Kept in Abeyance*)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

There will be two papers of 100 marks each

Paper-A Marks : 100
Time : 3 hours

1.	Grammar	50 marks
	Arabia Ka Maullim (Part-IV) by Abdul Sattar Khan is prescribed for the course.	
2.	Translation of Arabic into English/Hindi/Urdu	30 marks
3.	Writing a Para in Arabic	20 marks

Paper-B Marks : 100
Time : 3 hours

1. Prose — 60 marks

The following topics of Al-Abarat by Al-Manfluti :

1. Al-Zikra
2. Al-Hawiyah
3. Al-Jaza

2. Poetry — 40 marks

The following poets from the book 'Majmu' at

Min-al-Nazmw al Nathr.

1. Abdullah Basha Fikri
2. Al-Barudi
3. Al-Mutanabbi
4. Abu-Tammam

(All books can be had from Kutub Khana Rashidiyya, Urdu Bazar, Jama Masjid, Delhi-6)

FRENCH (Elective)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : Introduction to Drama, Poetry and Prose

Max. marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

I Literary Master Pieces Drama, La Femme Du, Boulanger by Marcel Pagnol. 25 Marks

Questions, Character sketches and Critical appreciation pertaining to the drama to be asked and answered in French.

II Selected Reading – Poetry 25 Marks
Questions, explanations of stanzas of poems and central ideas, critical appreciation and summaries of the poems to be asked and answered in French.

III Text Book : Version Originale-3 Méthode de Français/Livre de I' élève, Fabrice Barthélémy, Christine Kleszewski, Émilie Perrichone Sylvie Wuattier, Conseil pédagogique et révision : Christian Puren, Editions Maison des Langues, Paris.

(a) Questions to be based on the prescribed text. 20 Marks

(b) Questions on French civilization in the form of fill in the blanks, multiple choice and short answers of 1-2 sentences. 20 Marks

CHOICE TO BE GIVEN IN ALL QUESTIONS

Paper-B : Applied Grammar, Creative Writing and Viva-Voce

Max. Marks	:	100
Theory (60) + Viva-Voce (30)	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hrs.

1. Translation from English into French and from French into English of an unseen passage or of short sentences. 20 Marks

2. A composition of about 150 - 200 words. 10 Marks

3. Questions on applied grammar [Questions to be based on grammar section of Prose – Version Originale-3 Méthode de Français/Livre de l' élève, Fabrice Barthélémy, Christine Kleszewski, Émilie Perrichone Sylvie Wuattier, Conseil pédagogique et révision : Christian Puren, Editions Maison des Langues, Paris. 30 Marks

4. **Viva-Voce :** 30 Marks

1. Dictation of an unseen passage of about 100 words. (Easier than the prescribed text). 10 Marks

2. Conversation (general). 10 Marks

3. Reading (Unseen passage) 10 Marks

Courses of Reading :

POETRY : The following **Ten** poems are to be studied :

1.	Le Lac	Lamartine
2.	Le Pont Mirabeau	Apollinaire (Alcools)
3.	Horloge	Baudelaire (Fleurs du mal)
4.	Les animaux malades de la peste	La Fontaine
5.	La Grasse matinée	J. Prévert (Paroles)
6.	Le colloque sentimental	P. Verlaine
7.	Le manteau impérial	V. Hugo
8.	Le Déjeuner du Matin	J. Prévert
9.	La jeune Veuve	La Fontaine Fables (VI.21) 1968.
10.	La Vase brisé	Sully Prudhomme

Comprehension, explanation, literary, appreciation and criticism of the poems to be studied.

Note : 1. The latest syllabus should be strictly followed.
2. Choice should be given in questions.

Text Books for Reference :

1. French Poetry for students by – A.W. Bains.
2. Paroles (édition Folio) Jacques Prevert.
3. Le français et la Vie-Tome III G. Mauger/M.Bruezière.
4. Collection Littéraire XIX & XX Siècle-Nouveaux classique Larousse.
5. Poèmes choisis-A de. Musset-édition-Nouveaux classique Larousse.
6. Les Textes français - XIX et XX Siècles. R. Chevalier P. Audiat, Hachette.

Courses of Reading :

Drama - La Femme du Boulanger, Marcel Pagnol, Presses, Pocket.

Prose - Version Originale-3 Méthode de Français/Livre de I' élève, Fabrice Barthélémy, Christine Kleszewski, Émilie Perrichone Sylvie Wuattier, Conseil pédagogique et révision : Christian Puren, Editions Maison des Langues, Paris.

N.B. :
1. The latest syllabus to be followed strictly and the question paper should be of B.A. III level.
2. Choice in questions must be given.
3. Eight periods of one hour weekly – Six hours for text and two hours for compositions.
4. The composition and the unseen passages should be based on the vocabulary and grammar covered till B.A. III.
5. The paper will be set and answered in French (except Translation).

GERMAN (Elective)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : Written Paper of 90 marks

Time : 3 Hours

1.	Explanation and interpretation of a poem or stanzas from the prescribed book (German verse).	:	30 Marks
2.	Characterization/Literary question on the prescribed novel (Der Verdacht).	:	30 Marks
3.	Character Sketch/Literary question on the prescribed drama (Andorra).	:	30 Marks
4.	Internal Assessment (for regular students)	:	10 Marks

Note : The question paper shall be set in German.

Paper-B (I) : Theory and Practice of Translation/Written
(This paper is Applied in Nature)

Time : 3 Hours

1.	Translation of unseen texts from German into English. (Two out of three texts)	:	40 Marks
2.	Translation of unseen texts from English into German. (Two out of three texts)		20 Marks

II : Oral 30 Marks

1. Reading of text (s) and General conversation in German
2. Internal Assessment (for regular students) 10 Marks
3. The mode of evaluation for Internal Assessment is to be followed as per University Guidelines.

Prescribed Text Books :

1. German Verse by B.B. Kulkarni and R.N. Chapekar *The Following Poems :*

    1. Ermahnmung (Hofmannswaldau).
    2. Ein Traum ist unser Leben (Herder).
    3. Willkommen and Abschied (Goethe).
    4. Erlkonig (Goethe).
    5. Grenzen der Menschheit (Goethe).
    6. An die Freunde (Schiller).
    7. Hyperions Schiksalslied (Holderlin).
    8. Das Riesenspielzueg (Adelbert von Chamisso).
    9. Abschied (Eichendorff).
    10. Das Karussel (Rilke).

2. Der Verdacht by Friedrich Durrenmatt.
3. Andorra by Marx Frisch.
4. Deutsche Text zum ubersetzen (Max Hueber Verlag).
5. Moderne Welt 1 (Deutsch fur Fortgeschrittene) by Heinz Griesbach (Max Hueber Verlag, Munchen).
6. Texts to be supplied by the Teacher for Translation.

RUSSIAN (Elective)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : (Option-i) (General Group)

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 hours

1. Translation from Russian into English/Hindi/Panjabi (about 120 words) 35 marks
2. Translation from English/Hindi/Panjabi into Russian (about 90 words) 35 marks
3. Applied Grammar : 4 questions (out of seven) of 5 marks each. 20 marks

Note : **Dictionaries are allowed.**

OR

Paper-A: (Option-ii) (Scientific and Technical Translation)

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 hours

1. Translation from Russian (Scientific and Tech. material) into English/Hindi/Panjabi (about 120 words). 35 marks
2. Translation from English/Hindi/Panjabi into Russian (about 75 words). 35 marks
3. Applied Grammar : 4 questions (out of 7) of 5 marks each. 20 marks

Note : **Dictionaries are allowed.**

Paper-B

Theory : 60 marks
Internal Assessment : 10 marks
Time : 3 hours

1. Biographies of Russian writers from prescribed text books 16 marks
(2 questions of 8 marks each with internal choice)

2. (A) Russian Literature : (2 questions out of 4) on the following of 8 marks each : 16 marks

(i)	A.S. Pushkin	(a)	Nyane	Text-book
		(b)	V Sibir	
		(c)	Ya vas lubeel	
(ii)	L.N. Tolstoy	Posle Bala		
(iii)	A.P. Chekhov	Tolsti i Tonki, Spat' Khochitsa		
(iv)	N.V. Gogol	Revizor		

(B) 2 questions (out of 4) on the following of 9 marks each : 18 marks

(i)	M. Gorky	Mat' (extract)	
(ii)	Konstantin Paustovsky	Belaya Raduga (Page 76)	= DOROGY Moscow, 1979
(iii)	Yuri Kazakov	Goluboi Zeleony (Pages 20-29)	-do-
(iv)	Chingiz Aitmatov	Soldatyonok (Pages 55-59)	-do-

(C) Reference to the context : 10 marks
Prose/Poetry passages from the texts studied (one question out of two)

Paper-C: Oral 30 marks

Independent Translation of about 2500 words from Russian literary, socio-political, popular science texts into English/Hindi/Punjabi.

OR

Narration in Russian on any Two of the following topics :
1. India 2. Russia 3. Sports 4. Biography of any Indian/Russian personality 5. Native State 6. My favourite Russian literary work 7. My favourite Russian writer.

Books Prescribed :

1.	N.S. Burlakov & C.N. Chakravarti	:	A Chrestomathy of Russian Literature, 1970.
2.	Nikolai Bannikov	:	Three Centuries of Russian Literature, Progress Publishers, Moscow, 1980.
3.	T.E. Pecheritsa	:	DOROGI - Stories by Soviet writers, Russian Language, Moscow, 1970.
4.	Dictionaries		

BENGALI (Elective)
(*Kept in Abeyance*)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A

Marks : 100
Time : 3 hours

History of Literature	40 marks
Short Stories	40 marks
Essay	20 marks

History of Literature : An outline of the history of Bengali Literature.

Short Stories : Tara Shankar Bandopadyaya : Srestha Galpa (first six stories).

OR

Manik Bandopadhaya : Srestha Galpa (first six stories).

Essay : Students are expected to write on a literary topic.

Paper-B

Marks : 100
Time : 3 hours

Short Stories	40 marks
Biography/Travelogue	40 marks
Essay	20 marks

Short Stories : Sudhir Sarkar (Ed.), Katha Guchha

The following pieces only :

1. Kshudhita Pashan
2. Adarini
3. Abhagir Swarga
4. Bedeni
5. Pui Macha

6. Telini Pota Abiskar

7. Bhusandir Mathe

Travelogue/Biography

Mahaprasthaner Pathe-Prabodh Kumar Sanghal

Essay : An Essay on a literary topic.

TAMIL (Elective)
(*Kept in Abeyance*)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A Total Marks : 100 marks
Time : 3 hours

Applied Grammar 50 marks

Yappu & Ani 50 marks

Paper-B Total Marks : 100 marks
Time : 3 hours

Prose 50 marks

History of Tamil Literature (from Ancient time upto 9th Century)

Applied Grammar 50 marks

Tamil Syntax and Language Structure

Books Recommended

(i) Nadaiyiyal by Neethivanan and Sundaramurthy.

(ii) Sollilkkanam by Pulavar Arasu, Published by Kazhagam, 154, T.T.K. Road, Alwarpet, 18, Madras.

(iii) Yappilakkanam and (Aniyilakkanan), Published by Kazhagam, 154, T.T.K. Road, Alwarpet, Madras-108.

(iv) Yappiyal, Aniyial, Solliyal by Kalasagaram Rajagopal, Published by Star Prasuram, Madras.

Prose : Ilakkiyak Katchigal by Dr. C. Balasubramanian, Published by Paari Nilayam, 184, Braodway, Madras-108.

KANNADA (Elective)
(*Kept in Abeyance*)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A

Total Marks : 100
Time : 3 hours

1. *Prose :* 50 marks

   Samakalina Kannada Kavite Pages 91-150 (Bhaga :3), Published by Bangalore University, Banglore.

2. *Novel :* 50 marks

   Mula Mukhi by Anupama Niranjan, Published by D.V.K. Murthy, Krishnamurthy Puram, Mysore-4.

Total : 100 marks

Paper-B

Total Marks : 100
Time : 3 hours

1. Applied Grammar 50 marks

   Figures of speech, Idioms, Proverbs, Khyata, Kannada Kangalas, Some Sanskrit Viruttas, Scanning the Viruttas.

2. History of Kannada Literature (Ancient period) (9th to 12th century) 50 marks

Total : 100 marks

MALAYALAM (Elective)
(*Kept in Abeyance*)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A

Total Marks : 100

Time : 3 hours

1. *Poetry* (Collection of poems) — 50 marks

   'Navathalam' , Edited by Karassery, published by P.K. Brothers, Kozhikode.

2. *Novel :* — 50 marks

   Manase Nee Satchi by G.N. Panikkar, published by N.B.S., Kottayam.

Total : 100 marks

Paper-B

Total Marks : 100

Time : 3 hours

1. *Applied Grammer :* — 50 marks

   Idioms, Proverbs, Prosody and Poetics, Malayala Seili by Kutti-Krishna Marar, published by Marar Sahitya Prakashan, Calicut.

2. *History of Malayalam Literature* (Ancient period upto 15th Century). Recommended : Sahitya Sahiyam by A.R. Raajaraja-Verma, N.B.S., Kottayam. — 50 marks

Total : 100 marks

TELUGU (Elective)
(*Kept in Abeyance*)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A

Total Marks : 100

Time : 3 hours

1.	Applied Grammar	50 marks
2.	Proverbs, Idioms, Sandhis and Figures of Speech.	50 marks
	Total :	100 marks

Paper-B

Total Marks : 100

Time : 3 hours

1.	*Prose :* Kashmira Deepa Kalika by Nayani Krishna Kumari	50 marks
2.	*History of Telugu Literature* (Ancient Period Beginning to VIth Century)	50 marks
	Total :	100 marks

PHYSICAL EDUCATION

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

PAPER-A

Max. Marks	:	65
Theory	:	60 marks
Internal Assessment	:	05 marks
Time	:	3 hours

INSTRUCTIONS FOR THE PAPER-SETTER AND STUDENTS :

(a) There shall be nine questions in all, spread over five units.

(b) First question/unit is compulsory. It will contain six short answer type questions, spread over the whole syllabus to be answered in brief. It will carry 12 marks.

(c) Rest of the paper shall contain four units for descriptive questions. Each unit shall have two questions and the students shall be given internal choice i.e. the students shall attempt one question from each unit.

(d) All questions/units will carry equal marks.

(e) Private candidates and the students of the University School of Open Learning (USOL) will not be allowed to take this subject.

UNIT-I

12 Marks

Entire syllabus given in the Unit-II to V will be covered to set six short answer type questions in first question/unit of the question paper which is compulsory.

UNIT-II

12 Marks

Play:

- Meaning and Definition of Play.
- Various theories of play and their significance in Physical Education and Sports.

Recreation:

- Meaning, definition, characteristics, aim, objectives and types of recreation and recreational activities.
- Significance of recreation in the modern society.
- Recreation providing agencies.

UNIT-III

12 Marks

Competitions :

- Meaning, importance and conduct of intramural and extramural competitions.
- Meaning and types of tournament and their merits and demerits.
- Draw of fixtures of various tournaments.

Camps :

- Meaning, aim and objectives of the camp.
- Advantages of camping/outdoor education.
- Types and agencies promoting camping.
- Organization of camps and factors affecting its organization.
- Educative values of a camp.

Athletic Meet:

- Organization of an athletic meet.
- Importance/significance of an athletic meet.

UNIT-IV 12 Marks

Posture :

- Meaning, types and importance of a good posture.
- Causes, preventive and remedial measures of a poor posture.

Postural Deformities :

- Postural deformities (Kyphosis, Lordosis, Scoliosis and Flat foot), their causes, preventive and remedial measures.

Physical Activities and their effects on various physical parameters:

- Physical activities/training and their effects on aging, body composition, and obesity. General problems of obesity.
- Health related risk factors of obesity.
- Obesity and physical activity.
- Causes, preventive and remedial measures of obesity.

UNIT-V 12 Marks

Massage:

- Brief history of massage.
- Meaning and definition of massage.
- Principles/guidelines for massage.
- Types of massage and their benefits.
- Effects of massage on skin, blood circulation, nervous system and muscles.

Basics of Cricket :

- History of the game.
- Basic fundamentals.
- Equipment and specifications.
- Marking / layout of field.
- Rules and regulations (number of players, duration of game, number of officials required and general rules of play).
- Major tournaments and Arjuna awardees of the game.

References:

1. Butler, G.D. :*"Introduction to Community Recreation",* McGraw Hill Book Company, Inc. New York, 1959.

2. Garinder, M. Dena :*"The Principles of Exercise Therapy",* Fourth Edition, 1985, CBS Publishers and Distributors, Delhi.

3. Gladys Scott, M. :*"Analysis of Human Motion",* 2nd Edition, 1970, Eurasia Publishing House Pvt. Ltd., New Delhi.

4. Lee Joseph :*"Play in Education, National Recreation Association",* New York, 1942.

5. Meyer, Herold D., and Charles K. Brightbill : *"Recreation Administration : A Guide to its Practices",* Prentice Hall Inc., Englewood Cliffs, N.J., 1956.

6. Mitchell, Elmer D., and Bernard S. Mason *: "Theory of Play",* The Ronald Press Co., New York, 1948.

7. Nash, Jay B. : *"Philosophy of Recreation and Leisure",* The C.V. Mosby Co., St. Louis, 1953.

8. Parameswara Ram, N.: *"Kinesiology – Physical Education and Sports",* Parkash Brothers – Educational Publishers, 1988.

9. Piscopo, John and Baley, James A.: *"Kinesiology: The Science of Movement",* John Wiley & Sons, Inc., 1981.

10. Singh, Ajmer *et.al.* : *Modern Text Book of Physical Education, Health and Sports,* Kalyani Publishers, Ludhiana, 2000.

11. Struna, Monika and Church Connie : *"Self Massage : Touch Techniques of Relax, Soothe and Stimulate your Body",* Hutchinson & Co. Ltd., London, 1983.

12. Tirunarayanan, C. Hariharasarma, S. : *"Methods in Physical Education",* The South India Press, Karaikudi.

13. Wilmore, Jack H. :*"Atheltic Training and Physical Fitness",* 1977, Allyn and Bacon, Inc. 470, Atlantic Avenue, Boston Massachusetts.

14. Wood, Elizabeth C. and Becker, Paul D. :*"Beard's Massage",* Third Edition, 1981, W.B. Saunders Company, Philadelphia, P.A.

PAPER-B

Max. Marks		: 65
Theory	:	60 marks
Internal Assessment	:	05 marks
Time	:	3 hours

INSTRUCTIONS FOR THE PAPER-SETTER AND STUDENTS :

(a) There shall be nine questions in all, spread over five units.

(b) First question/Unit is compulsory. It will contain six short answer type questions, spread over the whole syllabus to be answered each in brief. It will carry 12 marks.

(c) Rest of the paper shall contain four units for descriptive questions. Each unit shall have two questions and the students shall be given internal choice i.e. the students shall attempt one question from each unit.

(d) All questions/units will carry equal marks.

(e) Private candidates and the students of the University School of Open Learning (USOL) will not be allowed to take this subject.

UNIT-I 12 Marks

Entire syllabus given in the Unit-II to V will be covered to set six short answer type questions in first question/unit of the question paper which is compulsory.

UNIT-II 12 Marks

Nervous System :

- Meaning of Nervous System.
- Main organs of Nervous System and their functions.
- Reflex action and Reciprocall Innervations.
- Functional classification of Nervous System.

Excretory System :

- Meaning of Excretory System.
- Main organs of Excretory System and their structure and functions.

Endocrine System :

- Meaning of Endocrine System.
- Meaning of Glands, their location and functions/Harmones produced by them.

UNIT-III 12 Marks

Sports Training :

- Meaning, definition, aim, objective, characteristics and principles of sports training.

General Physiological concept :

- Physiological concepts such as vital capacity, second wind, stitch in the side and its causes.
- Definition of oxygen debt/excess post exercise oxygen consumption (EPOC) and its implication.
- Meaning, definition and types of fatigue.
- Muscular contractions such as isotonic, isometric, eccentric and isokinetic.
- Meaning of Blood pressure, Hypertension : Its causes, effects and treatment, Exercise and Hypertension.

Effects of Physical Exercise/Training :

- Effects of Physical exercise/Training on muscular, respiratory and circulatory systems of the body.

UNIT-IV 12 Marks

Carrier aspects in Physical Education :

- Carrier options in Physical Eduction.
- Different avenues in Physical Education.
- Self assessment for carrier choices.
- Courses and institutions available for Physical Education profession.

UNIT-V 12 Marks

Coach :

- Coaching, coaching philosophy, definition of a coach.

- Qualification and characteristics of a coach.
- Responsibilities of a coach.

Basics of Table Tennis:

- History of the game.
- Basic fundamentals.
- Equipment and specifications.
- Marking/layout of T.T Table.
- Rules and regulations (number of players, duration of game, number of officials required and general rules of play).
- Major tournaments and Arjuna awardees of the game.

References :

1. Chaurasia, B.D. : *"Handbook of General Anatomy",* CBS Publishers and Distributors, First Edition, 1981.
2. Dandiya, P.C., Jafer, Z.Y.K. and Jafer Afifa: *"Health Education and Community Pharmacy",* Second Edition, 1996, Reprinted in 1998, Vallabh Prakashan, Pitampura, New Delhi.
3. Dick, Frank, W. : *"Sports Training Principles",* Lepus Books, London, 1980.
4. Jensen, C.R. and Fisher, A.G. : *"Scientific Basis of Athletic Conditioning",* Lea and Febiger, Philadelphia, 1979.
5. Matveyer, L. : *"Fundamentals of Sports Training",* Mir Publication, Moscow, 1981.
6. Mathews, D.K. and Fox, E.L. : *"The Physiological Basis of Physical Education & Athletics",* Second Edition, W.B. Saunders Co., Philadelphia, 1976.
7. McArdle, William D., Katch Frank I., and Katch, Victor, L.: *"Exercise Physiology : Energy, Nutrition and Human Performance",* Lea and Febiger, Philadelphia, 1981.
8. Morgan, R.E. and Adamson, G.T. : *"Circuit Training",* Bell and Sons, London, 1958.
9. Morehouse, L.E. and Miller, A.T.: *"Physiology of Exercise",* The C.V. Mosby Co., St. Louis, 1965.
10. Parrot, J.W. : *"Anatomy and Physiology for Physical Education Teachers",* Lend Edward Arnold Ltd., 1983.

11. Rose & Wilson : *"Foundations of Anatomy and Physiology*, 5th Edition, 1981, Reprinted in 1985.

12. Sampath, K. & Uma Maheshwar, B.: *"Human Anatomy & Physiology",* First Edition, 1999- 2000, Birla Publications, Delhi.

13. Shaphard, R.J.: *"The Fit Athlete",* Oxford University Press, 1978.

14. Shaver, Larry G. : *"Essential of Exercise Physiology",* Surjeet Publications, Delhi First Indian Print, 1982.

15. Singh, Ajmer, et.al. : *"Modern Text Book of Physical Education, Health and Sports",* Kalyani Publishers, Ludhiana, 2000.

16. Singh, Hardyal : *"Sports Training - General Theory and Methods",* Netaji Subhash National Institute of Sports, Patiala,1984.

PRACTICAL

Max. Marks	:	70
Practical	:	60
Internal Assessment	:	10

ATHLETICS

Throws (Shot-put or Discus Throw or Javelin Throw) and one event of the choice of the student.

(a) **Shot-put** (Holding the stance, the glide, the delivery and the reverse or the recovery).

OR

Discus Throw (The handhold, the initial stance, the preliminary swings, turn the delivery and the reverse or the recovery).

OR

Javelin Throw (The grip, the carry, the run way approach, the last five strides, the delivery, the reverse or the recovery).

(b) Measurements of equipment and the throwing circles or the approach run, the arc and the throwing area/the sectors.

GAMES

(Badminton or Hockey and any other one game of the choice of the student).

Badminton :

(a) Measurement (Badminton Court, Net, Racket and Shuttle cock) for singles and doubles.

(b) Number of players and officials.

(c) Holding the racket and shuttle cock.

(d) Types of Service : High and Low.

(e) Types of Strokes : fore hand, back hand, over head.

(f) Shots : Smash, Lob shot, net shot, dive shot.

(g) Rules and regulations of the game

OR

Hockey :

(a) Measurements (Hockey ground, goalpost, hockey stick, ball and flags) for men and women.

(b) Number and position of players and officials.

(c) Fundamental skills (grip, hitting, stopping, dribbling, pushing, scoop and flick).

(d) Rules and regulations of the game.

Physical Fitness Test : More emphasis shall be given on variety of physical exercises for the development of physical fitness. (Flexibility and Strength)

Test 1 - Flexibility : Sit and reach test

Test 2 - Strength : Medicine Ball Throw

Division of Practical Marks : *Marks for each activity shall be divided as under :*

Athletics 15 marks, games 15 marks, participation and achievement in sports/games 10 marks, physical fitness 10 marks, viva voce 10 marks and internal assessment 10 marks based on overall performance of a student during the current academic session which will be assessed by the teacher concerned.

Note : 1. Pole-vault, Hammer Throw, Hurdles, Relay races and steeple chase race are not included in the practical syllabus/course due to the fact that these events are highly technical. Moreover, in the absence of proper facilities required for the events mentioned above may prove to be injurious/fatal to the students.

2 12 periods per week (6 periods each for theory and practicals) shall be allotted to a class.

3 The theory and practical class shall consist of 60-80 students and 30-40 students respectively.

4 The theory (Paper A and Paper B) papers shall consist of 65 marks each and practical paper shall consist of 70 marks.

5 As per the Panjab University Calendar, Chapter XX (Page 298) Volume III, 1996, the maximum teaching work load for a Lecturer in Physical Education for B.A. Pass Course is 24 periods per week, which includes theory as well as practical.

6. The choice of games by the students shall be confined to the games approved by the Association of Indian Universities.

7. A student is required to prepare a practical note book of athletic events and games given in the syllabus.

EDUCATION

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : INDIAN EDUCATION

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 hours

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER :

The question paper will consist of five Units : I, II, III, IV and V. Units I, II, III and IV will have two questions from the respective unit of the syllabus and will carry 18 marks each. Unit V will consist of eight short answer type questions which will cover the entire syllabus uniformly and will carry 18 marks. The students are required to attempt 6 short answer type questions out of 8 in unit V which will be compulsory. The question paper should preferably carry internal division of marks for all the sub-questions of one main question. Preferably set the words limit for answer (300-350 words for units I, II, III, IV and 75 words for each short answer question in unit V).

GENERAL INSTRUCTIONS FOR THE CANDIDATE :

The students will be required to attempt one question each from Unit I, II, III and IV. Unit V will be compulsory. The students are required to attempt **6** short answer type questions out of **8** in Unit V. The words limit will be 300-350 words for unit I, II, III and IV and 75 words for each short answer question in unit V.

Objectives :

1. To enable the students to know about History of Indian Education.
2. To enable the students to understand the problems of pre-primary Education.
3. To make the students familiar with the concept of Universalization of Elementary Education and its problems.
4. To make the students familiar with the constitutional provisions of Education and role of different agencies in Education.

Course Contents :

UNIT-I : Education in the ancient and medieval period of Indian History.

UNIT-II : Current status and problems of pre-primary education. Public schools and their future.

UNIT-III : Importance of Elementary Education.
Universalization of Elementary Education. Problems of Elementary Education and Role of the State in Elementary Education.
Sarav Sikhiya Abhiyan – Concept, Objectives & Implementation.

UNIT-IV : Constitutional Provisions of Education.
Provisions for Education of Weaker Sections of Society.

Books Recommended :

1. Kaur, Kuldip : *Education in India (1781-1985), Policy, Planning and Implementation,* Chandigarh Centre for Research in Rural and Industrial Development, 1985.
2. Narula and Naik, J.I. : *Bharat Wich Vidyak Sikhya Da Itihas,* Patiala, Punjabi University.
3. Govt. of India, Ministry of Education : *Secondary Education Commission Report* (1952-53), New Delhi 1953.
4. Govt. of India, Ministry of Education : *Report of Education Commission* (1964-65), New Delhi, 1966.
5. Kochhar, S.K. : *Pivotal Issue of Education,* New Delhi, Sterling Publisher, 1984.
6. Govt. of India, Ministry of Human Resource Development : *National Policy of Education,* New Delhi, 1986.
7. Govt. of India : *Seventh Five Year Plan,* New Delhi.
8. Govt. of India : *Challenges of Education, A Policy Perspective,* New Delhi, 1986.
9. Sodhi, T.S. : *Bharti Sikhya Ate Us Diyan Samasiyavan,* Bawa Publications, Patiala.
10. Govt. of India, Ministry of Human Resource Development : *National Policy on Education, Programme of Action*, New Delhi, 1986.
11. Govt. of India, Ministry of Human Resource Development : *National Policy on Education,* 1986 (with modifications undertaken in 1992), New Delhi, 1992.

Paper-B : MODERN INDIAN EDUCATION

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 hours

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER :

The question paper will consist of five Units : I, II, III, IV and V. Units I, II, III and IV will have two questions from the respective Unit of the syllabus and will carry 18 marks each. Unit V will consist of eight short answer type questions which will cover the entire syllabus uniformly and will carry 18 marks. The students are required to attempt 6 short answer type questions out of 8 in unit V which will be compulsory. The question paper should preferably carry internal division of marks for all the sub-questions of one main question. Preferably set the words limit for answer (300-350 words for units I, II, III, IV and 75 words for each short answer question in unit V).

INSTRUCTIONS FOR THE CANDIDATE :

The students will be required to attempt one question each from Unit I, II, III and IV. Unit V will be compulsory. Students are required to attempt 6 short answer type questions out of 8 in unit V. The words limit will be 300-350 words for unit I, II, III and IV and 75 words for each short answer question in unit V.

Objectives :

1. To make the students familiar with the structure of Secondary Education in India.
2. To enable the students to know about different policies of education.
3. To make the students familiar with the problems of education for 21st century.
4. To enable the students to know about the need and importance of vocationalisation of education.
5. To enable the students to know about Adult, Continuing and Environmental Education.

Course Contents :

UNIT-I : Secondary Education – Structure, Recommendations of Kothari Education Commission and National Policy of Education.

RMSA(Rashtriya Madhmik Shiksha Abhiyan) – Concept Objectives and Implementation.

UNIT-II: Implementation of National Policy of Education of 1986 and 1992.

Education in the Twelveth Five Year Plan.

Problems and Prospects of Education for the 21st century.

UNIT-III : Vocationalisation of Education – Meaning, Need and Importance.

Aims, Objectives and Problems of Secondary Education.

Role of different Agencies – NCERT, NCTE AND DIET.

UNIT-IV : Adult, Continuing and Distance Education.

Environmental Education, ICT in Education.

Books Recommended :

1. Narula & Naik, J.P. : *Bharat Wich Vidyak Sikhya Da Ithas,* Patiala, Punjabi University.
2. Sharma, T.R. : *Bharti Sikhya Dian Samasiyan,* Patiala, Punjabi University.
3. Govt. of India, Ministry of Human Resource Development : *National Policy on Education,* 1986 (with modifications undertaken in 1992), New Delhi, 1992.
4. Govt. of India, Ministry of Education : *Secondary Education Commission Report* (1952-53), New Delhi, 1953.
5. Govt. of India, Ministry of Education : *Report of Education Commission* (1964-66), New Delhi, 1966.
6. Kochhar, S.K. : *Pivotal Issues in Education,* New Delhi, Sterling Publishers, 1984.
7. Kaur, K. : *Education in India (1781-1985), Policy, Planning and Implementation,* Chandigarh : Centre for Research in Rural and Industrial Development, 1985.
8. Govt. of India, Ministry of Human Resource Development : *National Policy on Education,* New Delhi, 1986.
9. Govt. of India : *Seventh Five Year Plan,* New Delhi.
10. Govt. of India : *Challenge of Education : A Policy Perspective,* New Delhi, 1986.
11. Govt. of India, Ministry of Human Resource Development : *National Policy of Education,* 1986 (with modifications undertaken in 1992), New Delhi, 1992.

ADULT EDUCATION

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Max. Marks	:	200
Theory	:	90 marks
Internal Assessment	:	10 marks
Practical	:	90 + 10 marks
Time	:	3 hours

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER :

The question paper will consist of five Units : I, II, III, IV and V. Units I, II, III and IV will have two questions from the respective Units of the syllabus and will carry 18 marks each. Unit V will consist of **Eight** short answer type questions which will cover the entire syllabus uniformly and will carry 18 marks in all. Each short question will carry 3 marks.

GENERAL INSTRUCTIONS FOR THE CANDIDATE :

The students will be required to attempt one question each from Units I, II, III and IV. The students are required to attempt **6** short answer type questions out of **8** in Unit V which will be **compulsory**. The question paper should preferably carry internal division of marks for all the sub-questions of one main question.

Objectives of the Course :

The main objectives of the paper are :

1. To expose students to the concept, importance and scope of continuing education.
2. To provide an overview of women empowerment.
3. To acquaint students with the basis of life skills education, its concept, meaning and its various forms.
4. To acquaint students with role of ICTs and mass media in adult education.
5. To expose students with life long learning and its future perspectives.
6. To equip students with role of universities in adult and continuing education programs.
7. To explain students extension as third dimension.

THEORY :

UNIT-I
(a) Continuing Education : Concept, Importance, Scope and Objectives; Types of Continuing Education Programs.
(b) Women Empowerment : Concept, Need and its Importance.

UNIT-II (a) Life Skills Education : Its meaning, Concept and its Various Forms.

1. Psycho-Social Skills
2. Communication Skills
3. Health Living Skills
4. Problem Solving and Decision Making Skills
5. Co-operation and Team Work Related Skills
6. Recreational Skills
7. Computer Related Skills
8. Work Related Skills
9. Disaster Management Skills
10. Entrepreneurial Skills

(b) Role of ICTs and Mass Media in Adult Education.

UNIT-III Life Long Learning : Meaning, Concept, Need, Scope, Strategies, Implications and Future Perspectives.

UNIT-IV (a) Extension as the Third Dimension : Major Thrust. Programs and Activities.

(b) Role of Universities in Adult & Continuing Education Programs.

PRACTICAL/FIELD WORK :

Undertaking any one of the following project and preparation of Report :

1. Organisation of two Awareness Generating Programs and Report Writing.
2. Organisation of two Extension Lectures in Slum Village and Report Writing.

Books Recommended :

1. Singh, Madan Mahanty : *Companion to Adult Educators – National Literacy Mission,* Directorate of Adult Education, Ministry of Human Resource Development, Department of Adult Education, New Delhi, 1999.
2. Suriakanta, V. : *Perspectives of Continuing Education,* Dept. of Adult, Continuing Education & Extension, Gandhi Gram Rural University, Gandhi Gram, 2007.

3.	Parthasarathy, K.	:	*Facets of Adult and Continuing Education,* School of Education, Centre for Adult, Continuing Education & Extension, Bharathidasan University, Trichirappatti, 2006.
4.	Parthasarathy, K.	:	*Population Education and Youth Development,* School of Education, Centre for Adult, Continuing Education and Extension, Bharathidasan University, Trichirappatti, 2007.
5.	Alan Rogers	:	*Teaching Adults,* Sterling Publishers Pvt. Ltd., New Delhi, 1989.
6.	Mohanty, J.	:	*Adult & Non Formal Education,* Deep Publications, New Delhi, 1993.
7.	U.G.C. Guidelines (X and XI Plan) for Adult, Continuing Education and Extension	:	*University Grants Commission,* New Delhi.
8.	Pillai, G.P.	:	*Continuing Education : A Case Study,* The Hindu, 1994.
9.	Sharma, S.N. & Prakash Ravi	:	*Adult Education and Social Growth,* Kanishka Publishers, Distributors, New Delhi, 1996.

MUSIC (Vocal)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

General Instructions :

1. In the case of private candidates, there will be no internal assessment and the marks obtained in the external assessment of the practical examination shall be proportionately increased.
2. There should not be more than eight students in a batch for practical examination.
3. Harmonium will not be allowed as accompaniment in Vocal Music. But harmonium can be used while singing Alankars.
4. The candidate can take Vocal Music along with Instrumental Music.
5. In all, nine questions will be set. The question paper will be divided into five units. Four units will contain two questions each. The candidates are required to attempt five questions selecting at least one question from each unit. In Unit Five, the ninth question based on the Notation shall be **compulsory**.
6. There should be up to eight students in one section in practical.

Outlines of Tests, Syllabi and Courses of Reading

Paper-A :	**THEORY** (3 hrs. duration) **(Duration 45 minutes 06 practical + 02 Theory periods per week)**		:	90 marks
Paper-B :	**PRACTICAL**		:	90 marks
	(i) Choice of Viva	:	70 marks	
	(ii) Harmonium	:	10 marks	
	(iii) Tabla	:	10 marks	
	Internal Assessment (Theory + Practical)	:	10+10 =	20 marks
		Total	**:**	**200 marks**

Paper-A : THEORY (Duration 45 minutes 02 Theory per week)

UNIT-I

1. General History of Indian Music i.e. from Bharat to Sharangdev.
2. Explain the following Gayan Shailies :

   Tappa, Thumri, Tarana, Dhamar, Sadra, Bhajan and Shabad.
3. Special features of time Theory of Indian Ragas.

UNIT-II

1. Topics of Essays :
   (i) Kanth-Sadhana.
   (ii) Lok Sangeet of Punjab.
   (iii) Role of Akashwani and Doordarshan towards the popularization of Indian Classical Music.
   (iv) Sansthagat Sangeet Shikshan Pranali.

UNIT-III

1. Knowledge of Ragang Paddhati.
2. Contribution in detail and life sketches in brief of the following :
   (i) Pt. Kumar Gandharv.
   (ii) Ustad Bade Gulam Ali Khan.
   (iii) Ustad Vilayat Hussain Khan.
   (iv) Pt. Onkar Nath Thakur (Sangeet Martand).

UNIT-IV

1. Description of the ragas and talas prescribed in the course :
   (i) Ragas : Puria, Jaunpuri, Vrindavani–Sarang, Madhuvanti and Rageshwari.
   (ii) Talas : Jhumra, Jhaptal, Deepchandi, Dhamar.
2. Knowledge of writing tala Jhaptal in Thah, Dugun, Tigun and Chaugun layakaries.
3. Non-detailed Ragas :
   Multani, Des, Asavari and Sohni.

UNIT-V

Knowledge of writing notation of Vilambit Khayals and Drut Khayals in the prescribed Ragas :
Durga, Vrindavani – Sarang, Madhuwanti, Jaunpuri, Rageshwari.

Paper-B : PRACTICAL (Duration 45 minutes 06 practical periods per week)

1. One Drut Khayal with Alaps & Tanas in each of the following Ragas :
   Durga, Vrindavani –Sarang, Madhuwanti, Jaunpuri and Rageshwari.
2. Two Vilambit Khayals in any two of the prescribed Ragas in the course with extempore –alaps and taanas.
3. Ability to sing one Dhamar in proper style.
4. One Tarana in any of the detailed Ragas prescribed in the course.
5. Ability to sing notations (in swaras) of Drut Khayals in each of the prescribed Ragas.
6. Ability to play Dhamar, Jhaptal on Tabla.
7. Ability to recite the following Talas in Thah, Dugun by hand :
   Jhumra, Jhaptal, Dhamar, Deepchandi.
8. Ability to sing any two Drut Khayals of your course on Harmonium.
9. Knowledge of non-detailed Ragas :
   Ability to sing their Arohas, Avrohas and Pakad with the help of Tanpura : Multani, Des, Sohni, Asavari.
10. Tuning of Tanpura.

Books Recommended :

1.	*Hamare Sangeet Ratna*	:	Sangeet Karyalayas, Hathras.
2.	*Sangeet Sar Part III*	:	Mrs. Veena Mankaran.
3.	*Folk Instruments of Punjab*	:	Prof. Anil Narula.
4.	*Sangeetanjali, I & IV*	:	Pt. O.N. Thakur.
5.	*Rag Vigyan Part III & IV*	:	Pt. V.N. Patwardhan.
6.	*Bhartiya Sangeet Ka Itihas*	:	Thakur Jaideva Singh.
7.	*Sangeet Shastra Vigyan*	:	Panna Lal Madan.

MUSIC (Instrumental)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

General Instructions :

1. In the case of private candidates, there will be no internal assessment and the marks obtained in the external assessment of the practical examination shall be proportionately increased.
2. There shall not be more than eight students in a batch for practical examination.
3. Harmonium will not be allowed as accompaniment in Vocal Music. But harmonium can be used while singing Alankars.
4. In Instrumental Music, the candidates have the option to take any one of the following instruments : Sitar, Sarangi, Veena, Sarod, Dilruba, Violin, Guitar, Bansuri, Shehnai.
5. The candidate can take Vocal Music along with Instrumental Music.
6. The candidate can also take Instrumental Music with Tabla.
7. In all, *nine* questions will be set. The question paper will be divided into five units. Four units will contain two questions each. The candidates are required to attempt Five questions selecting at least one question from each unit. In Unit Five, the *ninth* question based on the Notation shall be **compulsory**.
8. There will be up to eight students in one section in practial.

Paper-A : THEORY (Instrumental) (3 hrs. duration) : 90 marks
(Duration 45 minutes 06 practical + 02 Theory periods per week)

Paper-B : PRACTICAL (20 minutes duration) : 90 marks

(i) Viva	:	70 marks	
(ii) Gayan	:	10 marks	
(iii) Tabla	:	10 marks	
Internal Assessment (Theory + Practical)	:	10 +10 =	20 marks
		Total :	**200 marks**

Paper-A : THEORY (Duration 45 minutes 02 theory periods per week)

UNIT-I

1 General History of Indian Music i.e. from Bharata to Sharangdeva.
2. Explanation of different Vadan Shailies (Styles) of your own instrument.
3. Historical development of Indian Musical scale.

UNIT-II

1. Importance of Laya and Tala in Music.
2. Varities of Taanas (Toras).
3. Notation System : Origin and development.

UNIT-III

1. The life sketches and contributions of the following musicians :
   (i) Ustad Bismillah Khan
   (ii) Ustad Hafiz Ali Khan Sahib
   (iii) Pt. Nikhil Banerji

2. Essays :
   (i) Chitrpat Sangeet Mein Vadhyon Ki Bhoomika.
   (ii) Role of Radio and T.V. in popularising Instrumental Music.
   (iii) Folk Instruments of Punjab.
   (iv) Future of Instrumental Music.

UNIT-IV

1. Description of detailed ragas and talas prescribed in the course :
   Ragas : Madhuwanti, Mian-Ki-Todi, Bhimplasi, Jai-Jaiwanti and Bageshwari.
   Talas : Dhamar, Adachautal, Tilwara, Sultal.

2. Description of the following non-detailed Ragas :
   Multani, Gujri-Todi, Patdeep, Tilak-Kamod, Rageshwari.

UNIT-V

Knowledge of writing notation of Vilambit Gat and Drut Gat in the prescribed Ragas : Madhuwanti, Mian-ki-Todi, Bhimplasi, Jai-Jaiwanti and Bageshwari.

Paper-B : PRACTICAL (Duration 45 minutes 06 practical periods per week)

1. One Razakhani Drut Gat with Alaps and Toras & Jhalas in each of the following Ragas : Madhuwanti, Mian-Ki-Todi, Bageshwari, Bhimplasi, Jai-Jaiwanti.
2. Two Maseet Khani (Vilambit) Gats with Alap-Jod and Toras in any of the prescribed ragas.
3. Knowledge of the following non-detailed ragas : Multani, Gujri-Todi, Rageshwari, Patdeep.
4. Ability to demonstrate by hands in Ekgun and Dugun layakaries of the following talas : Dhamar, Adachautal, Tilwara, Sultal.
5. Ability to play Ektal on Tabla.
6. One gat in Madhya laya other than teen taal with toras in any prescribed ragas.
7. Ability to play techniques of your Instruments: Meend, Kan, Krintan.
8. Ability to sing shudh, komal and tivra swaras with the help of harmonium.
9. Tuning of your Instrument.

Books Recommended :

1.	*Rag Parichaya Part III & IV*	:	H.C. Shrivastava.
2.	*Folk Instruments of Punjab*	:	Prof. Anil Narula (Published by Punjabi University, Patiala).
3.	*Hamare Sangeet Ratna*	:	Sangeet Karayalaya, Hathras.
4.	*Sangeet Granth atey Bharti Sangeet Da Itihas*	:	Chander Kanta, Khosla.
5.	*Sangeet SAR Part III*	:	Mrs. Veena Mankaran.

TABLA (Instrumental Music)

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

General Instructions :

1. In the case of private candidates, there will be no internal assessment and the marks obtained in the external assessment of the practical examination shall be proportionately increased.
2. There will be upto eight students in one section.
3. Harmonium will not be allowed as accompaniment in Vocal Music, but Harmonium can be used while singing Alankars.
4. Candidates can take any other instrument alongwith Tabla out of Sitar, Sarangi, Veena, Sarod, Dilruba, Violin, Guitar, Bansuri, Shehnai.
5. The candidate can take Vocal Music along with Instrumental Music.
6. The candidate can also take Instrumental Music with Tabla.
7. In all, 9 questions will be set. The question paper will be divided into five units. Four units contain 2 questions each. The candidates are required to attempt 4 questions selecting at least one question from each unit. The 5th unit based on the notation shall be **compulsory**.

Paper-A : THEORY (Instrumental) (3 hrs. duration) : 90 marks

Paper-B : PRACTICAL (20 minutes duration) : 90 marks

(i)	Viva	:	60 marks
(ii)	Harmonium	:	10 marks
(iii)	Tabla	:	10 marks
(iv)	Padhant on Hand	:	10 marks

Internal Assessment (Theory + Practical) (10 +10) : 20 marks

Total : 200 marks

Paper-A : THEORY

UNIT-I

(a) Classification of Musical Instruments.

(b) Folk Taal instruments of Punjab.

(c) Brief study of following instruments :
Mridangam, Ghatam, Tabil, Morchang, Kanjeera, Manjeera.

UNIT-II

(a) Origin and development of Tabla Instrument.

(b) History of Pakhawaj.

(c) Playing technique of Tabla and Pakhawaj.

(d) Basic Principals of Tabla/ Pakhawaj accompaniment.

UNIT-III

(a) Comparison between Uttari and Dakshini Taal System.

(b) Comparative study of Bhatkhande and Paluskar Taal Notation System.

(c) Comparative study of different talas having equal matras :

(i) Teental Tilwada – Panjabi Taal.

(ii) Deepchandi – Jhoomra.

(iii) Roopak – Tivra.

(iv) Jhaptal – Sooltal.

(v) Ektal – Chautal.

UNIT-IV

(a) Significance of Tabla Gharanas in present time.

(b) History and playing techniques of Banaras, Punjab, Delhi, Ajrada, Farrakkhabad & Lakhnow Gharana.

(c) Life sketches and contribution of the following :

Mian Kadar Baksh.

Ustad Allarakha Khan.

Pt. Ram Sahai.

Pt. Kishan Maharaj.

Pt. Anokhe Lal Mishra.

UNIT-V

Ability to write in notation of Quaida, Paran, Gat, Tihais & Peshkar in Pancham Sawari, Adachautal, Trital, Jhaptal, Roopak Taal.

Paper-B : PRACTICAL

(a) Talas prescribed Ada Chautal, Tilwara, Dhamar, Basant, Pancham Sawari and Tappa Talas including the Talas prescribed in the previous classes.

(b) Proper Badhat of Ada–Chautal, Dhamar, Pancham Sawari, Talas.

(c) Knowledge of playing Dhamar, Chautal.

(d) Playing of all the prescribed Talas with Vocal and Instrumental performance as well as solo item.

(e) Tuning of Tabla.

(f) Practical knowledge of the following in the prescribed Talas :

(i) Dhamar – Simple Paran, Chakardar Paran, Farmaishi Paran, Tukra, Mukhra, Mohra, Uthan.

(ii) Sawari (15 matras) Peshkar, Quaida, Gat, Paran.

(iii) Tilwara – Vilambit, Theka with accompaniment in vocal.

(iv) Basant Taal (9 matras), Tappa, Theka on Tabla.

(v) Ada-chautal-Peshkar, Quaida, Palta, Rela, Paran.

Books Recommended :

1. *Tala Parichya Part I, II & III* : G.C. Srivastava.
2. *Bhartiya Taalon Ka Shastriya Vivechan* : Arun Kumar Sen.
3. *Taal Maartand* : B.S. Sharma.
4. *Bhartiya Sangeet Vadya* : L.M. Mishra.
5. *Hamare Sangeet Ratna* : Sangeet Karyalaya, Hathras.
6. *Taal Vadya Shastra* : Manohar Bhalchand Marathe.

INDIAN CLASSICAL DANCE

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

General Instructions :

1. In the case of private candidates, there will be no Internal Assessment and the marks obtained in the External Assessment of the Practical Examination shall be proportionately increased.

2. There should not be more than eight students in a batch for practical examination.

3. In all, *nine* questions will be set. The question paper will be divided into five units. Four units will contain two questions each. The candidates are required to attempt five questions selecting at least one question from each unit. In unit Five, the ninth question based on Notation shall be **compulsory.**

4. Accompaniment of Tabla, Harmonium/ Sarangi will be allowed.

5. The practical paper shall be set from the syllabus prescribed for Paper-B (Practical).

6. There would be upto eight students in one section in practical.

Paper-A : THEORY (3 hours duration)			: 90 marks
(Duration 45 minutes 06 practical + 02 Theory periods per week)			
Paper-B : PRACTICAL (20 minutes duration)			: 90 marks
(i) Viva	:	60 marks	
(ii) Harmonium	:	10 marks	
(iii) Tabla	:	10 marks	
(iv) Padhant on Hand	:	10 marks	
Internal Assessment (Theory + Practical)	: 10 + 10		= 20 marks
		Total :	**200 marks**

Paper-A: THEORY (Duration 45 minutes 02 theory periods per week)

Unit-I

(a) Knowledge of techniques of Indian Ballet.

(b) Origin of dance and its development upto Mughal period.

(c) Study of Rasa and its importance.

Unit-II

(a) Origin of Tala and its various aspects.

(b) Brief study of Nayak – Nayika Bhed.

(c) Knowledge of Dakshini Tala system.

Unit-III

(a) Survey and essential characteristics of Manipuri dance.

(b) Important characteristics of Kathakali dance.

(c) Survey and essential characteristic of Kathak dance upto 20th century.

Unit-IV

(a) Dance and other fine arts.

(b) Dance and religion.

(c) A brief study of Natya Shastra.

Unit-V

Description of Talas prescribed in the course :

(i) Sawari-Taal & Teen-Taal (15 Matra).

(ii) Notation of Aamad Toda Paran etc. in priscribed taals.

(iii) Notation of Nagma in Teental & Sawari.

PAPER–B : PRACTICAL (Duration 45 minutes 06 practical periods per week)

(a) Swari Tal (Matra 15)

Tatkars in Single, Dugun and Chaugun Layakaries :

Thaat	:	1
Amad	:	1
Tora	:	6
Paran	:	2
Chakardar Paran	:	1
Kavit	:	2

(b) Teen-Taal – 1

(i) Tatkar with Paltas Thaat	:	1
(ii) Amad	:	1
(iii) Chakardar Tora	:	1
(iv) Paran	:	2
(v) Chakardar Paran	:	2
(vi Tisrajati Paran	:	1
(vii) Kavit	:	2

(c) Demonstrate Dhamar (14 matra) with Tatkar, Tukras and Toras.

(d) Anyone Gat Bhava of the following :
Panghat Ki Chhed-Chhad.

(e) Practical demonstration of Asmyukta and Sanyukta hand gestures according to Abhinaya Darpan.

(f) Ability to Demostrate practically the folk dances of Haryana.

Books Recommended :

1.	*Kathak Darpan*	:	T.R. Azad.
2.	*Kathak Nritya Shiksha Part I & II*	:	Puru Dadhich
3.	*Kathak Shingar*	:	T.R. Azad.
4.	*Nritya Prashan Panjika*	:	J.N. Pathak.

FINE ARTS

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : Theory (History of Art) Max. Marks : 60

Written Paper	:	54 Marks
Internal Assessment	:	06 Marks
Total	:	**60 Marks**

Paper-B : Practical

This paper consists of three sections :

		Max. Marks	*Max. Time*
1..	Landscape Composition	30	5 hours
2.	Life Sketching	30	5 hours
3.	Illustration in Advertising or Clay Modelling	30	5 hours
SESSIONAL MARKS: (Based on work related equally to 3 sections)		50	
	Total	**140**	

Note: Minimum of 9 hours teaching per week be assigned and out of nine hours, six hours be earmarked for practical classes and three hours for theory classes (per week)

THEORY
Paper-A : History of Art

Max. Marks: 54
Time: 3 Hrs.

Note:

1. Each paper carries 54 Marks.
3. The paper-setter is required to set **9 questions** in all. **The Candidate is to attempt 5 questions as per the instructions given in the question paper.**
4. The first question shall be of short answer type containing **9 questions** spread over the whole syllabus. Each question is to be answered in about 25 to 30 words. It shall carry **18 Marks** and shall be a **Compulsory question.**
5. 8 questions are to be set from the entire syllabus consisting of 4 units. Two questions will be set from each unit and the candidates shall be given internal choice i.e. a candidate shall attempt one question from each unit. So in all, the candidate shall attemp 4 questions in all out of 8 questions. **Each question** would be of **9 marks.**

Objectives :

The aim of the paper is to introduce to the students various schools, styles and phases of the developments in painting and sculpture in India and the West. The emphasis will be to make them aware of the different terms, concepts, forms and subject matter of these works.

UNIT-I : History of Indian Painting

- Rajasthani Painting – Chavand Ragmala from Mewar, Raga-Ragini from Bundi, Nayika from Kishangarh.
- Pahari Painting – Basohli Ramayana, Gita-Govinda from Kangra.
- Company Painting.
- Folk Style – Kalighat and Maithili.

UNIT-II : History of Indian Sculpture

- Mahabalipuram - Mahishasuramardini
- Elephanta – Ardhanarishavara.
- Chola Bronzes – Nataraja and Parvati images.
- Khajuraho – Mother and Child.
- Konarak – Musician figures.
- Sravanbelgola – Bahubali.
- Terracotta Sculptures from the Medieval Period.

UNIT-III : History of Western Art

- Beginning of Modern Painting – Manet
- Impressionism – Monet
- Neo Impressionism – Seurat
- Post Impressionism – Van Gogh
- Expressionism – Edvard Munch

UNIT-IV : Definition of Key Terms, General Concepts and Techniques

- Form and Content, Art and Soul, Art and Society, Tradition, Modernity, Beauty, Rasa and Bhava, Gestures, Postures and Movements, Stained glass, Cire-perdue (lost-wax casting)

Pedagogy :

The students are expected to familiarize themselves with the art forms as seen from the books, slides and related films.

Suggested Readings

1. Majumdar, R.C. (editor) : *The History and Culture of Indian People,* Vols. I, II and III (Sculpture and Painting Sections only), Bhartiya Vidya Bhawan, Bombay, 1988, 1990, 1996.

2. Goswamy, B.N. & Fischer, E. : *Pahari Masters : Court Painters of Northern India,* Museum Reitberg, Zurich, Switzerland, 1992.

3. Khandalwala, Karl : *Pahari Miniature Painting,* The New Book Co. Pvt. Ltd., Bombay, 1958.

4. Rowland, Benjamin : *The Art and Architecture of India,* Penguin Books, Great Britain, 1959.

5. Saraswati, S.K. : *A Survey of Indian Sculpture,* Munshiram Manoharlal Publishers Pvt. Ltd., New Delhi, 1975.

6. Ray, N.R. : *An Approach to Indian Art,* Publication Bureau, Panjab University, Chandigarh, 1974.

7. Barrett, D. and Gray, B. : *Painting of India,* The World Publishing Co., Ohio, 1963.

8. Archer, W.G. : *India and Modern Art,* George Allen & Unwin Limited, London, 1959.

9. Aggarwala,V.S. : *Heritage of Indian Art,* Publications Division, Ministry of Information & Broadcasting, Govt. of India, New Delhi, 1976.

10. Aggarwala,V.S. : *Indian Art* (English), Varanasi, 1965.

11. Aggarwala,V.S. : *Bhartiya Kala* (Hindi), Prithvi Prakashan, 1977.

12. Read, Herbert : *Meaning of Art,* Faber & Faber, London, 1972.

13. Janson, H.W. : *History of Art,* Thames & Hudson, London, 2001.

14. Gardener, Helen : *Art Through the Ages,* Harcourt Brace & Co., U.S.A., 1991.

15. Gombrich, E.H. : *The Story of Art,* Phaidon Press Limited, New York, 1995.

16. Arnason, H.H. : *A History of Modern Art,* Thames & Hudson, London, 1988.

PRACTICAL

Paper-B : Practical

This paper consists of three sections :

		Max. Marks	*Max. Time*
1.	Landscape Composition	30	5 hours
2.	Life Sketching	30	5 hours
3.	Illustration in Advertising or Clay Modelling	30	5 hours
	Total :	**90**	

Note: The same paper setter will be requested to set papers in all the options in Section-III.

SECTION-I

Landscape Composition

Landscape Composition: By Imagination or on the spot. Emphasis should be on linear perspective, aerial perspective, colour, tones and textures as visible.

Medium : Oil, Water or Pastel colours
Size : ½ Imperial sheet

SECTION-II

Life Sketching

Life Sketching : From live model, coloured in any medium on ½ Imperial Sheet.

Approach to work should be academic relating to proportion, structure, tonal values, volume and perspective and Individuality of the model.

SECTION-III

Illustration in Advertising or Clay Modelling

Illustration in Advertising :

To launch and promote a product and campaign advertisement, understanding of modern technique.

Concept of lettering : Block lettering, Roman lettering, Script lettering and free hand brush lettering. To design. book cover with illustration and title, author's name etc. in three colours (excluding the background colour)

Medium : Poster & Ink
Size : Half Imperial sheet

OR

CLAY MODELLING:

To render animal or human forms in clay in relief. Minimum number of three forms to be composed. size of slab minimum 30 cms. × 25 cms. Creative Pottery – Creative pot making in clay – any size.

SESSIONAL MARKS: 50 (based on work related equally to 3 sections)

Sessional marks will be given on the basis of work done during the session in all the three sections. At least five works will be submitted in each section. Sessional marks shall be given by external and internal examiners jointly. In case of difference of opinion, marking may be done separately by each examiner giving marks out of 50% of the aggregate of the sessional marks.

Note: 1. Choice of option to be offered would depend on the facility available in each Institution concerned.

2. Minimum of 9 hours teaching per week be assigned to the subjects and out of nine hours, six hours be earmarked for practical classes and three hours for theory classes (per week).

HISTORY OF ART

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : History of Indian Painting (from ca. 1800 to the present times) and Sculpture (from ca. 600 to 1300 A.D.) Max. Marks : 100

Paper-B : History of European Painting and Sculpture (from ca. 1850 A.D. onwards) and Theory and Principles of Art Appreciation. Max. Marks : 100

Paper-A : History of Indian Painting (from ca. 1800 to the present times) and Sculpture (from ca. 600 to 1300 A.D)

Max. Marks : 100

Time : 3 Hrs.

Note :

1. The paper-setter is required to set 9 questions in all. The candidate is to attempt 5 questions as per the instructions given in the question paper.

2. The first question shall be of short answer type containing 14 questions spread over the whole syllabus. Each question is to be answered in about 25 to 30 words. It shall carry 28 marks and shall be a **Compulsory** question.

3. 8 questions are to be set from the entire syllabus consisting of 4 units. Two questions will be set from each unit and the candidates shall be given internal choice i.e. a candidate shall attempt one question from each unit. So in all, the candidate shall attempt 4 questions in all out of 8 questions. Each question would be of 18 marks.

Objectives :

The aim of the paper is to introduce to the students various schools, styles and phases of the developments in painting and sculpture in India. The emphasis will be to make them aware of the different terms, concepts, forms and subject matter of these works.

History of Indian Painting :

Unit-I

- Company painting.
- Early oil painters – Raja Ravi Verma.
- Bengal School with special reference to Abanindranath Tagore

Unit-II

- New Trends : Nandlal Bose, Rabindranath Tagore, Gaganendranath Tagore, Jamini Roy, Amrita Shergil, D.P. Roy Chowdhury, Sobha Singh, M.F. Hussain, Satish Gujral.

History of Indian Sculpture

Unit-III

- Sculptures of Pala and Sena Period – Bengal, Bihar, Orissa.

Unit-IV

- Pratihara Sculpture of Central and Western India.
- Chola Sculpture in Stone & Bronze

Pedagogy :

The students are expected to familiarize themselves with the art forms as seen from the books, slides and related films.

Suggested Readings :

1. Archer, W.G. : *India and Modern Art,* George Allen & Unwin Limited, London, 1959.
2. Archer, M. & Archer, W.G. : *Indian Painting for the British,* 1770-1880, Oxford University Press, London, 1955.
3. Chaitanya, Krishna : *A History of Indian Painting, The Modern Period,* Abhinav Publications, New Delhi, 1994.
4. Appasamy, Jaya : *Abanindranath Tagore and the Art of His Times,* Lalit Kala Akademi, New Delhi, 1968.
5. Mago, P.N. : *Contemporary Art in India,* A Perspective, National Book Trust of India, New Delhi, 2000.
6. …… : *Lalit Kala Monographs,* Lalit Kala Akademi, Delhi.
7. Journals and Periodicals : *Lalit Kala Contemporary,* Roopa-Lekha, Marg.
8. Parimoo, Ratan : *The Paintings of the Three Tagores,* Maharaja Sayajirao University, Baroda, 1973.

9. Majumdar, R.C. (editor) : *The History and Culture of Indian People,* Vol. I, II and III. (Sculpture and Painting sections only) Bhartiya Vidya Bhawan, Bombay, 1996, 1990, 1988.

10. Rowland, Benjamin : *The Art and Architecture of India,* Penguin Books, Great Britain, 1959.

11. Saraswati, S.K. : *A Survey of Indian Sculpture,* Munshiram Manoharlal Publishers Pvt. Ltd., New Delhi, 1975.

12. Aggarwala, V.S. : *Heritage of Indian Art,* Publications Division, Ministry of Information & Broadcasting, Govt. of India, New Delhi, 1976.

13. Aggarwala, V.S. : *Indian Art* (English), Varanasi, 1965.

14. Aggarwala, V.S. : *Bhartiya Kala* (Hindi), Prithvi Prakashan, 1977.

Paper-B : History of European Painting and Sculpture (from ca. 1850 A.D. onwards) and Theory and Principles of Art Appreciation.

Max. Marks : 100

Time : 3 Hrs

Note :

1. The paper-setter is required to set 9 questions in all. The candidate is to attempt 5 questions as per the instructions given in the question paper.

2. The first question shall be of short answer type containing 14 questions spread over the whole syllabus. Each question is to be answered in about 25 to 30 words. It shall carry 28 marks and shall be **Compulsory** question.

3. 8 questions are to be set from the entire syllabus consisting of 4 units. Two questions will be set from each unit and the candidates shall be given internal choice i.e. a candidate shall attempt one question from each unit. So in all, the candidate shall attempt 4 questions in all out of 8 questions. Each question would be of 18 marks.

Objectives :

The aim of the paper is to introduce to the students various schools, styles and phases of the developments in painting and sculpture in the west. The emphasis will be to make them aware of the different terms, concepts, forms and subject matter of these works.

History of European Painting and Sculpture :

Unit-I

- Impressionism – Monet, Degas, Renoir.
- Post-impressionism – Van Gogh, Cezanne, Gauguin.

Unit-II

- Cubism – Picasso, Braque.
- Expressionism – Munch, Nolde.
- Abstract Art – Kandinsky.
- Abstract Expressionism – Jackson Pollock.

Theory and Principles of Art Appreciation.

Unit-III

- Function of Art.
- A brief study of Indian and Western approaches to Art.

Unit-IV

- Explanation of the term
- Form, Content, Abstraction, Modernity, Contemporaneity, Pointilism, Collage, Lithograph, Etching and Ready-made with the help of relevant examples.

Pedagogy :

The students are expected to familiarize themselves with the art forms as seen from the books, slides and related films.

Suggested Readings :

1. Janson, H.W. : *History of Art,* Thames & Hudson, London, 2001.
2. Gardener, Helen : *Art Through the Ages,* Harcourt Brace & Co., U.S.A., 1991.
3. Gombrich, E.H. : *The Story of Art,* Phaidon Press Limited, New York, 1995.
4. Arnason, H.H. : *A History of Modern Art,* Thames & Hudson, London, 1988.
5. Ray, Niharanjan : *An Approach to Indian Art,* Publication Bureau, Panjab University, Chandigarh, 1974.
6. Aggarwala,V.S. : *Heritage of Indian Art,* Publications Division, Ministry of Information & Broadcasting, Govt. of India, New Delhi, 1976.
7. Aggarwala,V.S. : *Indian Art* (English), Varanasi, 1965.
8. Aggarwala,V.S. : Bhartiya Kala (Hindi), Prithvi Prakashan, 1977.
9. Read, Herbert : *Meaning of Art,* Faber & Faber, London, 1972.

ANCIENT INDIAN HISTORY, CULTURE AND ARCHAEOLOGY

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Outlines of Tests, Syllabi & Courses of Reading

Paper-A (VOCATIONAL) : EXCAVATIONS, MONUMENTS AND SCULPTURES IN STONE AND BRONZE

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objectives :

The primary objective of this paper is to prepare the students to become professional archaeologists through the study of various excavated archaeological sites, monuments and antiquities such as stone and bronze sculptures. The study of this paper is also designed for preparing students to take higher and advanced study in the subject.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

1. The paper setter is required to set 9 questions in all. All questions shall carry equal marks.
2. The first question shall be short answer type containing 15 short questions spread over the entire syllabus. The candidate is required to answer any 9 short answer type questions. Each question shall be of 2 marks to be answered in 25-30 words each. **OR** A question on map. The map work shall consist of 12 marks for the map and 06 marks for the explanatory notes.
3. The map question shall have the following topics :
   (a) Location of important archaeological sites mentioned in Unit I.
   (b) Location of important monuments mentioned in Unit II.
   (c) Location/Provenance of important sculptures mentioned in Units III and IV.
4. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. Each unit shall have two questions and the candidate shall be given internal choice, i.e. the candidate shall attempt one question from each unit. Each question shall carry 18 marks.
5. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper setter must put note (5) in the question paper.

Unit-I : Excavations : (Note : Instructions will be confined only to the location of the site and important results obtained). Kalibangan, Lothal Mitathal, Sugh, Sanghol, Inamgaon, Atranjikhera, Sh: supalgarh, Nagarjunkonda.

Unit-II : Monuments : Asokan Pillars, Stupa of Sanchi, Rock-cut-architecture of Ajanta and Ellora, Temples : Kandariya, Mahadev, Lingaraja, Jagannath, Konark, Hoyasalesvara, and Brihadesvara.

Unit-III : Sculptures : Stone-Harappan, Rampurva Asokan Bull, Didarganj Yakshi, Katra Buddha, Sarnath Seated Buddha, Gomatesvara at Sravanavelagola, Maheshamurti, (Elephanta), Krishna, holding Govardhana (Halebid), Sun Image of Konark, Kasia reclining Buddha (Mahaparinirvana).

Unit-IV : Bronze : Sultanganj Buddha, Nalanda Image of Balarama and Buddha, Kurkihar Avalokitesvara, Indra and Padmapani from Nepal, Tanjore (Chola), Nataraja, Balakrishna Kaliyadamana and Somaskanda Murti.

Pedagogy of the Course Work :

It is expected to familiarize students with brief outline of the topics with the help of visual aids like slides and transparencies. Field work trips to museums and sites may also be undertaken.

Essential Readings :

1. Agrawala, P.K. : *Bharitya Kala Avam Vastu,* Prithvi Prakashan, Varanasi.
2. Agrawala, V.S. : *Indian Art,* Prithvi Prakashan, Varanasi, 1965.
3. Allchin, B. and Allchin, F.R. : *The Rise of Civilization in India and Pakistan,* Cambridge University, Cambridge, 1982.
4. Bajpai, K.D. : *Five Phases of Indian Art,* Vidya Prakashan, Jodhpur, 1991.
5. Brown, Percy : *Indian Architecture (Buddhist and Hindu Period),* Vol-1, rpt. D.B. Tara Gorevala sons, Bombay, 1971.
6. Grover, Satish : *The Architecture of India : Buddhist and Hindu,* Delhi, 1980.
7. Lal, B.B. : *The Early Civilization of South Asia,* Aryan Books, New Delhi, 1997.
8. Saraswati, S.K. : *A Survey of Indian Sculpture, 2nd ed.* Munshiram Manoharlal, New Delhi, 1957.

Paper-B : EPIGRAPHY AND NUMISMATICS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objectives :

This course makes the students aware of major scripts of Ancient India and their origin and development upto 6th century A.D. Antiquity and art treasure laws are also taught. It also provides knowledge about the origin and antiquity of Punchmarked coins; tribal coins; Yaudheyas, Kunindas, Agra, Audumbaras and Malavas; Kushana and Gupta coins.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

1. The paper setter is required to set 9 questions in all. All questions shall carry equal marks.
2. The first question shall be short answer type containing 15 short questions spread over the entire syllabus. The candidate is required to answer any 9 short answer type questions. Each question shall be of 2 marks to be answered in 25-30 words each. **OR** A question on map. The question on map shall consist of the location of the sites of important inscriptions of Asoka.
3. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. Each unit shall have two questions and the candidate shall be given internal choice i.e. the candidate will attempt one question from each unit. Each question shall carry 18 marks.
4. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
   The paper setter must put note (4) in the question paper.

Unit-I : (a) Epigraphy : Major scripts of Ancient India.
(b) Their origin and development up to 6th century A.D.

Unit-II : Rock and pillar edicts of Asoka.

Unit-III : (a) Origin and Antiquity of Punchmarked coins.
(b) Origin and date of Tribal coins : Yaudheyas, Kunindas, Agra, Audumbaras and Malavas.

Unit-IV : (a) Kushana Coins.
(b) Gupta Coins.

Pedagogy of the Course Work :

The students are to be taught with the help of slides, photographs and maps. In addition to it, special lectures, workshops, seminars, written assignments, class discussions, and term papers, should be included in the teaching work.

Essential Readings :

1. Allan, John : *Catalogue of Coins of Ancient India,* Vol.I, rpt. Delhi, 1974.
2. Altekar, A.S. : *Coinage of the Gupta Empire,* Numismatic Society of India, BHU, Varanasi,1957.
3. Chattopadhya, B. : *The Age of Kushanas,* A Numismatic Study, Calcutta, 1970.
4. Gupta, P.L. : *Coins,* rpt., National Book Trust, New Delhi, 2006.
5. Hultzsch, E. : *Corpus Inscriptionum Indicarum,* Vol.I, rpt. Archaeological Survey of India, Delhi, 1991.
6. Pandey, R.B. : *Indian Palaeography,* Munshiram Manoharlal, Banaras, 1952.

DEFENCE & STRATEGIC STUDIES

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

INSTRUCTIONS FOR THE PAPER SETTER AND THE STUDENTS :

1. There will be two theory papers A and B carrying 70 marks each. The internal assessment will be of 10 marks for each of the two papers. 40 marks are kept for the practical test. Each theory paper will have one compulsory short answer type question containing 15 short answer type questions of 2 marks each covering the entire syllabus. The candidates will be required to attempt any 10 short answer type questions. In addition, there will be four sections of the question paper. The candidates will be required to attempt one question from each of these sections carrying each 12 ½ marks. Each theory paper will be of three hours duration. The practical test will be of 1 ½ hours duration.

2. Practical examination will be **compulsory** for regular, USOL and private candidates. The serving armed forces and para-military personnel will be exempted from practical examination and marks secured by such candidates out of 140 will be proportionately raised out of 200.

3. The persons appearing as private candidates (except serving armed forces and para-military personnel) shall have to complete the requirement of attending the practical at the Department of Defence & National Security Studies, Panjab University, Chandigarh, in January every year, after paying the required fee as prescribed by the University from time to time. Private candidates shall have to attend practical classes for 10 hours spread over two to three days for which the certificate shall be issued by the Chairman, Department of Defence & National Security Studies, Panjab University only.

4. For reappearing candidates, who have not been assessed earlier for internal assessment and practical examination, their marks will be increased proportionately upto 100 marks for each of two papers.

Paper-A : NATIONAL SECURITY, CONCEPTIONAL ASPECTS

Max. Marks : 70
Time : 3 Hours

Objective :

This paper deals with the conceptual aspects of national security and the role of economy & military organizations in furthering national pursuits.

SECTION-I

1. *National Security :* Definition and Concept.

2. Elements of National Security, Geography, Mineral Resources, Economic Stability, Military Preparedness, Socio-Political Factors, Science and Technology.

SECTION-II

3. *Collective Security Arrangements :* Security Relevance of UN in the modern context; Role of NATO in the Post Cold War era.

SECTION-III

4. *Regional Groupings :* SAARC, ASEAN and BIMSTEC (Aims and Objectives of the three Alliances to be discussed).

SECTION-IV

5. *Non-aligned Movement and National Security :* History, Features, Problems and Achievements.

Books Recommended

1. Abhyankar, M.G. : *Defence Principles and Organisation,* Usha Publications, Longmans, Michigan, 1966.
2. Aditya, Chibber : *National Security Doctrine,* Lancers International, New Delhi, 1990.
3. Alva, Myrdal : *The Game of Disarmament,* Pantheon Books, New York, 1982.
4. Bandhyopadhyay, J. : *India's Foreign Policy Making,* Allied Publishers Private Ltd., Mumbai, 1970.
5. Choudhary, Subrata Roy : *Military Alliances and Neutrality in War and Peace,* Orient Longmans, Bombay, 1966.
6. Hadley, Arthur T. : *The Nation's Safety and Arms Control,* Viking Press, University of Michigan, 1961.
7. Hans, J. Margenthau : *Politics Among Nations,* McGraw-Hill, New York, 1993.
8. Misra, K.P. : *Studies in Indian Foreign Policy,* Vikas Publications, Delhi, 1969.
9. Rana, A.P. : *The Imperative of Non-Alignment,* McMillan Cop. of India, 1976.
10. Shah, A.B. : *India's Defence and Foreign Policies,* Manaktalas, 1966.
11. Singh, Jasbir : *Indian Defence Year Book, 2012-13,* Natraj Publishers, Dehradun, 2001.

Paper-B : NATIONAL SECURITY OF INDIA

Max. Marks : 70

Time : 3 Hours

Objective : This paper covers the various factors related to National Security in India.

SECTION-I

1. India's security problems since 1947 :

   (a) Geo-political effects of partition – boundaries and frontiers.

   (b) Integration of States : J. & K., Junagarh, Hyderabad and Goa.

   (c) Security problems related to Pakistan and China with particular reference to 1947, 1965, 1971 and 1999 Indo-Pak Wars; 1962 Sino-Indian War and subsequent developments.

SECTION-II

2. Indian Ocean and India's Maritime Security.

3. Role of Air Warfare—India's perspective.

4. Nuclear Policy of India: History, Evolution, Policy options.

SECTION-III

5. Internal Dimension of India's National Security with particular reference to Insurgency, Terrorism and Low Intensity Conflict. Role of Para-Military Forces.

6. Disaster Management and the role of Civil Defence.

SECTION-IV

7. Economic Mobilization for National Defence.

8. Planning and Production for National Defence with particular reference to India's Defence Production and DRDO.

Books Recommended :

1. Barnawal, S.P. : *Military Year Book,* A Guide Publication, New Delhi, 2012
2. Chopra, Maharaj, K. : *India the Search for Power: India and the Indian Ocean,* Lalwani Publication House, Delhi, 1969
3. Cohen, Stephen, P. : *The Indian Army,* Oxford University Press, New Delhi, 1990
4. Kaul, Ravi : *India's Strategic Spectrum,* Chanakya Publishing House, Delhi, 1969.
5. Kavic, L.J. : *India's Quest for Security,* Cambridge University Press, London, 1967
6. Klaus, Knorr : *War Potential of the Nation,* Greenwood Press, Northwestern University, 1979.
7. Khera, S.S. : *India's Defence Problems,* Orient Longmans, University of California, 1968.
8. Kohli, S.N. : *Sea Power in the Indian Ocean,*Tata McGraw Hill, Delhi, 1978.
9. Misra, R.N. : *Indian Ocean and India's Security,* Mittal Publications, Delhi, 1986.
10. Pathak, K.K. : *Nuclear Policy of India,* Gitanjali Prakashan, Delhi, 1980
11. Panikkar, K.M. : (i) *Defence Problems of India.*
    (ii) *India and the Indian Ocean,* G. Allen & Unwin, Ltd., 1945.
12. Rao, PVR : *Defence Without Drift,* Popular Prakashan, Delhi, 1970.
13. Rao, Rama : (i) Self Reliance and Security, Radiant, University of Michigan, 1984.
    (ii) *Role of Defence Production.* Radiant, University of Michigan, 1984.
14. Singh, Bhupinder : *Textbook of Defence Studies* (Part III Punjabi), B.C. Publishers, Patiala, 1983.
15. Singh, Jasbir : *Defence Year Book, 2012-13*, Natraj Publication, Dehradun, 2001.
16. Singh, Jasjit : *Nuclear India,* Knowledge World, Delhi, 1998.
17. Singh, Jaswant : *Defending India, Hamisphire,* McMillan, Basingstoke, 1998.
18. Thomas, Raju, G.C. : *The Defence of India (A Budgetary Perspectives),* McMillan, Delhi, 1970.
19. Subrahmanyam, K. : (i) *Indian Security Perspectives,* ABC Publishing House, New Delhi, 1982.
    (ii) *Planning for Defence.*

Paper –C : PRACTICAL

Total Marks : 40
Time : $1\frac{1}{2}$ Hours

Note :
1. Practical Examination will be compulsory for regular, private and USOL students.
2. There will be five questions in all and candidates will be required to attempt any three questions of ten marks each.
3. Examiners are required to set the question paper atleast half an hour before the examination.
4. There will be 3 hours of teaching per week for practical.
5. For practical classes, the number of students in one group shall not exceed fifteen.
6. Practical exercises should be carried out on drawing sheets with explanatory notes or on computer.

SECTION–A : Practical Test : 30 Marks

1. Relief features and their representation on map.
2. Degree of slopes, gradients and inter visibility of points.
3. Study of field craft with reference to the following :
   (a) Ground
   (b) Cover
   (c) Camouflage
   (d) Observation
4. Tactical Formations : Section and Platoon.
5. Application of Fire : Fire control, fire control orders and sequence of fire control orders.
6. Military Messages : Verbal and written.

SECTION–B : Practical Record : 5 Marks

SECTION–C : Viva-Voce : 5 Marks

HISTORY

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

INSTRUCTIONS FOR THE PAPER-SETTERS AND CANDIDATES (For papers A & B) :

1. *The syllabus has been divided into four units.*
   There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any **9** short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

2. For the private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper-setter must put note (2) in the question paper.

3. One question from Unit IV shall be set on the map.

Explanation :

1. Each essay type question would cover about one-third to one half of a topic detailed in the syllabus.

2. The distribution of marks of the map question would be as under :

   Map : 10 marks

   Explanatory Note : 08 marks

   Note : In case, a paper-setter chooses to set a question of map on important historical places, the paper-setter will be required to ask the students to mark 10 places on map of 1 mark each and write explanatory note on any four of 2 marks each.

3. The paper-setter would avoid repetition between different types of questions within one question paper.

Paper A : WORLD HISTORY 1500-1870 A.D.

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hrs.

Objectives : To introduce the students to the history of the Modern World.

Pedagogy : Lectures, library work and discussions.

Unit-I

1. *World Around 1500 :* Polity; Economy; Society.
2. *Rise of Absolute States :* France and Prussia.
3. *Rise of Parliamentary Government :* The Glorious Revolution and its effects.

Unit-II

4. *The American Revolution :* Its social, political and economic causes; its consequences.
5. *The French Revolution :* Causes and impact of the French Revolution.
6. *Napoleon Bonapart :* Reforms of Napoleon Bonapart; his continental system.

Unit-III

7. *Congress of Vienna (1815) :* Motives, working, principles, provisions and significance of Congress of Vienna.
8. *The Industrial Era :* The Industrial Revolution (1750-1850); causes for its origins in England; New inventions; spread to Europe; impact on society.
9. *Eastern Question :* Greek War of Independence; Mehmat Ali and Egypt; Crimean War.

Unit-IV

10. *Unification of Italy :* Different stages in unification of Italy; role of Mazzini, Cavour and Garibaldi.
11. *Unification of Germany :* Rise of Nationalism and role of Bismarck in the unification of Germany.
12. *Map :*
    - (a) *Important Places :* Paris, London, Rome, Berlin, Frankfurt, Vienna, Waterloo, Moscow, Crimea, Trafagar, Venice, Bonn and Budapest.
    - (b) Unification of Germany.
    - (c) Unification of Italy.

Suggested Readings :

1. Davis, H.A. : *An Outline History of The World,* OUP, 1964, 4th Edition.
2. Garraty, John A. & Peter Gay (ed.) : *The Columbia History of the World,* United Kingdom : New Orchard Edition, 1985.

3. Gupta, P.S. : *Adhunik Paschim Ka Uday,* Hindi Madhyam Karanvaya Nideshalaya, Dilli : Dilli Vishavidhalaya, 1997 (Hindi Medium).

4. Gupta, P.S. : *Europe Ka Itihas,* Hindi Madhyam Karanvaya Nideshalaya, Dilli : Dilli Vishavidhalaya, 1996 (Hindi Medium).

5. Hinsley, F.H. (ed.) : *Material Progress and World Wide Problems, 1870-1898,* Cambridge University Press, 1976.

6. Stavrianos, L.S. : *The World Since 1500,* Delhi : Prentice Hall of India, 1965.

Paper B : WORLD HISTORY (1871 to 1956 A.D.)

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hrs.

Objectives : To introduce the students to the history of the Modern World in the period of European Domination.

Pedagogy : Lectures, library work and discussions.

Unit-I

1. *New Imperialism 1871-1914 :* Main features; Partition of Africa – Causes, colonization and impact.

2. *Congress of Berlin :* Circumstances, provisions and significance of the Congress of Berlin 1878.

3. *Diplomatic Developments in Europe :* Circumstances leading to the formation of Triple Alliance of 1882 and Triple Entente.

Unit-II

4. *World War I :* Division of Europe into two blocks; causes of the First World War.

5. *Paris Peace Conference :* Treaty of Versailles 1919; Provisions, major defects and their impact.

6. *Russian Revolution :* Causes and impact of the Russian Revolution of 1917.

Unit-III

7. *Nationalism and Communism in China :* Causes for the nationalist revolution of 1911 and its results; circumstances leading to the revolution of 1949 and its results.

8. *Modernization in Japan :* Meji restoration and modernization in Japan.

9. *The Great Depression of 1929 :* Causes of the Great Depression in the USA; its spread to Germany, France and Britain; its impact; Roosevelt's New Deal.

Unit-IV

10. *Fascism and Nazism :* Circumstances responsible for rise of Fascism under Mussolini in Italy; Nazism in Germany under Adolf Hitler.

11. *World War :* Causes responsible for the Second World War; Modernization of Turkey under Mastafa Kamal Pasha.

12. *Map :*
    (a) *Important Places :* Geneva, Washington, Tokyo, Versailles, Constantinople, Peking, Beijing, Manchuria, Nanking, Frankfurt, Johannesberg, Cairo, Jerusalem, Nagasaki, Warsaw and Stalinguard.
    (b) Europe on the eve of World War I.
    (c) Europe on the eve of World War II.

Suggested Readings :

1. Ashworth, W. : *A Short History of the International Economy, 1850-1950,* London : Longmans, 1954.
2. Brecher, Michael : *The Foreign Policy System of Israel, Toronto* : Oxford University Press, 1972.
3. Chabod, Fredrico : *History of Italian Fascism,* London : Weidenfeld, 1961.
4. Gupta, P.S. : *Adhunik Paschim Ka Uday,* Hindi Madhyam Karanvaya Nideshalaya, Dilli : Dilli Vishavidhalaya, 1997 (Hindi Medium).
5. Gupta, P.S. : *Europe Ka Itihas,* Hindi Madhyam Karanvaya Nideshalaya, Dilli : Dilli Vishavidhalaya, 1996 (Hindi Medium).
6. Garraty John, A. & Peter Gay (ed.) : *The Columbia History of the World,* United Kingdom : New Orchard Edition, 1985.

POLITICAL SCIENCE

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : COMPARATIVE POLITICAL SYSTEMS (UK AND USA)

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

The purpose of this paper is to serve as an introduction to the field of comparative politics. It provides a broad overview of the field of comparative politics and examines some key approaches. The major part of the paper is devoted to understanding and analyzing the origins and working of two political systems, the UK and the USA. The student will not only become familiar with the working of these two political systems but also understand how the concepts of comparative politics can be used to understand real world politics.

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

1. *The syllabus has been divided into four units :*
   There shall be **9** questions in all. The first question is compulsory and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions, and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

2. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
   The paper setter must put note (2) in the question paper.

Unit-I : Theoretical Framework

(i) Meaning, Nature and Scope of Comparative Government and Politics.

(ii) Comparative Method.

Unit-II : U.K.

(iii) The British Political tradition.

(iv) Parliamentary Government, Monarchy, Cabinet, Parliament.

(v) Judicial System and Rule of Law.

Unit-III : U.S.A.

(vi) Evolution of American political system - War of Independence, Framing of the Constitution, Bill of Rights.

(vii) The Constitutional Framework.

(viii) Federal System.

Unit-IV

(ix) Political Parties and Interest/Pressure Groups of U.K.

(x) Political Parties and Interest/Pressure Groups of U.S.A.

Books Recommended :

1. J.C. Johari : *Major Modern Political Systems,* Vishal Publication, Delhi.
2. J.C. Johari : *Comparative Politics,* Sterling Publishers, New Delhi.
3. A.C. Kapoor : *Select Constitutions,* S. Chand and Company, New Delhi.
4. V.N. Khanna : *Comparative Study of Government and Politics,* S. Chand and Comp., New Delhi.
5. Vishnu Bhagwan and Vidya Bhushan : *World Constitutions,* Sterling Publisher, New Delhi.
6. K.R. Bombwall : *Major Contemporary Constitutional Systems,* Modern Publication, Ambala Cantt.
7. H.S. Deol : *Adhunik Sarkrana* (Punjabi) Publication Bureau, Punjabi University, Patiala.
8. Andrew Heywood : *Politics,* Macmillan Palgrave, New York, 1997.
9. Mackintosh : *The Government and Politics of Britain,* Hutchinson and Comp., London, 1977.
10. C.O. Johnson : *Government in the United States,* any edition, New Delhi.
11. Herman Finer : *Theory and Practice of Modern Government,* Theuen and Comp., London, 1963.
12. Hague and Harrop : *Comparative Government and Politics,* Palgrave, New York, 2001.

Paper-B : INTERNATIONAL POLITICS : THEORY AND PRACTICE

Max. Marks		: 100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives :

This paper provides students with an overview of the broad theories and concepts used to understand international politics. It also examines key issues in contemporary global history from an international politics perspective.

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

1. *The syllabus has been divided into four units :*
   There shall be **9** questions in all. The first question is compulsory and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions, and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

2. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
   The paper setter mut put note (2) in the question paper.

Unit-I

1. Meaning, Nature and Scope of International Politics.
2. Realist and Idealist approaches to International Politics.

Unit-II

3. National Power : Its Elements.
4. Balance of Power and Collective Security.

Unit-III

5. Cold War and Post Cold War era of International Politics.
6. Bipolar, Unipolar and Nature of Emerging World Order.

Unit-IV

7. United Nations : Aims, Objectives and Principles.

8. Regional Organisations : SAARC and EU.

9. New International Economic Order (NIEO).

Books Recommended :

1. H. Bull, *The Anarchical Society : A Study of Order in World Politics,* London, Macmillan.

2. E.H. Carr, *The Twenty Year Crisis,* London, Macmillan, 1939.

3. –do--, *Conditions of Peace,* New York, The Macmillan Company, 1944.

4. J. Frankel, *The Making of Foreign Policy,* London, Oxford University Press, 1963.

5. S.H. Hoffman (ed.), *Contemporary Theory in International Relations,* Massachusetts, Addison-Wesley, 1979.

6. A. Hurell, "Collective Security and International Order Revision", *International Relations*, Vol. II, No. 1, April.

7. H.J. Morgenthau, *Politics Among Nations : The Struggle for Power and Peace*, 6th Edn., revised by K.W. Thompson, New York, Alfred Knopf, 1985.

8. N.D. Palmer and H. Perkins, *International Relations,* Calcutta Scientific Book Company, 1971.

9. A. Roberts, "The UN and International Security"*, Survival*, Vol. 35, No. 1, Spring.

10. S.P. Verma, *International System and the Third World,* New Delhi, Vikas, 1988.

11. John Baylis and Steve Smith, *The Globalisation of World Politics,* OUP, Oxford, 2001.

ECONOMICS

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper–A : ECONOMICS OF DEVELOPMENT

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objective :

The primary course objective is to introduce the students to the basic features, determinants, and theories and strategies of development of underdeveloped economies. It also introduces students to the theory of how control and direction of economic activity by a central public authority can be used as an alternative to market by the underdeveloped economies.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 09 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

Unit-I

Economic Growth and Development. Main Features of an Underdeveloped Economy.

Determinants of Economic Development. Capital Formation : Its Source, Nurkse's Thesis of Disguised Unemployment. Lewis Theory of Unlimited Supply of Labour.

Unit-II

Dualism : Social and Technological. Classical Model of Growth; Harrod Domar Model of Economic Growth; One Sector Neo-classical Model of Growth.

Unit-III

Strategies of Economic Development : Balanced Vs. Unbalanced Growth. Leibenstein's Critical Minimum Efforts Thesis. Theory of Big Push. Rostow's Theory of Stages of Growth.

Unit-IV

Theory of Planning in Developing Countries : Need, Objectives, Strategies and Problems of Planning, Price Mechanism and Planning. Investment Criteria. Choice of Technique. Export Promotion and Import Substitution Strategy.

Books Recommended :

1. G. M. Meier (Eds.) : *Leading Issues in Economic Development,* Oxford University Press, New York, 1995.
2. W. W. Rostow : *Stages of Economic Growth,* Cambridge University Press.
3. Benjamin Higgins : *Economic Development : Principles, Problems and Policies,* Universal Book Stall, New Delhi, 1994.
4. G. M. Meier and James E. Rauch : (Eds.) : *Leading Issues in Economic Development,* Seventh Edition, Oxford University Press.
5. R. Nurkse : *Problems of Capital Formation in Underdeveloped Countries,* Oxford University Press.

Supplementary Readings :

Michael Todaro : Economic Development in the Third World, Orient Longman, London.

Note : Four to Five lectures of 45 minutes each per week are required to complete the syllabi.

Paper–B : QUANTITATIVE METHODS

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objective :

The objective of the course is to train the students in the use of basic mathematical and statistical tools in analyzing various economic phenomenons. It deals with the design of how data is presented, the analysis of the data, and the drawing of conclusions from the data. The course aims to improve decision-making accuracy of the students and enabling them to test new ideas.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

1. There shall be **9** questions in all. All questions carry equal marks. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 09 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all.

2. Use of simple calculator is allowed.

3. The paper setter may in general stick to the distribution of marks of 1/3 to theory and 2/3 to numericals.

Unit-I

Elementary Idea of Sets and Function : Simple, Derivative, Differentiation of Simple functions – Polynomial (x), Exponential functions. Maxima and Minima of functions of one variable only. Simple Application of Economics.

Unit-II

Matrices : Definition and Types, Operations (Sum, Difference, Product and Transpose), Adjoint and Inverse of a matrix (upto 3 x 3), Solution of Equations (upto 3) by Matrix Methods and Crammer's rule.

Measures of Central Tendency : Mean, Median, Partition Values, Mode, Measures of Dispersion, Skewness.

Unit-III

Correlation Analysis—Karl Pearson's (except grouped data) and Spearman's formula, Simple Regression Analysis.

Interpolation – Binomial, Expansion, Newton's (Advancing Difference Method) and Lagrange's Method.

Unit-IV

Index Numbers : Concepts, Problems and Importance; Simple Index Number, Lespeyre's and Fisher's Index Numbers only (among weighted index numbers), Reversibility Tests.

Time Series Analysis : Components of Time Series, Determination of Trend, Least Square and Moving Average Method.

Books Recommended :

1. Archibald, G. C. and Lipsey, R. G. : *An Introduction to a Mathematical Treatment of Economics,* 1977, English Language Book Society, Weidenfeld and Nicolson.

2. Sanchati, D. C. & Kapoor, V. K. : *Business Mathematics,* Sultan Chand & Sons, 1993, New Delhi.

3. Gupta, S. C. : *Fundamentals of Statistics,* Mumbai, Himalaya Publishing House.

Note : Four to Five lectures of 45 minutes each per week are required to complete the syllabi.

SOCIOLOGY

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper–A : SOCIETY IN INDIA

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER SETTER AND THE CANDIDATES :

(i) For written paper, the students will be required to attempt **five** questions in all. Question No. 1 will be **compulsory** comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students are required to attempt **nine** short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

In addition to it, Question Nos. II to IX will consist of long answer (essay type) questions, two questions from each unit with internal choice carrying 18 marks each i.e. $4 \times 18 = 72$ marks. The students are required to attempt one question from each of the four units.

(ii) On an average, 15 hours are to be devoted to each unit.

(iii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper–setter must put note (iii) in the question paper.

Objective :

The objective of this paper is to present a comprehensive view of Indian Society. The students are exposed to the tribal, rural and urban societies and are presented with the social structure and social institutions to understand these segments of Indian Society. Through this paper, the students are also introduced to the problems of the underprivileged in Indian Society.

Course Content :

Unit-I

Tribal Society : Meaning, Characteristics; Classification of tribes.

Institutional Features: Family, Marriage, Economy; The Changing trends.

Unit-II

Rural Society : Meaning; Characteristics.

Institutional Features : Family, Marriage; Economy and Polity (Village Panchayat); Changing Trends.

Unit-III

Urban Society : Meaning and Characteristics, Concepts of Urbanization and Urbanism; Institutional features; Urban family - features and changes; Economy; Voluntary associations; Slums.

Unit-IV

Under-privileged Sections—Women, Scheduled Castes; Scheduled Tribes and the physically Disabled; their problems and measures to improve their status.

Essential Readings :

1. Bose, N.K. (1980) : *Tribal Life in India,* National Book Trust.
2. Desai, A.R. (1969) : *Rural Sociology in India,* Bombay : Popular Prakashan.
3. Lal, S.N. & Nahar, U.R. (1992) : *Rural Social Transformations,* Jaipur : Rawat.
4. Madan & Majumdar (1989) : *An Introduction to Social Anthropology,* New Delhi : National Publications (Hindi & English).
5. Madhurima (2009) : *Readings in Sociology-Part-3*, Jalandhar : New Academic Publishing House [All Mediums].
6. Mandelbaum, G. (1970) : *Society in India,* Bombay : Popular Prakashan (Hindi & English).
7. Shah, Ghanshyam (ed.) (2002) : *Caste & Democratic Politics in India,* New Delhi : Permanent Black.
8. Sharma, R.K. (1997) : *Indian Society : Institutions & Change,* New Delhi : Atlantic Publications.
9. Singh, K.S. (2002) : *The Scheduled Castes*, New Delhi : Oxford University Press.

Further Readings :

1. Ahuja, Ram (1981) : *Social Problems in India,* Ghaziabad, Vikas Publications.
2. Mahajan, Sanjeev (2004) : *Gramin Samajshastra,* New Delhi : Arjun Publishing House.
3. Punit, A.E. (1978) : *Social Systems in Rural India,* New Delhi : Sterling Publications.

4. Rao, M.S., Chandrashekar Bhatt, L.N. Kelkar (eds.) (1991) : *A Reader in Urban Sociology,* New Delhi : Orient Longman.

5. Shah, D.S. & Sisodia, Y.S. (ed.) (2004) : *Tribal Issues in India,* Jaipur : Rawat Publications.

6. Singh, K. Suresh (2001) : *The Scheduled Tribes,* New Delhi : Oxford University Press.

7. Vardhan, Ranjay (2008) : *Single Women : A Study of Spinsters*, Delhi : Indian Publishers.

Paper-B : SOCIAL DISORGANISATION AND EMERGING PROBLEMS

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER SETTER AND THE CANDIDATES :

(i) For written paper, the students will be required to attempt **five** questions in all. Question No. 1 will be **compulsory** comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students are required to attempt **nine** short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

In addition to it, Question Nos. II to IX will consist of long answer (essay type) questions, two questions from each unit with internal choice carrying 18 marks each i.e. $4 \times 18 = 72$ marks. The students are required to attempt one question from each of the four units.

(ii) On an average, 15 hours are to be devoted to each unit.

(iii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper-setter must put note (iii) in the question paper.

Objective :

This paper exposes the students to social disorganization, its levels and current problems. It helps students to understand social realities and also equips them to utilize their knowledge in various theoretical and practical exercises.

Unit-I

Social Disorganization : Concepts and Levels –Personal, Familial and Societal .

Unit-II

Personal Problems : Problems of Adolescence; Alcoholism; Drug Addiction, Suicide.

Unit-III

Familial Problems : Girl child, Working Women, Female Headed, Household, Issues relating to Domestic Violence.

Unit-IV

Societal Problems : Poverty; Corruption;Communal Conflicts, Problems of the Aged.

Essential Readings :

1. Ahuja, Ram (1981) : *Social Problems in India,* Ghaziabad : Vikas Publications.
2. Elliot, H.V. & Merill, F. (1950) : *Social Disorganization,* New York : Harper Brothers.
3. Gill, S.S. (1998) : *Pathology of Corruption,* New Delhi : Harper Collins.
4. Macionis, John J. (2005) : *Social Problems,* New York : Prentice Hall.
5. Madan, G.R. (1978) : *Indian Social Problems,* New Delhi : Allied Publishers.
6. Mitra, S.M. (2005) : *Indian Problems,* New Delhi : Eastern Book Corporation.
7. Mohan, J. & Sehgal, M. (2004) : *Youth & Drugs,* New Delhi : Abhijit Publications.
8. Narasaiah, M.L. (2005) : *Poverty & Environment,* New Delhi : Discovery Publishing House.
9. Rai, Bhartiya (2004) : *Samajik Samasyaein,* New Delhi : Arjun Publishing House.
10. Vardhan, Ranjay (1999) *Female Headed Households in Patriarchal Society : A Sociological Study,* Delhi : Indian Publishers Distributors.

Further Readings :

1. Chakraborty, Rajgopal Dhar (2004) : *The Greying of India : Population, Ageing in the Context of Asia,* New Delhi : Sage.

2. Mahajan, Amarjit & Madhurima (1994) : *Family Violence and Abuse in India,* New Delhi : Deep and Deep Publications.

3. Natrajan, P.S. (2003) : "A Theory of Indian Communalism" in *Think India Quarterly,* Vol. 6, No. 3, July-Sept.

4. Pavrala Vinod (1996) : *Interpreting Corruption,* New Delhi : Sage.

5. Phandaris, Urmila (1989) : *Ethnicity and Nation Building in South Asia,* New Delhi : Sage.

6. Shankar Rao, C.N. (2007) : *Indian Society,* Delhi : S. Chand and Company.

PUBLIC ADMINISTRATION

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Outline of Tests, Syllabi and Courses of Reading

Paper–A : LOCAL GOVERNMENT (with special reference to Punjab)

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objective of the Paper :

The objective of the paper is to give the student an understanding of the concept, significance and evolution of local government in India. It would also acquaint them with the pattern and working of divisional and district administration. The key areas covered are the types, structure, functions, finances and personnel of rural and urban local governments. It would also include the concept of state control over local bodies, provincialisation and rural-urban relationship with reference to Punjab.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

1. For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.

   The paper-setter must put a note in question paper in this regard.

2. The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one. Each question will carry 18 marks.

Unit-I

Meaning and Significance of Local Government.

Evolution of Local Government since 1882.

Role of Deputy Commissioner.

Role of Divisional Commissioner.

Unit-II

The 73rd Constitutional Amendment – Provisions and its impact.

Gram Sabha – Composition, Functions and Powers.

Panchayati Raj Institutions in Punjab – Structure, Functions, Sources of Finances and Personnel.

Unit-III

The 74th Constitutional Amendment –Provisions and its Impact.

Urban Local Bodies –Structure, Functions, and Sources of Finance.

Mayor – Position, Functions and Powers.

Municipal Commissioner – Position, Functions and Powers.

Unit-IV

State Control over Local Bodies.

State Finance Commission : Composition, Functions and Role.

Provincialisation of Municipal Services.

Rural–Urban Relationship – Challenges and Remedies.

Essential Readings

1. S.R. Maheshwari : *Local Government in India* (Agra : L.N. Aggarwal, 2002).
2. Sahib Singh & Swinder Singh : *Local Government in India* (Jalandhar : New Academic Publishers, 2000).
3. K.K. Puri & G.S. Barara : *Local Government in India* (Jalandhar : Bharat Prakashan, 2008).
4. S.L. Goel & Shalini, Rajneesh : *Panchayati Raj in India* (New Delhi : Deep & Deep, Publishers, 2003).

Further Readings

1. B.S. Khanna : *Rural Developments in South Asia* (New Delhi : Deep & Deep Publications, 1991).
2. H.B. Singh : *Revitalised Urban Administration in India* (Delhi : Kalpaz Publishers, 2003).

Paper–B : DEVELOPMENT ADMINISTRATION (with special reference to Punjab)

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objective of the Paper :

The objective of the paper is to give the student an in-depth understanding about the concept & significance of development administration, features of developed & developing countries, planning machinery at Centre & State level and the emergence of India as a welfare state. The paper would also give an understanding about the concept, forms, role and problems of public enterprises as well as the working of select Union Ministries and agencies in Welfare and Development Administration.

INSTRUCTIONS FOR PAPER-SETTER AND CANDIDATES :

1. For Private/University School of Open Learning (USOL) students, who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.

   The paper-setter must put a note in question paper in this regard.

2. The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one. Each question will carry 18 marks.

Unit-I

Development : Meaning, Features and Aspects.

Development Administration : Meaning, Nature, Scope and Significance.

Features of developed and developing countries.

Unit-II

India as a Welfare State

Planning : Meaning, Objectives and Significance.

Planning Machinery : National Development Council (NDC), Planning Commission (PC), State Planning Boards (SPB), District Planning Committees(DPC).

Unit-III

Public Enterprises : Concept and Forms.

Role of Public Enterprises in Economic Development.

Problems of Public Enterprises.

Privatisation : Concept, Rationale; Cases of BALCO & Modern Bakery.

Unit-IV

Union Ministry of Human Resource Development - Organisation and Functions.

University Grants Commission – Composition and Functions.

Union Ministry of Health and Family Welfare – Organisation and functions.

Essential Readings

1. Sahib Singh and Swinder Singh : *Public Administration : Development and Local Administration* (Jalandhar : New Academic Publishers, 2010).
2. B.P. Mathur : *Public Enterprise Management (* New Delhi: Macmillan India 1999).
3. R.K. Sapru : *Development Administration* (New Delhi : Sterling Publishers, 2nd Edition, 2003).

Further Readings

1. Vijay Rattan : *Women and Child Development Programme Administration* (New Delhi : S. Chand, 1997).
2. R.B. Jain : *Public Administration in India – 21st Century Challenges for Good Governance* (Delhi : Deep & Deep Publications Pvt. Ltd., 2002).
3. S.P. Srivastava (ed.) : *The Development Debate : Critical Perspectives* (Jaipur & New Delhi : Rawat Publications, 1998).
4. India Development Report, 2002 S.Parikh, and R. Radhakrishna, (eds.) : *Indira Gandhi Institute of Development Research* (New Delhi : Oxford University Press, 2002).
5. S.L. Goel : *Advanced Public Administration* (New Delhi : Deep & Deep Publications, 2003).

PHILOSOPHY

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Outlines of tests, syllabi and courses of reading

Paper-I : INDIAN EPISTEMOLOGY AND METAPHYSICS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours
Lectures	:	75

Aims and Objectives :

This paper discusses the main epistemological and metaphysical issues as discussed in the various Indian Philosophical Systems.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

1. There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any **9** short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions and the candidate shall be given internal choice of attempting one question from each unit – 4 in all. Each question will carry 18 marks.

2. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper setter must put note (2) in the question paper.

Unit–I

1. Salient features of Indian Epistemology and Metaphysics.
2. Concept of Prama, Prameya and Pramana with special reference to Nyaya.
3. Pramanas : General account of Pratyaksa, Anumana, Sabda, Upmana, Arthapatti and Anupalabdhi.

Unit–II

4. Materialism of Charvakas.
5. Anekantavada of Jainism.
6. Arya satyas, Pratityasamutpada of Buddhism.

Unit–III

7. Concept of Self and Consciousness with special reference to Mandukya Upanishads.
8. Advaita Vedanta : Sankara on Brahman and Maya.

Unit–IV

9. Samkhya : Theory of Causation.
10. Vaiseika : Nature and kinds of Padartha.
11. Sikhism : Nature of Ultimate Reality, Man and the World.

Essential Readings :

1. Chatterjee, S.C. & Datta, D.M. : *An Introduction to Indian Philosophy* (Calcutta University, Calcutta, 1993).
2. Hiriyanna, M. : *Outlines of Indian Philosophy* (Allen and Unwin, London, 1999) (Hindi Translation Available), London : Allen.
3. Tarka-Samgraha of Annambhatta (Poona-Bhandarkar Oriental Research Institute, Poone, 1992).

Further Readings :

1. Ananikana : *A Nyaya Manual,* Ed. Sri. Krishna Sarma, Adyar Library and Research, Chennai, 1991.
2. Chatterjee, S.C. : *The Nyaya Theory of Knowledge* (University of Calcutta, Calcutta, 1995).
3. Panchapagesha Sastri : *The Philosophy of Aesthetic Pleasure* (Annamalai University, 1999).
4. Complete Works of Swami Vivekananda (relevant portions), (Advaita Ashram, Calcutta), 1957.

Paper–II : WESTERN EPISTEMOLOGY AND METAPHYSICS

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours
Lectures	:	75

Aims and Objectives :

This paper aims at exposing the students to main epistemological and metaphysical theories and problems of western philosophy. It also deals with basic themes of existentialism, logical positivism and analytical philosophy.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

The syllabus has been divided into four units.

1. There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any **9** short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions and the candidates shall be given internal choice of attempting one question from each unit – 4 in all. Each question will carry 18 marks.

2. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper setter must put note (2) in the question paper.

Unit-I

1. Nature of Philosophy with special reference to Metaphysics and Epistemology (introductory).
2. *Materialism :* Mechanical and Dialectical.
3. *Idealism :* Objective (Plato), Subjective (Berkeley).

Unit-II

1. *Nature of Knowledge :* Knowing subject, Act of knowing and object of Knowledge.
2. *Theories of Truth.*
3. *Foundationalism and Anti-Foundationalism.*

Unit-III

4. *Theories of Knowledge :*
   (a) Rationalism.
   (b) Empiricism.
   (c) Transcendentalism of Kant.

Unit-IV

5. Universal and Particulars : Concept of Being (Parmenides) and Becoming (Heraclites).
6. Substance (Spinoza) Causality (Hume).
7. Existentialism : Being in itself; Being for itself; Choice and Freedom.

Essential Readings :

1. Ewing, A.C. : *Fundamental Questions of Philosophy* (London : Routledge & Kegan Paul, 1999).
2. Titus, H.H. : *Living Issues in Philosophy,* Prentice Hall Pub. (New Delhi : Eurasia Pub. House, 4th ed., 1999).
3. Macquire, John : *Existentialism, Philosophical Library,* New York, 1949.

Further Readings :

1. Ayer, A.J. : *Twentieth Century Philosophy,* Penguin Books, Harmondsworth, 2002.
2. --do-- : *Language, Truth and Logic, Soft Cover ISBN 0486200108,* Dover Publications, 1969.
3. Korner, Stephen : Fundamental Questions of Philosophy (Penguin), Harmondsworth, 2002.

PSYCHOLOGY

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Objectives :

(I) This course will enable the students to get an introductory knowledge about abnormal psychology with emphasis on the dynamics of some of the behavioural disorders and therapies. Students will also have some knowledge about stress and coping; and will get acquainted with elementary inferential statistics.

(II) Pedagogy of the Course Work :

80 % Lectures (including expert lectures).

20% assignments, discussion and seminars.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

The syllabus has been divided into four units.

(a) There shall be **9** questions in all. First question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 7 short answer type questions i.e. 2 marks of each. It shall carry 14 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all with 14 marks for each.

(b) The practical will be of 40 marks.

Paper A : ABNORMAL PSYCHOLOGY

Theory	:	70 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

UNIT-I : *Viewpoints Regarding Abnormality :* Historical, Psychodynamic, Behavioural, Cognitive Behavioural, Humanistics and Interpersonal; Current Classifications of Abnormality.

UNIT-II : *Stress:* Concept of Stress; Categories of Stressors; Dynamics of Stress; Coping Strategies, Problem Focussed and Emotion Focussed.

UNIT-III : *Causes of Abnormal Behaviour :* Biological, Psychological and Sociocultural Causes.

UNIT-IV : *Significance of Statistics :* Mean, Standard Deviation, Correlation.
Significance of Difference Between Means (Correlated and Uncorrelated).

Note : The use of non-programmable calculators and statistical tables is allowed in the examination.

Paper B : BEHAVIOURAL DISORDERS

Theory : 70 marks
Internal Assessment : 10 marks
Time : 3 Hours

UNIT-I : Generalized Anxiety Disorders, Conversion Disorders, Dissociative Disorders : Types, Symptoms and Etiology.

UNIT-II : *Mood Disorders :* Types, Symptoms and Etiology.

UNIT-III : *Schizophrenia Types :* Symptoms and Etiology.

UNIT-IV : *Psychotherapies :* Psychodynamic Therapy, Basic of Behavioural Therapy, Cognitive-Behaviour Therapy, Humanistic Therapy.

PSYCHOLOGY PRACTICALS

Max. Marks : 40
Time : 3 Hours.

Eight practicals have to be performed out of the following :

1. Interview.
2. TAT.
3. Depression Inventory.
4. Locus of Control.
5. The use of Biofeedback.
6. Presumptive Stressful Life Event Scale.
7. Adjustment Inventory.
8. Self Concept.
9. Mental Health Inventory.
10. Rosenzweig's Picture Frustration Study.

11. Rorschach Inkblot Test.
12. Test of Anxiety.

Books Recommended :

Essential Readings :

1. Carson, R.C.; Butcher, J.N.; and Mineka, S. (2003) : *Abnormal Psychology and Modern Life,* New York : Pearson Education.
2. Davison, G.C. and Neale, J.M. (1998) : *Abnormal Psychology,* New York : John Wiley and Sons.
3. Garrett, H. E. (1966) : *Statistics in Psychology and Education,* New Delhi : Vakils, Feffer, and Simons.

Reference Books :

1. Feldman, R.S. (1996) : *Understanding Psychology,* New Delhi : Tata McGraw Hill.
2. Guilford, J.P., and Fruchter, B. (1981) : *Fundamental Statistics in Psychology and Education*. Singapore: McGraw Hill.
3. Sarason, B.R. and Sarason, I.G. : *Abnormal Psychology : The Problem of Maladaptive Behavior (11/e).* Prentice Hall of India.

GEOGRAPHY

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper–A : WORLD REGIONAL GEOGRAPHY- I

Max. Marks	:	75
Theory	:	65 marks
Internal Assessment	:	10 marks

Objectives :

To provide an understanding of the concept of world regions with respect to Land, People, Polity and Economy. The physical and human resource base and their interface with economic development, development problems and prospects.

Course Content :

Study of the following regions of the world in terms of constituent countries, strategic location, salient physical, demographic and economic features, cultural patterns, resource base, economic development, problems, prospects and issues related to Regional Groupings (European Union, North Atlantic Treaty Organization, North American Free Trade Agreement and Commonwealth of Independent States).

UNIT-I

(i) Anglo-America (15 Lectures)

UNIT-II

(ii) Latin America (12 Lectures)

UNIT-III

(iii) Europe (15 Lectures)

UNIT-IV

(iv) Russia and Commonwealth of Independent States (18 Lectures)

(v) Australia

Note : 1. Questions will be put on region(s) as a whole and not on individual country. The questions should focus on regional perspective.

2. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 10 parts in about 25-30 words each. Each part will carry 1½ marks (Total 15 marks).

3. The whole syllabus has been divided into 4 units. Eight questions will be set out of the whole syllabus i.e. 2 from each unit. The students will be required to attempt one question from each unit. Each question will carry 12 ½ marks. These will be in addition to the compulsory question at serial number 1.

4. Special credit will be given to suitable use of maps and diagrams. Use of unmarked stencils and colours are allowed.

5. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

6. For USOL/reappear/improvement candidates, who have not been assessed earlier for internal assessment, the marks secured by them in the theory paper will be proportionately increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note 6 in the question paper

Books Recommended :

Essential Readings :

1. Baerwald, T. J. and C. Fraser : *World Geography: A Global Perspective,* Prentice Hall, New Jersy, 1995.

2. Blij, Harm J. de Peter, O. Muller : *Geography : Regions and Concepts,* John Wiley, New York, 1993.

3. Cairns, G. O. et. al. : *Australia,* Macmillan Co., New York, 1962.

4. Coysh, A. W. and M. E. Tomlinson : *The Southern Continents,* University Tutorial Press, London, 1980.

5. English, Paul Ward & James, A. Miller : *World Regional Geography: A Question of Place,* John Wiley, New York, 1989.

6. Gilbert, Alan : *Latin American Development : A Geographical Perspective,* Harmondsworth Penguin Books Middlesex, London, 1974.

7. Gottmann, Jean : *A Geography of Europe,* Harrape & Co., London, 1969.

8. Gregory, J. S. ed. : *The Geography of USSR,* Novosti Press Agency, Moscow, 1985.

9. Hudson, F. S. : *North America,* Mc. Donald & Evans, Fly Mouth, 1975.

10. Jackson, Richard H. & Lloyd E. Hudman : *World Regional Geography: Issues for Today,* John Wiley, New York, 1991.

11.	Kromm, D. E.	:	*World Regional Geography,* Saunders Publishing, New York, 1980.
12.	Malmstorm, V. H.	:	*Geography of Europe: A Regional Analysis,* Englewood Cliffs, Prentice Hall, New York, 1991.
13.	Patterson, J. H.	:	*North America,* Oxford University Press, New York, 1995.
14.	Salter, C. L., J.J. Hobbs et. al.	:	*Essentials of World Regional Geography,* Saunders College Publishing and Harcourt Brace College Publishers, Orlando, 1998.

Further Readings :

1.	Don R. Hoy (Ed.)	:	*Essentials of Geography and Development,* Macmillan, New York, 1980.
2.	Hussain, Majid	:	*World Regional Geography,* Rawat Publications, Jaipur, 2009.
3.	Mankoo, Darshan Singh	:	*A Regional Geography of the World,* Kalyani Publishers, Ludhiana, 2009.
4.	Singh, Malkit	:	*World Regional Geography,* Rasmeet Prakashan, Jalandhar, 2009.
5.	Tikkha, R. N., Bali, P.K. and Sekhon, M. S.	:	*World Regional Geography,* New Academic Publishers, 2007.

Pedagogy :

The teacher should involve maximum use of detailed maps of the countries and continents. Students should be encouraged to use atlas in classrooms. Video shows about culture, physiography and economy of these countries may be arranged if possible.

Paper-B : WORLD REGIONAL GEOGRAPHY- II

Max. Marks	:	75
Theory	:	65
Internal Assessment	:	10

Objectives:

To provide an understanding of the concept of world regions with respect to Land, People, Polity and Economy. The physical and human resource base and their interface with economic development, development problems and prospects.

Course Content :

Study of the following regions of the world in terms of constituent countries, strategic location, salient physical, demographic and economic features, cultural patterns, resource base, economic development, problems, prospects and issues related to Regional Groupings (South Asian Association of Regional Co-operation, Association of South East Asian Nations, Organization of Petroleum Exporting Countries and Organization of African Unity).

UNIT-I

(i) East Asia (14 lectures)

UNIT-II

(ii) South East Asia (20 lectures)

(iii) South Asia.

UNIT-III

(iv) Middle East and North Africa (14 lectures)

UNIT-IV

(v) Africa South of Sahara (12 lectures)

Note :

1. Questions will be put on region(s) as a whole and not on individual country. The questions should focus on regional perspective.
2. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 10 parts in about 25-30 words each. Each part will carry 1½ marks (Total 15 marks).
3. The whole syllabus will be divided into 4 units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit. Each question will carry 12.5 marks. These will be in addition to the compulsory question at serial number 1.
4. Special credit will be given to suitable use of maps and diagrams. Use of unmarked stencils and colours are allowed.
5. Internal assessment may include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.
6. For USOL/reappear/improvement candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will be proportionately increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note 6 in the question paper.

Books Recommended :

Essential Readings:

1. Baerwald, T. J. and C. Fraser : *World Geography : A Global Perspective*, Prentice Hall, New Jersy, 1995.

2. Blij, Harm J. de Peter, O. Muller : *Geography : Regions and Concepts,* John Wiley, New York, 1993.

3. Cressy, G. B. : *Asia's Land and People,* McGraw Hill, London, 1963.

4. Dobby, E. H. G. : *South East Asia,* University of London Press, London, 1980.

5. English, Paul Ward & James, A. Miller : *World Regional Geography : A Question of Place,* John Wiley, New York, 1989.

6. Fisher, W. B. : *The Middle East,* Methuen, London, 1978.

7. Ginsburg, Norton Ed. : *The Pattern of Asia,* Englewood Cliffs, New Jersey, 1958.

8. Hance, William, A. : *The Geography of Modern Africa,* Columbia University Press, London, 1965.

9. Jackson, Richard H. & Lloyd E. Hudman : *World Regional Geography : Issues for Today,* John Wiley, New York, 1991.

10. Kromm, D. E. : *World Regional Geography,* Saunders Publishing, New York, 1980.

11. Pritchard, J. M. : *Africa,* Longman, London, 1981.

12. Salter, C. L., J.J. Hobbs et. al. : *Essentials of World Regional Geography,* Saunders College Publishing and Harcourt Brace College Publishers, Orlando, 1998.

13. Spencer, J. E. and L. T. William : *Asia, East by South : A Cultural Geography,* John Willey & Sons, New York, 1971.

14. Spate, O. H. K. and A. T. A. Learmonth : *India and Pakistan: A General and Regional Geography,* Methuen, London, 1967.

Further Readings :

1. Don R. Hoy (Ed.) : *Essentials of Geography and Development,* Macmillan, New York, 1980.
2. Hussain, Majid : *World Regional Geography,* Rawat Publications, Jaipur, 2009.
3. Mankoo, Darshan Singh : *A Regional Geography of the World,* Kalyani Publishers, Ludhiana, 2009.
4. Singh, Malkit : *World Regional Geography,* Rasmeet Prakashan, Jalandhar, 2009.
5. Tikkha, R. N., Bali, P.K. and Sekhon, M. S. : *World Regional Geography,* New Academic Publishers, 2007.

Pedagogy:

The teacher should involve maximum use of detailed maps of the countries and continents. Students should be encouraged to use atlases in classrooms. Video shows about culture, physiography and economy of these countries may be arranged if possible.

Paper-C: MAP PROJECTIONS AND FIELD REPORT

Max. Marks : 50
Time: 3 hours

Distribution of Marks :

(i)	Written paper of three hours covering all the units to be held for students of University School of Open Learning For others, it shall be at the respective colleges.	:	20 marks
(ii)	Practical Record relating to Map Projections	:	10 marks
(iii)	Viva voce on Practical Record relating to Map Projections	:	5 marks
(iv)	Field Report	:	10 marks
(v)	Viva Voce on Field report	:	5 marks

Objectives:

- To provide an analytical understanding of the use of common map projections.
- To acquaint the students with the importance of field work as one of the methodologies in geography.
- To sensitize the students about pre-field work and post-field work i.e. data processing and analysis and writing of field work report.

Course Content :

UNIT-I

Map Projections :

General introduction, classification of projections, general principles of identification and choice of projections.

(5 lectures)

UNIT-II

Construction, properties and limitations of the following projections :

Cylindrical : Simple, Equal area, Mercators

Conical : One Standard Parallel, Two Standard Parallels, Bonne's, Polyconic and International

(10 lectures, 10 lab. session)

UNIT-III

Zenithals : Gnomonic, Stereographic, Orthographic, Equidistant and Equal Area (Polar cases only).

Conventional : Sinusoidal and Molleweide's (normal case only).

Introduction to Elementary Remote Sensing

(10 lectures, 15 lab. session)

UNIT-IV

Fieldwork (Theory) : (i) Role of fieldwork in geography.

(ii) Scale of study and fieldwork methodology.

(iii) Methods of collecting primary data, questionnaire, observation and measurement.

(6 lectures)

Fieldwork (Practical) : A field report of minimum 10 pages will be prepared based on primary data on problems such as (a) local market survey, (b) service area of school/or hospital; (c) traffic flow, and (d) socio-economic characteristics of student's village/ mohalla/ sector.

(15 lab. sessions)

Note :

1. The written and practical examination including viva-voce shall be conducted at the respective college itself except for USOL students. However, the format of the question paper shall be uniform. A separate paper for 20 marks shall be prepared for colleges by the University from the prescribed syllabus.

2. Practical exam. at the respective colleges shall be conducted by one internal and one external examiner. The external examiner shall be appointed by the Principal of the respective colleges in consultation with the senior most teacher of geography in the college.

3. For students of USOL, a written theory paper for 20 marks shall be conducted by the University along with the University examinations.

4. A compulsory question containing 12 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 8 parts. The answer of each part should not exceed 25 words. Each part will carry 1/2 mark (Total 4 marks).

5. The whole syllabus has been divided into 4 units. Eight questions will be set out of the whole syllabus i.e. 2 from each unit. The students will be required to attempt one question from each unit. These will be in addition to the compulsory question at serial number 1.

6. Evaluation of Practical Record will be done at the time of viva-voce examination. A minimum of 20 sheets are to be prepared by the students. There will be no laboratory exercise at that time.

7. There will be no viva-voce examination for the candidates appearing through USOL. They will be required to submit their Practical Note Book (Practical files) with the University School of Open Learning (Department of Geography) at least 10 days before the commencement of their examination. Their Note Books (Practical files) will be evaluated by two examiners (including at least one from USOL).

8. For the students of USOL, there will be an internal assessment of 10 marks in lieu of the viva-voce examination in practical record and field report. The marks obtained by the candidate will be added to the marks awarded by the internal and external examiners evaluating the practical record and field report.

9. A fresh practical note book/field report shall be prepared by failed/improvement candidates.

10. For practical classes, the number of students in one group shall not exceed fifteen.

11. There will be 3 hours of teaching per week for this paper.

12. For USOL/reappear/improvement candidate(s) who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.

The paper setter must put note 12 in the question paper.

Books Recommended :

Essential Readings

1. Jones, P.A. : *Fieldwork in Geography*, Longman, London, 1968.
2. Kellaway, George P. : *Map Projections*, Methuen and Co., London.
3. Singh, Gopal : *Mapwork and Practical Geography*, Surjeet Book Depot, Delhi, 1993.
4. Steers, J.A. : *Map Projections*, University of London Press, London.

Further Readings

1. Archer, J.E. & Dalton, T.H. : *Fieldwork in Geography,* E.T. Bastford Ltd., London, 1968.
2. Hudson, F.S. : *A Geography of Settlements*, MacDonald, London, 1970.
3. Singh, L.R. & Singh, Raghunandan : *Mapwork and Practical Geography*, Central Book Depot, Allahabad, 1993, Reprint.

Pedagogy :

Basic fundamentals of map projections are introduced by demonstration of construction exercises in class. The students need to be trained to collect primary data, its processing and cartographic representation through taking up field exercises.

GANDHIAN STUDIES

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

PAPER-A : ECONOMIC THOUGHT OF MAHATMA GANDHI

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objectives :

The paper is designed to acquaint the students with the Economic Thought of Mahatma Gandhi.

Pedagogy of the Course Work :

90% Lectures (including expert lectures)

10% Unit Tests, Snap Tests, assignments, attendance and class room participation.

Note :

1. The syllabus has been divided into four (4) units.
2. There shall be 9 questions in all.
3. The first question is **compulsory** and shall be short answer type containing 15 short answer type questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question.
4. Rest of the paper shall contain four (4) units and each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each unit - 4 in all. All questions shall carry 18 marks.
5. For the private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper setter must put note (5) in the question paper

UNIT-I

1. Influences on Gandhi's Economic Thought.
2. Fundamental Principles of Gandhi's Economic Thought

UNIT-II

3. Critique of Machinery
4. Labour Capital Relations
5. Theory of Trusteeship

UNIT-III

6. Concept of Sarvodaya & Bread Labour
7. Concept of Wantlessness
8. Doctrine of Swadeshi

UNIT-IV

9. Gandhi on Communism
10. Gandhi on Socialism
11. Gandhi on Capitalism
12. Contemporary Relevance

I Essential Readings:

1. Diwan, Romesh and Lutz, Mark : *Essays in Gandhian Economics,* New Delhi: Gandhi Peace Foundation, 1985.
2. Gupta, S.S. : *Economic Thought of Mahatma Gandhi,* New Delhi, Concept, 1994.
3. Kumarappa, J.C. : *Gandhian Economic Thought,* Varanasi, Sarva Seva Sangh, 1962.
4. Maharajan, M. : *Economic Philosophy of Mahatma Gandhi,* New Delhi: APH, 1996.
5. Mashruwala, K.G. : *Towards Sarvodaya Order,* Ahmedabad: Navajivan Publishing House, 1971.
6. Mishra, A.D. : *Reading Gandhi;* New Delhi : Dorling Kindersley, Pearson, 2012

II Further Readings:

1. Gandhi, M.K. : *Hind Swaraj,* Ahmedabad: Navajivan Publishing House, 1985.
2. Schumacher, E.F. : *Small is Beautiful,* London, Abacus, 1989.

PAPER-B : GANDHI AND HIS CONTEMPORARIES

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objectives:

The paper is designed to acquaint the students with the interaction Gandhi had with his contemporaries.

Pedagogy of the Course Work :

90% Lectures (including expert lectures)

10% Unit Tests, Snap Tests, assignments, attendance and class room participation

Note :

1. The syllabus has been divided into four (4) units.
2. There shall be 9 questions in all.
3. The first question is **compulsory** and shall be short answer type containing 15 short answer type questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question.
4. Rest of the paper shall contain four (4) units and each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each unit – 4 in all. All questions shall carry 18 marks.
5. For the private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper setter must put note (5) in the question paper

UNIT-I

1. Gandhi & G .K. Gokhale
2. Gandhi & Vinoba Bhave

UNIT-II

3. Gandhi & Subhash Chander Bose
4. Gandhi & Bhagat Singh

UNIT-III

5. Gandhi & Ambedkar
6. Gandhi & Lala Lajpat Rai

UNIT-IV

7. Gandhi & M.A. Jinnah
8. Gandhi & Jawahar Lal Nehru

I Essential Readings:

1. Bharill, Chandra : *Social & Political Ideas of B.R. Ambedkar : A Study of His Life, Services, Social and Political Ideas,* Jaipur, Aalekh, 1977.
2. Bhave, Vinoba : *Democratic Values,* Kashi: Sarva Seva Sangh Prakashan, 1966.
3. Merriam, Allen Hayes : *Gandhi Vs Jinnah : The Debate over the Partition of India,* Calcutta: Minerva, 1980.
4. Nagar, Purshottam : *Lala Lajpat Rai : The Man and His Ideas,* New Delhi : Manohar Book Service, 1977.
5. Nanda, B.R. : *Three Statesmen: Gokhale, Gandhi & Nehru,* New Delhi, Oxford, 2004.
6. Ram, Suresh : *Vinoba and His Mission,* Kashi: Akhil Bharat Sarva Seva Sangh, March, 1958.
7. Suda, J.P. : *Main Currents of Social & Political Thought in Modern India,* Vols. I, II, III, Meerut: K. Nath & Co., 1973.
8. Varma, V.P. : *Modern Indian Political Thought,* Agra: Lakshmi Narain Aggarwal, 1994.

II Further Readings:

1. Goyal, O.P. : *Moderate and Extremists,* Allahabad: Kitab Mahal, 1977.
2. Kulkarni, V.B. : *The Indian Trium Virate: A Political Biography of Mahatma Gandhi, Sardar Patel and Pandit Nehru,* Bombay: Bhartiya Vidya Bhawan, 1969.

JOURNALISM & MASS COMMUNICATION

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Objectives :

(I) The course will introduce the students to the basic structure of media organizations and the regulatory framework of the journalistic profession. It will also make them conversant with strategic communication by imparting training in writing advertising copy and press releases amongst other tools.

(II) Pedagogy of the Course Work:

80% Lectures (including expert lectures)

20% assignments, discussion and seminars.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES:

There shall be **9** questions in all. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 7 short answer type questions i.e. of 2 marks each. It shall carry 14 marks and shall be a **Compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all. Each question will carry 14 marks.

Paper-A : MEDIA MANAGEMENT, ETHICS AND LAWS

Max. Marks	:	80 marks
Theory	:	70 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Unit I : Types of ownership patterns; structure of newspapers, radio and television organizations.

Unit II : Basic principles of Management given by Henry Fayol.

Unit III : The Indian Constitution and Freedom of Speech and Expression; Self regulation and code of ethics; common ethical violations; role, responsibility and accountability of a journalist.

Unit IV : Overview of Media Laws: Right to Information Act, Copyright Act, Contempt of Court, Contempt of Legislature, Law of Defamation, Law of Obscenity.

Paper-B : ADVERTISING & PUBLIC RELATIONS

Max. Marks	:	80 marks
Theory	:	70 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Unit I : Concepts & types of Public in PR; Public Relations in Public & Private Sector; PR Agency; Structure & Functions.

Unit II : Brief introduction to PR tools and media relations.

Unit III : Concept & types of advertising; Advertising in marketing mix; advertising agency, structure and functions.

Unit IV : Basic principles of writing advertising copy; concept of USP, AIDA formula.

PRACTICALS

Max. Marks : 40 marks

1.	To prepare a file of stories illustrating ethical code violations	5 marks
2.	Project on the management structure of any one media organisation	5 marks
3.	Compilation of five press releases pertaining to campus events	10 marks
4.	Portfolio of five self-designed advertisements	10 marks
5.	Performance report of internship in an advertising or PR agency	10 marks

Essential Readings :

1. Harrison, Toni : *A Handbook of Advertising Techniques,* Beckman Books, 1989.
2. Jefkins Frank : *Copywriting and its Presentation,* International Textbook Co., London, 1977.
3. Kumar, Niraj : *Communication and Management,* Gyan Publishing House, 1997.
4. Venkateswaran, K.S. : *Mass Media Laws and Regulations in India, Asian Mass Communication Research and Information Center,* Singapore, 1993.
5. Padhy, K.S. : *Battle for Freedom of Press in India,* Academic Foundation, N.D., 1991.

6. Basu, Durga Das : *Law of the Press,* Prentice Hall, India, N.D., 1996.

7. Williams, Herbert Lee : *Newspaper Organization and Management,* Surjeet Publications, 1978.

8. Lafrance Pierre : *Fundamental Concepts in Communication,* Prentice Hall, New Delhi, 1992.

9. Ravindran, R.K. : *Handbook of Mass Communication,* Anmol Publications, N.D., 1999.

10. Ravindran, R.K. : *Handbook of Press Laws and Ethics,* Anmol Publications, N.D., 1999.

11. Trikha, N.K : *The Press Council – A Self Regulatory Mechanism for the Press,* Somaiya Publications, N.D., 1986.

12. Clampitt, Phillip. G. : *Communicating for Managerial Effectiveness,* Sage Publications, N.D., 1991.

13. Thayer, Frank : *Newspaper Business Management,* Prentice Hall, N.Y., 1990.

14. Sindhwani, Trilok, N. : *Newspaper Economics and Management,* Ankur Publishing House, N.D., 1979.

15. Olen, Jeffrey : *Ethics in Journalism,* Prentice Hall, New Jersey, 1988.

16. Christians, Clifford and Traber, Michael : *Communication Ethics and Universal Values,* Sage Publications, N.D., 1997.

17. Sharma, B.R. : *Freedom of Press – Under the Indian Constitution,* Deep and Deep Publications, N.D., 1993.

18. Darbe, Arnold & Watson, S. : *Advertising: Its Role in Modern Marketing,* Mc Graw Hill, New York, 1964.

19. Scott, Cutlip & Genter, A.H. : *Effective Public Relations,* Prentice Hall, 1964.

20. Mehta, D.S. : *Handbook of Public Relations in India,* Allied Publishers, 1972.

21. Black, Sam : *Role of Public Relations in Management,* Pitman, London, 1972.

22. Chiranjeev, Avinash : *Electronic Media Management,* Authors Press, New Delhi, 2000.

23. Panigrahy, Dibakar : *Media Management in India,* Kanishka Publishing House, New Delhi, 1993.

24. Jackquette, D. : *Journalistic Ethics: Moral Responsibility in the Media,* Pearson Education, 2007.

25. Batra, R. : *Advertising Management,* Pearson Education, 1996.

26. Wells, William : *Advertising Principles and Practice,* Pearson Education, 2007.

27. Jefkins, F. : *Advertising,* Pearson Education, 2000.

28. Cutlip, S. : *Effective Public Relations,* Pearson Education, 2000.

Reference Books :

1. Bhatia, Sita : *Freedom of Press – Politico-Legal Aspects of Press Legislations in India,* Rawat Publications, Jaipur, 1997.

2. D'Souza, Y.K. : *Freedom of Press: Constitution and Media Responsibility,* Commonwealth Publishers, N.D., 1998.

3. Chauhan, Swati and Chandra, Navin : *Journalism Today: Principles, Practices and Challenges,* Kanishka Publishers, N.D., 1997.

4. Sharma, S.R. : *Democracy and the Press,* Radha Publications, N.D., 1996.

5. Joseph, M.K. : *Freedom of the Press,* Anmol Publications, N.D., 1997.

6. Padhy, K.S. : *Indian Press: Role and Responsibility,* Ashish Publishing House, N.D., 1984.

7. Padhy, K.S. : *The Muzzled Press,* Kanishka Publishers, N.D., 1984.

8. Aggarwal, S.K. : *Media and Ethics,* Sipra Publications, N.D., 1993.

9. Grover, A.N. : *Press and the Law,* Vikas Publishing House, N.D., 1990.

10. Haywood, Roger : *All About PR,* Mc Graw Hill, 1987.

11. Pavlik, John : *Public Relations Handbook,* Sage, London, 1987.

12. Ogilvy, David : *Ogilvy on Advertising,* Pan Books, London, 1988.

13. Cohen, Dorothy : *Advertising,* Wiley, New York, 1972.

14. Padhye, Prabhakar : *Principles of Journalism,* Popular Prakashan, Mumbai, 1991.

15. Karkhanis, Sharad : *Indian Politics and Role of the Press,* Vikas Publishers, N.D., 1981.

16. Mudgal, Rahul : *Contemporary Issues in Journalism,* Sarup and Sons, New Delhi, 1998.

17. Kamath, M.V. : *Journalists Handbook,* Vikas Publishing House, New Delhi, 1983.

18. Clampitt, Phillip. G. : *Communicating for Managerial Effectiveness,* Sage Publications, N.D., 1991.

POLICE ADMINISTRATION

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Outline of Tests, Syllabi and Courses of Reading

Paper-A : Orgnisation Behaviour with Special Reference to Police Administration

(A) Course Objectives:

The objective of this course is to familiarize the students with the concept, nature and significance of organizational behavior with special reference to police administration. The course also aims to discuss the foundations and models of organisational behavior. In particular, the students would be taught the concepts such as motivation, morale, leadership, communication, decision-making, and transactional analysis. Further, the inputs regarding the concept and rationale of organizational change and organizational development would be imparted to the students.

(B) Pedagogy of the Course Work:

90 per cent of the Course Content would be delivered through Lecture Method and rest 10 per cent would comprise of two internal examinations and attendance.

(C) Instructions for Paper Setters and Candidates:

- The maximum marks for the paper will be 100. The question paper will be of 90 marks and internal assessment of 10 marks.
- Time allowed will be 3 hours.
- There shall be 9 questions in all.
- The first question shall be compulsory and be short answer type containing 12 short questions spread over the whole syllabus and to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions carrying 2 marks each (9×2 = 18 marks).
- Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall have two questions and the candidates shall attempt one question from each unit i.e. four questions in all. Each question will carry 18 marks (4×18 = 72 marks).

(D) Course Content :

Unit-I

Organisational Behaviour: Concept, Nature and Significance. Foundations of Organisational Behaviour – Scientific Management, Human Relations, Systems and Contingency. Models of Organisational Behaviour.

Unit-II

Motivation: Concept and Theories – Maslow's Need Hierarchy and McGregor's Theory X & Y. Morale: Concept and Factors to Build-up Morale in Indian Police. Leadership: Concept; Theories – Trait and Situational; and Qualities.

Unit-III

Communication: Concept, Process, Types and Barriers. Communication Modes in Indian Police. Decision-Making: Concept; Types; and Simon's Rational Comprehensive Theory. Transactional Analysis: Meaning; and Types of Transactions.

Unit-IV

Organisational Change: Concept; Rationale; and Resistance to Change. Organisational Development: Concept and Rationale.

Essential Readings :

1. Prasad, L.M. : *Organisation Behaviour,* Sultan Chand & Sons, New Delhi, 2008.

2. Aswathappa, K. : *Organisational Behaviour,* 7th Edition, Himalaya Publishing House, New Delhi, 2007.

3. Moorhead, Gregory & Ricky W. Griffin : *Organizational Behaviour: Managing People & Organizations,* Biztantra, New Delhi, 2005.

4. Robbins, Stephen P. : *Organizational Behaviour,* Prentice-Hall of India Private Limited, New Delhi, 2001.

5. Sharma, K.K. & Sahni, Pradeep : *Organisation Behaviour,* Deep & Deep Publishers, New Delhi, 1988.

6. Dubin, Robert : *Human Relations in Administration,* Prentice-Hall of India Private Limited, New Delhi, 1968.

7. George, C. S. : *History of Management Thought,* Prentice-Hall of India Private Limited, New Delhi.

8. Simon, Herbert A. : *Administrative Behaviour: A Study of Decision Making Process in Administrative Organisation,* Macmillan, New York, 1976.

9. Sapru, R. K. : *Theories of Administration,* S. Chand, New Delhi, 1996.

10. Maheshwari, S. R. : *Administrative Thinkers,* MacMillan, New Delhi, Latest Edition.

11. Saiyadain, Mirza S. : *Organisational Behaviour,* Tata McGraw-Hill Publishing Company Ltd., New Delhi, 2006.

12. McShane, Steven L. Glinow, Mary Ann Von & Sharma, Radha R. : *Organisational Behaviour,* Tata McGraw-Hill, Publishing Company Ltd., New Delhi, 2006.

Further Readings :

1. Argyris, Chris : *Personality and Organisation: The Conflict Between System and the Individual,* Harper and Row, New York, 1957.

2. Argyris, Chris : *Integrating Individual and Organisation,* Wiley, New York, 1964.

3. Mayo, Elton : *The Social Problems of an Industrial Civilisation,* Routledge and Kegan Paul Limited, London, 1957.

4. Hicks, Herbert G. : *The Management of Organisation : A System and Human Resource Approach,* McGraw Hill, New York,1972.

5. Vadackumchery, James : *Police Leadership: The Inside Story,* APH Publishing Corporation, New Delhi, 1999.

6. Pugh, D. S. (ed.) : *Organisation Theory,* Penguin Books, Harmondsworth, 2nd Edition, 1984.

Paper-B : LAW AND POLICE ADMINISTRATION

(A) Course Objectives :

The police system in India has to work within the ambit of legal framework laid down by the Constitution and by the enacted laws. The major responsibility of the police is to ensure the implementation of such laws. The course has been designed to impart knowledge to the students on the laws governing the 'prevention and detection of crime' which is laid down as the primary duty in the Indian Police Act 1861. The endeavour of the course is to familiarize the students with the main provisions of the Indian Penal Code 1860, the offences under it and the offences affecting the human body. In addition, meaning and definition of terms covered under Section 2 relevant to the police administration along with the powers of the police officer have been discussed.

(B) Pedagogy of the Course Work :

90 per cent of the Course Content would be delivered through Lecture Method and rest 10 per cent would comprise of two internal examinations and attendance.

(C) Instructions for Paper Setters and Candidates :

- The maximum marks for the paper will be 100. The question paper will be of 90 marks and internal assessment of 10 marks.
- Time allowed will be 3 hours.
- There shall be 9 questions in all.
- The first question shall be compulsory and be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 9 short answer type questions carrying 2 marks (9×2 = 18 marks).
- Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall have two questions and the candidates shall attempt one question from each unit i.e. four questions in all. Each question will carry 18 marks (4×18 = 72 marks).

(D) Course Content:

Unit-I

The Indian Penal Code 1860 : Main Provisions

(i) Jurisdiction (Sections 1-5)

(ii) General Explanations (Sections 6-52-A)

(iii) Punishments (Sections 53, 53A, 54, 55, 60, 63, 73)

(iv) Joint & Constructive Liability Section 34 & Section 149.

Unit-II

Offences under the Indian Penal Code 1860

(i) Abetment (Section 107 & 108)

(ii) Criminal Conspiracy (Sections 120-A, 120-B)

(iii) Offences against the State (Sections 121, 121-A, 124-A)

(iv) Offences Against Public Tranquility (Sections 141-147)

Unit-III

Offences Affecting Human Body

- Culpable Homicide (Section 299) & Murder (Section 300)
- Causing death by rash and negligent act (Section 304A)
- Dowry Death (Section 304B)
- Hurt (Section 319)
- Grievous Hurt (Section 320)
- Wrongful Restraint and Wrongful Confinement (Sections 339, 340)
- Assault (Section 351)
- Kidnapping and Abduction (Sections 359-362)
- Rape (Sections 375, 376)
- Murder (Sections 354A, 354B, 354C & 354D)

Unit-IV

- Bailable and non-bailable offence
- Cognizable and non-cognizable offence
- Summons Case
- Warrant Case
- Public Prosecutor

Powers of Police Officer :

- Lodging of First Information Report (FIR) (Section 154) & Effects of Refusal to lodge FIR
- Power of Police to investigate
- Arrest (Section 41) & Rights of Arrested Persons
- Search (Section 165)

Essential Readings :

1. Tandon, Mahesh Prasad & Tandon, Rajesh : *The Indian Penal Code,* Allahabad Law Agency, Latest Edition.
2. Gandhi, B. M. : *Indian Penal Code,* Eastern Book Company, 2nd Edition.
3. Bhattacharya, T. : *Indian Penal Code,* Central Law Agency, Allahabad, 2007.
4. Mishra, S.N. : *Indian Penal Code,* Central Law Publications, Allahabad, 2007.
5. The Indian Penal Code : *Bare Act with Short Notes,* Universal Law Publishing Co. Pvt. Ltd., Delhi, 2008.

Further Readings :

1. Lal, Rattan and Dhiraj Lal : *Indian Penal Code.*

2. Vadackumchery, James : *Indian Police and Miscarriage of Justice,* A.P.H. Publishing Corporation, New Delhi, 1997.

3. Dutta, K.K. : *Some Aspects of Criminal Law,* A.P.H. Publishing Corporation, New Delhi, 1998.

WOMEN'S STUDIES

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : NATIONAL AND INTERNATIONAL INITIATIVES FOR WOMEN

Max. Marks	:	100 marks
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objective :

This course aims to conscientise the students about the efforts made by national and international organizations, both governmental and non-governmental, towards women's empowerment and increased participation in social, economic and political life. The course exposes the student particularly to the U.N. initiatives in empowering women, putting an end to age old discrimination against them.

INSTRUCTIONS FOR THE PAPER SETTER AND THE CANDIDATES:

- In this paper, the candidate will be assessed for 90 marks on the basis of a written examination and for 10 marks internal assessment.
- There shall be 9 questions in all. The first question shall be compulsory containing 12 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidate is required to attempt any 09 short answer type questions carrying 2 marks each (18 marks). Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidate shall attempt one question from each unit - 4 in all. Each question will carry 18 marks.

Course Contents :

Unit-I - Constitution of India and Women's Rights :

(a) Fundamental rights and women's rights: de jure and de facto position.

(b) Directive Principles of State Policy and women's rights with special reference to Articles 39, 42, 43 and 44.

(c) Factors inhibiting effective implementation of Constitutional provisions.

Unit-II - Women and Law in India :

(a) Dowry Prohibition Act, 1961 (as amended in 1984 & 1986).

(b) Protection of Women from Domestic Violence Act, 2005.

(c) Women's Rights in Property as Coparcener (as per 2005 Amendment of Hindu Succession Act, 1956).

Unit-III - UN Conferences on Women :

(a) Mexico (1975)

(b) Copenhagen (1980)

(c) Nairobi (1985)

(d) Beijing (1995)

(e) Beijing +5 (2000)

(f) Beijing + 10 (2005)

Unit-IV - UN Conventions for Women and Children :

(a) Convention on the Political Rights of Women, 1952.

(b) Convention on the Elimination of all Forms of Discrimination against Women (CEDAW), 1979.

(c) Convention on the Rights of the Child (CRC), 1990.

Essential Readings :

1. Aggarwal, Nomita : *Women and Law in India,* New Century Publication, Delhi, 2002.

2. Basu, D.D. : *Introduction to the Constitution of India,* Wadhwa & Co., Agra, 2001.

3. Jain, Devaki : *Women Development and the UN : A Sixty Year Quest for Equality and Justice,* Orient Longman, Delhi, 2006.

4. Shukla, Surinder : *United Nations and Empowerment of Women,* Arun Pub. House, Chandigarh, 2009.

5. *United Nations and the Advancement of Women (1945-1996),* United Nations Publication, 1996.

Further Readings :

1. Diwan, Paras : *Dowry and Protection to Married Women,* Deep and Deep Publication, New Delhi, 1987.

2. Ghosh, S. K. : *Women in Changing Society,* Ashish Publishing House, New Delhi, 1984.

3. Ratra, Amiteshwar; Parveen Kaur & Chhikara, Sudha : *Marriage and Family : In Diverse and Changing Scenario,* Deep & Deep, Publication New Delhi, 2006.

4. Singh, Indu Prakash : *Women, Law and Social Change in India,* Radiant, New Delhi, 1989.

Readings in Hindi :

1. Anthony, M. J. : *Kanooni Salaha,* Hind Pocket Books, New Delhi, 2006.
2. Bansal, Bandna : *Panchayati Raj Mein Mahila Bhagyadari,* Kalpaz Pub., New Delhi, 2004.
3. Gupta, Kamlesh Kumar : *Bharatiya Mahilayon ka Shoshan, Utpidan avam Adhikar,* Book Enclave, Jaipur, 2005.
4. Natani, Prakash Narayan : *Mahila Jagriti aur Kanoon,* Aavishkar Pub., Jaipur, 2002.
5. Natani, Prakash Narayan : *Manavadhikar avam Mahilayen,* Sublime Pub., Delhi, 2005.
6. Sanyukt Rashtr, *Karavahi Hatu Ghushana-Patra Tatha Beijing Ghoshana,* Campewell Publication, New Delhi, 1995.
7. Mathew, D. P. : *Hindu Vivaha Aur Talaq,* Indian Social Institute, New Delhi, 1986.
8. Mathew, D. P. : *Atam Raksha,* New Delhi, 1984.

Paper-B : WOMEN'S DEVELOPMENT AND EMPOWERMENT IN INDIA

Max. Marks	:	100 marks
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objectives :

This paper focuses on the issues related to the process of development and its impact on women, particularly in the context of a developing nation like India. The attempt is to equip the student to understand the major aspects of critique of development through a gender lens in its global and local contexts.

INSTRUCTIONS FOR THE PAPER SETTER AND THE CANDIDATES:

- In this paper, the candidate will be assessed for 90 marks on the basis of a written examination and for 10 marks internal assessment.
- There shall be 9 questions in all. The first question shall be compulsory containing 12 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidate is required to attempt any 09 short answer type questions carrying 2 marks each (18 marks). Rest of the paper shall contain 4 units. Each Unit shall have two questions and the candidate shall attempt one question from each unit - 4 in all. Each question will carry 18 marks.

Course Contents

Unit-I - Concept of Development:

- Changing concept from Growth to Development;
- Human Centric Development;
- Why is gender a development issue?
- Indices of development: Human Development Index (HDI), Gender Development Index (GDI), Gender Empowerment Measure (GEM).

Unit-II - Strategies for Women's Development:

- Women in Development (WID); Women and Development (WAD); and Gender and Development (GAD);
- Empowerment and Sustainable Development;
- Microcredit.

Unit-III - Policy Initiatives:

- Changing approaches from welfare to development to empowerment with special reference to Five Year Plans (First to Eleventh Plan).
- National Policy for Empowerment of Women.

Unit-IV - Implementation Machinery:

- Social Welfare Boards,
- National Commission for Women (NCW),
- Ministry of Women and Child Development (MOWCD),
- National Institute of Public Cooperation and Child Development (NIPCCD).

Essential Readings :

1. Ahooja-Patel, Krishna : *Women and Development,* Ashish Publishing House, New Delhi, 1995.
2. *Govt. of India, Five Year Plans (1st to 11th),* Planning Commission, New Delhi.
3. Gupte, Shakuntla : *Women Development in India,* Anmol Pub., New Delhi, 2005.
4. Mathu, Anuradha (ed.) : *Facets of Women's Development,* Kalpaz Pub., Delhi, 2006.
5. *South Asia Human Development Reports 2000 and 2004,* Mahbub ul Haq Human Development Centre.
6. *UNDP Human Development Reports,* 1995-2007.

Further Readings :

1. Heyzer, Noeleen and Sen, Gita (ed.) : *Gender, Economic Growth and Poverty,* Kali for Women, New Delhi, 1994.
2. Jain, Devaki : *Development as if Women Mattered,* Monograph, ISS, New Delhi, 1983.
3. Krishanaraj, Maitreyi : *Women and Development: The Indian Experience,* Subhada Publishers.
4. Rajput, Pam & Swarup, Hemlata (ed.) : *Women and Globalization,* Ashish Pub. House, New Delhi, 1994.
5. Reddy, V. Narayana et al (eds.), : *Women in Development: Challenges and Achievements,* Serials Pub., Delhi, 2005.
6. Singh, Gopal, et al (eds.): *Economic Empowerment of Rural Women in India,* RBSA Pub., Jaipur, 2003.
7. Verma, S.B, Jiloka, S.K., Kushwah, K.J. (eds.) : *Rural Women Empowerment,* Cambridge University Press, U.K., 2005.

Readings in Hindi :

1. Kumar, Raj (ed.) : *Bhartiya Mahila,* Arjun Publishing, New Delhi, 2003.
2. Sarawat, Sawapnal : *Mahila Vikas Ak Paridrashay,* Nayanprakashan, New Delhi, 2005.

HUMAN RIGHTS AND DUTIES

B.A. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A : INTERNATIONAL HUMAN RIGHTS RESPONSES TO SELECT PROBLEMS

Max. Marks	:	100 marks
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objective :

The paper deals with a few selected problems that have been addressed internationally through the UN Conventions and by means of regional human rights standards and mechanisms for the protection of human rights.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

Note : (i) For written paper, the students will be required to attempt five questions in all. Question No. 1 will be compulsory comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students shall be required to attempt 9 short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

(ii) In addition to it, Question Nos. II to IX will consist of long answer (essay) type questions i.e. two questions from each unit with internal choice carrying 18 marks of each i.e. $4 \times 18 = 72$ marks.

UNIT-I

Racial Discrimination :

(a) International Convention on Elimination of All Forms of Racial Discrimination, 1965.

(b) UNESCO Declaration on Race and Racial Prejudice, 1978.

UNIT-II

Freedom from Torture :

(a) Convention on the Prevention and Punishment of the Crime of Genocide, 1948.

(b) Convention against Torture and Other Cruel and Inhuman or Degrading Treatment or Punishment, 1984.

UNIT-III

Rights of Women and Children :

(a) Convention on the Elimination of All Forms of Discrimination against Women, 1979.

(b) Convention on the Rights of the Child, 1989.

UNIT-IV

Regional Human Rights Standards and Mechanisms :

(a) European Convention on the Protection of Human Rights, 1950.

(b) Latin American Standards and Mechanisms for Protection of Human Rights.

(c) African Standards and Mechanisms for Protection of Human Rights.

(d) Developments in Asia.

REFERENCES :

Suggested Readings :

1. Aggarwal, H.O. : *International Law and Human Rights,* Central Law Publications, New Delhi, 2010.

2. Aggarwal, H.O. : *Human Rights,* Central Law Publications, Allahabad, 2010.

3. Kapoor, S.K. : *Human Rights Under International Law,* Central Law Publications, Allahabad, 2009.

4. Jacobs, Francis G. and R.C.A. White : *The European Convention on Human Rights,* Oxford : Clarendon University Press, 1996.

5. Lauterpacht, Hersch : *An International Bill of the Rights of Man,* New York : Columbia University Press, 1945.

6. South Asian Human Rights Documentation Centre : *Human Rights and Humanitarian Law,* South Asian Human Rights Documentation Centre, New Delhi, 2008.

Further Readings :

1. Langley, Winstone E. : *Encyclopedia of Human Rights Issues Since 1945,* London : Fitzroy Dearborn Publishers, 1999.
2. Rao, Mamta : *Law Relating to Women and Children,* Eastern Book Company, Lucknow, 2008.
3. Verma, R.S. : *Human Rights : Burning Issues of the World,* Volumes I, II and III. Delhi : Radiant Publishers, 2000.

Paper-B : SPECIFIC HUMAN RIGHTS PROBLEMS AND RESPONSES : INDIA

Max. Marks	:	100 marks
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objective :

The paper deals with a few specific problems that are peculiar to India along with some key responses of the Indian State in the form of legislative enactments.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

Note : (i) For written paper, the students will be required to attempt five questions in all. Question No. 1 will be compulsory comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students shall be required to attempt 9 short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

(ii) In addition to it, Question Nos. II to IX will consist of long answer (essay) type questions i.e. two questions from each unit with internal choice carrying 18 marks of each i.e. $4 \times 18 = 72$ marks.

UNIT-I

Structural Problems :

(a) Communal and Caste Conflicts

(b) Discrimination and Violence against Women

(c) Discrimination and Violence against Children

(d) Problems of Disabled and Aged Populace

UNIT-II

Systemic Problems :

(a) Slum Dwellers

(b) Refugees

(c) Internally Displaced Persons

(d) Human Trafficking

UNIT-III

Institutional Problems :

(a) Problems of the Accused

(b) Denial of Rights to Victims

(c) Custodial Violence

(d) Corruption

UNIT-IV

Responses to Contemporary Problems :

(a) Right to Food

(b) Right to Clean Environment (As a part of right to Life)

(c) Right to Education

(d) Right to Information

REFERENCES :

Suggested Readings :

1. Ahuja, Ram : *Social Problems in India,* Rawat. Jaipur, 2009.

2. Alam, Aftab (ed.) : *Human Rights in India : Issues and Challenges,* New Delhi : Raj Publications, 1999.

3. Begum, S.M. (ed.) : *Human Rights in India : Issues and Perspectives,* New Delhi : APH Publishing Co., 2000.

4. Biswal, Tapan : *Human Rights, Gender and Environment,* New Delhi : Viva, 2006.

5. Centre for Development and Human Rights : *The Right to Development - A Primer,* New Delhi : Sage Pub., 2004.

6. Paul, R.C. : *Situation of Human Rights in India,* New Delhi : Commonwealth Publishers, 2000.

7. Rao, Ashiwini : *Status of Human Rights in India,* New Delhi, Pacfic, 2010.

8. Sehgal, B.P. Singh (ed.) : *Human Rights in India : Problems and Perspectives,* New Delhi : Deep and Deep, 1999.

Further Readings :

1. Khanna, S.K. : *Children and the Human Rights,* New Delhi : Commonwealth, 1998.

2. Kumar, Bindal : *Problems of Working Children,* New Delhi : APH Publications, 2000.

3. Nirmal, C.J. : *Human Rights in India,* New Delhi : Oxford, 1999.

4. Pachauri, S.K. : *Children and Human Rights,* New Delhi : APH Publications, 1999.

5. Pachauri, S.K. : *Women and Human Rights,* New Delhi : APH Publications, 1999.

6. Karna, G.N. : *Disability Studies in India : Retrospect and Prospect,* New Delhi : Gyan Publishing House, 2001.

7. Karna, G.N. : *United Nations and the Rights of Disabled Persons : A Study in Indian Perspective,* New Delhi : APH Publishing Co., 1999.

COMPUTER SCIENCE

B.A./B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Scheme of Examination

					Exam. Hrs.	Ext.	*Int.*	*Max.Marks*
Paper A	:	Theory	:	Business Data Processing and Database Management Software	3	65	10	75
Paper B	:	Theory	:	Web Programming	3	65	10	75
Paper C	:	Practical	:	Practical based on Paper A and Paper B	3	50	-	50

Note : Practical marks will include the appropriate weightage for proper maintenance of lab. record.

SYLLABUS AND COURSES OF READING

Paper-A : BUSINESS DATA PROCESSING AND DATA BASE MANAGEMENT

Objective :

The course enables the students to understand the basics of date processing, files, DBMS (concept & design). Any RDBMS tools such as Oracle/mySQL/MS-SQL may be used.

Note : (i) The question paper will consist of Four sections.

(ii) Examiner will set total of ***nine*** questions comprising ***two*** questions from each Section and ***one*** compulsory question of short answer type covering the whole syllabi.

(iii) The students are required to attempt ***one*** question from each Section and the entire Compulsory question.

(iv) All questions carry equal marks unless specified.

SECTION-A

Introduction: Database vs File Oriented Approach, Advantages and Disadvantages of DBMS, Data models: Hierarchical, Network and Relational models, Entity Relationship model, Client-Server Architecture, Distributed databases, Data Independence.

SECTION-B

Relational Algebra, Calculus and Normalization: Storage Organization for Relations, Relational Algebra: Operations - union, intersection, difference, Cartesian product, projection, selection, division and relational algebra queries; Relational Calculus: Tuple oriented and domain oriented relational calculus and its operations, Normalization, Database Integrity, Security, Concurrency, Recovery.

SECTION-C

SQL I: Data Definition Language(DDL): Create, rename, truncate, alter and drop tables. Data manipulation Language(DML): Insert, update and delete commands. Various forms of SELECT- simple, using special operators, aggregate functions, arithmetic functions, character functions, date functions, group by clause, Nested Queries, Joins.

SECTION-D

SQL II: Introduction to VIEWs, Manipulating the Base table(s) through VIEWs, Rules of DML Statements on Join Views, Dropping a VIEW, Inline Views, Materialized Views. Database Security and Privileges, GRANT Command, REVOKE Command, Application Privileges Management, Sequences, Introduction to INDEXES, Creating simple, unique and composite INDEX, dropping INDEX, COMMIT and ROLLBACK, Backup and Recovery of the database.

Note : Any RDBMS such as MS-Access/My-SQL/MS-SQL/ORACLE may be used. The examiner may set the Questions independent of any above mentioned RDBMS tools.

References

1. Date, C.J., 1981 : *An Introduction to Data Base Systems,* Vols. I and II, Addison-Wesley.

2. Ullman, Jeffrey D., 1982 : *Principles of Data Base Systems,* 2nd Edition, Galgotia Publ. Pvt. Ltd.

3. Whittington, R.P. : *Data Base System Engineering,* Clavendon Press.

4. Pratt, P., 1987 : *Data Base Systems Management and Design,* Boyd and Fraser Publ. Comp.

5. Kroenke, D.M. : *Data Base Processing : Fundamentals, Design, Implementation*, 2nd Edition, Galgotia Publ. Pvt. Ltd.

6. Powell, Gavin : *Beginning Database Design.*

7. Jain, V.K. : *Database Management Systems.*

8. Inmon, W. H., 2002 : *Building the Data Warehouse, John Wiley.*

9. Vieira, Robert : *Beginning SQL Server 2005 Programming.*

10. Black Book : *Oracle 10g Database Administration Little Book.*

Paper B : WEB PROGRAMMING

Objective :

The course provides fundamentals of Internet & Web Programming using HTML. It also focuses Front Page, Java script & Java .

Note : (i) The question paper will consist of four sections.

(ii) Examiner will set ***nine*** questions comprising of ***two*** questions from each Section and ***one*** compulsory question of short answer type covering the whole syllabi.

(iii) The students are required to attempt ***one*** question from each Section and the entire Compulsory question.

(iv) All questions carry equal marks unless specified.

SECTION-A

1. ***Internet :*** Evolution of Internet, Future of Internet, Services provided on the Internet, Internet Access Methods.

2. ***World Wide Web :*** Evolution of www, Future of www, Fundamentals of web.

3. ***Installing Netscape Communicator :*** Browising Internet using Netscape, Netscape Messenger.

4. ***Hypertext Markup Language :*** Introduction to HTML, Building Blocks of HTML, HTML Lists, HTML Links, Images in HTML.

SECTION-B

5. ***Advanced HTML :*** HTML Tables, Frames, Layers, Forms, Editors.

6. ***Cascading Style Sheets :*** Introduction to CSS, Limitation of HTML, CSS Positioning.

7. ***Front Page :*** Installing Front Page, Front Page Editor, Create a Sample Website, Frames in Front Page, Front Page Components, Forms, Database pages.

SECTION-C

8. ***Dynamic HTML :*** Moving elements and images, changing colours and hiding elements, moving between layers, mouse rollovers.

9. ***Java Scripting :*** Features, tokens, data types, variables, operators, control structs, strings, arrays, functions, core language objects, client side objects, event handling, application related to client side from validation.

SECTION-D

10. ***Fundamentals of Java Programming Language :*** Java vs. C++, bytecode, java virtual machine, constants, variables, data types, operators, expressions, control structures, defining class, creating objects, accessing class members, constructors, method overloading.

References

1. Kraynak & Hubraken : *Internet 6 in 1,* Prentice Hall of India, 2000
2. Kasser : *Using the Internet,* PHI, 4th ed, 2000
3. WALL : *Using the World Wide Web,* PHI, 2nd ed., 2000
4. Phillips : *Using HTML, PHI,* 4th ed., 2000
5. Randall & Jones : *Special Edition Using Front Page 2000,* PHI, 2000
6. Liang : *An Introduction to Java Programming,* PHI, 2000
7. : *DHTML (Any Book).*
8. : *Java Script (Any Book).*
9. Duckelt, John : *Beginning Web Programming.*
10. Keogh, Jim : *Java Programming Fundamentals.*
11. Wilton, Paul : *Beginning Java Script, II ed.* Wilby India Pvt. Ltd.

PAPER-C : PRACTICAL : Practical based on Paper A and Paper B.

ENVIRONMENT CONSERVATION
Mode of Instruction- Punjabi/Hindi/English
B.A./B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Note: The practical will include survey and its project reports carrying 15 marks, 35 marks will be allotted to laboratory practicals. The number of hours for theory and practical per week shall be 5 hours and 4 hours respectively.

There will be 9 questions in all, two each from section I to 1V. Question No. I (17 marks) will be short answer type & will cover the whole syllabus. Candidates will attempt five questions in all, selecting one each (12 marks) from section I to IV and the first compulsory question.

Paper-A **Biodiversity and Conservation**

Total Marks	:	75 marks
Theory Marks	:	65 marks
Internal Assessment	:	10 marks
Time	:	3 hours

Section-I

Biodiversity: Definition, levels, importance, threats to biodiversity, hot spots of biodiversity in India. Environment Protection Act, 1986, Water (Prevention and Control of Pollution Act) 1974, Forest (Conservation) Act, 1980.

Section-II

Conservation: Causes of extinction of biodiversity**,** Strategies of conservation (in-situ and ex-situ).

Protected Areas (National Parks, Sanctuaries and Biosphere reserves). Wild Life Protection Act, 1972.

Section-III

Environmental Ethics: Role of religion in Environment Protection. Role of Indian tradition in environment conservation, Sustainable agriculture.

Different possible measures to make aware localties about environmental hazards and its remedies. Public Interest Litigation(PIL).

Section-IV

Cultivation Practices: Role of tree & plants in conservation; use of medicinal plants; cultivation methods *(Alovere, Calotropis (AK), Mentha, Ocimum*, Neem, *Ricinus, Jatropa, Accacia, Khair* (Katha).

Paper-B Public Awareness and Environmental Issues

Total Marks	:	75 marks
Theory Marks	:	65 marks
Internal Assessment	:	10 marks
Time	:	3 hours

Section-I

Current Environmental Issues:

Rain water harvesting.
Reclamation of mining areas.
Impact of green revolution on environment with special reference to Punjab.
Joint Forest Management (JFM).
Tehri dam.
Narmada Project.
Bhopal Gas Tragedy.
River Cleaning Project of Sant B.S. Seechewal (Punjab).

Section-II

Environmental Protection:

Role of Non - Governmental Organizations (NGOs) in environmental protection (Chipko movement, For a Living Ganga by WWF, Transformation DTC fleet to CNG driven transport, Earth Hour, Green Peace (Nitrate Pollution in Punjab), KVM, Jaito).
Role of individual in environment protection.
Role of environmental education.

Section-III

Environmental Agencies:

Role of International & National Agencies: UNEP, UNDP, WWF, MOEF, CPCB, in environment conservation and management.

A brief account on Ramsar convention, CITES (Convention on International Trade in Endangered Species), UNFCC, Montreal Protocol, Kyoto Protocol, Copenhagen Summit.

Section-IV

RS and GIS: Definition, Application in Environment**.**

Biostat: Mean mode, median, standard deviation, co-relation, Regression.

Practical

Total Marks	:	50 marks
Practical Marks	:	50 marks
Time	:	3 hours

Part-A

Biodiversity and Conservation

1. Collection of local flora and fauna.
2. Study of vegetation of local area/college campus.
3. Collection of ten pests.
4. Identification of weeds.
5. Study of vegetation density by quadrat method
6. Plantation of medicinal plants**.**
7. Adaptation features of plants and animals.
8. Assignments/project as assigned by the teacher.

Part-B

Public Awareness and Environment Issues

1. Public/Mass awareness programmes in different villages and their reports.
2. Survey report on different diseases based theory.
3. Visit to a dam to study its construction and working.
4. To visit any NGO working for environment.
5. To study various indoor pollutants in houses like LPG, AC, Paints, Polish, Cosmetics, Detergents, Shampoo, Hair dyes.
6. To study the different tools of stat in data interpretation.
7. Assignments/project as assigned by the teacher.

Books Recommended :

1. Jaswal & Jaswal : Environmental Law, APH Publications, New Delhi, 2006.
2. Khitoliya, R. K. : Environment Protection and Law, APH Publications, 2009.
3. Trivedi, R.K. : Handbook of Environmental Laws, Acts, Guidelines, Compliance and Standards, Vol. I & II, B.S. Publications, Hyderabad. 2004.

4.	Srivastava, A.K.	:	Environmental Ethics, APH Publications, 2004.
5.	Lillesand, T.M. and Kiefer, R.W.	:	Remote Sensing and Image Interpretation, John Wiley, Cambridge, 1987.
6.	Fazal, S.	:	Remote Sensing Basics, Kalyani Pub. 2008.
7.	Patel, A.N. & Singh, S.	:	Remote Sensing: Principles and Applications, Scientific Books, Jodhpur, 2009.
8.	Jensen, J.	:	Remote Sensing of the Environment, Pearson Education, Prentice Hall, 2007.
9.	CSE, Delhi	:	A Water Harvesting Manual, 2010.
10.	Arora, P.N.	:	Biostatistics, Himalayan Pub. House, 2008.
11.	Spencere, J.	:	Biodiversity: Beginners Guide, One World Publication, Oxford, 2009.
12.	Chaudhari, A. B.	:	Biodiversity Endangered, Scientific Books, 2009.
13.	Singh, Oraon & Prasad.	:	Medicinal Plants, APH Publications, 2009.

HOME SCIENCE

B.A./B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015
Scheme of Examination

	Theory					Practical				
Sr. No.	*Name of Paper*	*No. of Papers*	*Time in hrs.*	*Marks allotted*	*Int. Ass.*	*No. of Papers*	*Time in hrs.*	*Marks Allotted*	*Int. Ass.*	*Total Marks*
1.	Foods and Nutrition	1	3	65	10	1	3	40	10	125
2.	Child Development	1	3	65	10	--	--	--	--	75
									Total :	**200**

Note : 1. Practical examination will be held before the Theory Examinations.
2. Each Practical group will have 12-15 students.

Paper A : FOODS AND NUTRITION (Theory)

Max. Marks : 75
Theory : 65
Int. Ass. : 10
Periods : 3 Hours/Week

INSTRUCTIONS FOR THE PAPER SETTER :

The question paper will consist of five Sections : A, B, C, D and E. Sections A, B, C and D will have **two** questions from the respective sections of the syllabus and will carry 13 marks each. Section E will consist of 13 objective type questions covering the entire syllabus uniformly and will carry 13 marks.

INSTRUCTIONS FOR THE CANDIDATES :
Candidates are required to attempt one question each from the Sections A, B, C and D of the question paper and the entire section E.

SECTION-A

I. *Importance and Functions of Food :*

(a) Physiological; (b) Psychological; (c) Social

II. *Food Constituents :* Carbohydrates, Proteins and Fats—Functions, sources, requirements and deficiency, digestion process in brief.

III. *Methods of Cooking :* Boiling, Steaming, Frying, Baking, Roasting and Microwave cooking.

SECTION-B

IV. Functions, recommended allowances, deficiency, excess and sources of the following food nutrients :

(a) Vitamins–A, B_1, B_2, Niacin, Vit. C, Vit. D.; (b) Minerals–Calcium, Phosphorous, Iron, Sodium, Iodine.

V. *Food Preservation :*

(a) Definition, Importance & Principles.
(b) Causes of food spoilage.
(c) Household methods of food preservation—sun drying, use of salt, oil, spices, sugar & chemical preservatives.

SECTION-C

VI. *Food Adulteration :*

(a) Definition; (b) Common Adulterants; (c) Food standards.

VII. *Concept of balanced diet.*

Classification of food based on the five/seven food groups.

VIII. *Principles of meal planning.*

(a) Pre-school children
(b) School going
(c) Adolescents
(d) Adult—male and female (only moderate worker)

SECTION-D

IX. *Therapeutic Diets & Modification of Normal Diets :*

(a) Principles of therapeutic diets; (b) Concepts of soft, bland, liquid diets with examples.

X. *Therapeutic diets in the following conditions with principles involved :*

(a) Fever; (b) Constipation; (c) Diarrhea; (d) High blood pressure/ Hypertension; (e) Cholesterolemia (f) Diabetes Mellitus.

Paper B : CHILD DEVELOPMENT

Max. Marks : 75
Theory : 65
Int. Ass. : 10
Periods : 3 Hours/Weak

INSTRUCTIONS FOR THE PAPER SETTER :

The question paper will consist of five Sections: A, B, C, D and E. Sections A, B, C and D will have **two** questions from the respective sections of the syllabus and will carry 13 marks each. Section E will consist of 13 objective type questions covering the entire syllabus uniformly and will carry 13 marks.

INSTRUCTIONS FOR THE CANDIDATES :

Candidates are required to attempt one question each from the Sections A, B, C and D and the compulsory question section E.

SECTION-A

I. Pregnancy
   (a) Signs and symptoms of pregnancy.
   (b) Discomforts.
   (c) Complications.
   (d) Care during pregnancy.
   (e) Methods of family planning in brief.

II. Pre-natal Development
   (a) Stages of pre-natal development.
   (b) Factors affecting pre-natal development.

III. Feeding of the infant
   (a) Importance and technique of breast feeding.
   (b) Bottle feeding.
   (c) Weaning.
   (d) Different kinds of weaning foods and its importance.

SECTION-B

IV. Definition and importance of Child Development.
   (a) Differences between growth and development; (b) Principles of development.

V. Physical development of the child from infancy to late childhood and factors affecting the same.

SECTION-C

VI. Motor Development from infancy to late childhood.
   (a) Pattern of motor development.
   (b) Factors affecting motor development.

VII. Emotional Development of the child from infancy to late childhood.
   (a) Characteristics of children's emotions.
   (b) Common childhood emotions—Fear, angers, jealousy, love and affection, anxiety and curiosity.

VIII. Language Development.

(a) Stages of language development.

(b) Factors affecting language development.

SECTION-D

IX. Play

(a) Significance of play.

(b) Types of play.

(c) Play materials/equipment required for various age groups.

X. Common behavioural problems and their remedies—Bed wetting, thumb sucking, nail biting, temper tantrums.

PRACTICAL

Paper-A : FOODS AND NUTRITION

Max. Marks	:	50
Practical	:	40
Int. Ass.	:	10
Time	:	3 Hours
Teaching Periods	:	3 Periods/Week

Section A

I. Preparation of minimum of three dishes by using various methods of cooking (e.g. boiling, steaming), frying (deep/shallow); using cereal pule combinations, fermentation and sprouting.

II. Planning & Preparation of diets for the following :

(a) Pre-School; (b) School going/packed lunch; (c) Adolescence; (d) Adult (Man & Woman) moderate worker; (e) Pregnancy and lactation.

III. Cooking and serving of the following : Soft, liquid, fluid diets.

IV. Hot and cold beverages (at least two each).

V. Enhancing nutritive value.

VI. Preservation of one pickle, jam, chutney and squash.

VII. Project

- Survey of common feeding practices
- Survey of weaning foods
- Safety measures of play equipment

Reports to be submitted

List of Equipments for a practical group of 15 students :

1.	Gas Burners	15
2.	Cooking Range	01
3.	Ovens	05
4.	Mixers and Grinders	05
5.	Weighing Scales (for food)	05
6.	Gas Lighters	15
7.	Dustbin-Small	15-Big-1
8.	Vegetable Racks	02
9.	Plate Racks	15
10.	Storage Jars and Containers	25
11.	Refrigerator	01
12.	Icing Sets	05

Cooking Utensils

1.	Pressure Cooker	15
2.	Patila with Lid	30
3.	Kadahai	15
4.	Parat	15
5.	Tawa	15
6.	Chakla-Belna	15
7.	Grinding Stone	15
8.	Saucepans	15
9.	Karchhi	15
10.	Palta	15

11.	Poni	15
12.	Soup Strainers	15
13.	Sieves	15
14.	Enamel Bowls	15
15.	Baking Trays and Tins	15
16.	Cookie Trays	15
17.	Serving Trays	15
18.	Cutting Knives	15
19.	Peelers	15
20.	Jelly Moulds	15

Crockery and Cutlery

1.	Full Plates	30
2.	Half Plates	30
3.	Quarter Plates	30
4.	Cups & Saucers	30
5.	Soup Bowls	30
6.	Glasses	36
7.	Katoris Vegetables Bowls	30
8.	Dongas	30
9.	Forks	30
10.	Table Knives	30
11.	Table Spoons	48
12.	Tea Spoons	48
13.	Serving Spoons	24
14.	Tea Sets	05
15.	Dinner Sets	05
16.	Borosil Bowls	15
17.	Casseroies	15

Reference Books :

1. *Applied Nutrition,* R. Rajalakshmi, Oxford & IBH Publishing Co. Pvt. Ltd., N. Delhi.

2. *Principles of Nutrition-Dietetics,* Dr. M. Swaminathan,The Bangalore Printing and Publishing Co. Ltd. 88, Mysore Road, Bangalore.

3. *Food & Nutrition, Educational Planning Group,* Arya Publishing House, Karol Bagh, New Delhi-5.

4. *Normal and Therapeutic Nutrition,* Br Corinne, H. Robinson, Marlya R. Lowler, Macmillan Publishing Co., New York, Collier Macmillan Publishers, London.

5. *Nutritive Value of Indian Foods,* G. Copalen, B.V. Rama Sastri, and S.C. Balasubramaniam, National Institute of Nutrition, Indian Council of Medical Research, Hyderabad, India.

6. *Human Development,* Graing J. Graig, 5th Edition, 1989/Prentice Hall, Englewood Cliffs, New Jersey, 07632.

7. *The Modern Parents Guide to Baby & Child Care-Violet,* Broadribb, R. N.H.S. & Henry F. Loc. M.E., 1973 Macdonald's and Jane's London.

8. *Good House Keeping's Baby Book V—The Good Housekeeping,* 12th ed., 1959.

9. *These are your children—Dadys,* Gardner Jenkins and Helen Shacter, 4th ed., Scott, Foresman and Co., Glenview Illinios.

10. Hurlock, E.B., *Child Development,* 6th ed., McGraw Hill International Book Company, 1978.

AGRICULTURE

B.A./B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

PAPER-A

Max. Marks : 75
Theory : 70 Marks
Int. Ass. : 05 Marks
Time : 3 Hours

Theory

Period per week:
1. Theory – Three of 45 minutes duration
2. Practical – One of three hours duration

INSTRUCTIONS FOR THE PAPER SETTER AND THE STUDENTS :

1. Question paper shall have five sections.
2. Section A shall comprise of 10 short answer type questions covering the whole syllabus and will be compulsory. Each question will carry 1 (one) mark.
3. Section B, C, D and E shall have two questions each from respective section, out of which one from each Unit is to be permitted. Each question will be of 15(fifteen) marks.
4. Total five questions are to be attempted.

Unit – I

History of Agroforestry
Agroecological Zonification
Socioeconomic Aspects of Agroforestry
Agroforestry System for Small Holdings
Arid Land Agroforestry

Unit – II

Establishment of Orchard
Basic Cultural Practices
Elemental Role and Needs of Nutrients
Propagation - Principles and Techniques
Stock- Scion Relationship and their incompatibility
Fruit Physiology

Unit – III

Respiration – Glycolysis, Citric Acid Cycle, Photorespiration
Photosynthesis – Light Reaction, Dark Reaction, C_4-Cylce, CAM Plants
Growth Harmones and their role in Agriculture
Enzymes and Vitamins
Vernalization
Photoperiodism

Unit – IV

Intellectual Property Right
Informatics in Agriculture

Seed Production and Technology
Indigenous Technical Knowledge in Agriculture
Crop Biotechnology
Agrobiodiversity

PRACTICAL

Max. Marks : 25
Practical : 20 Marks
Int. Ass. : 05 Marks

Planning an orchard
Study of Agroforestry of Local Area
Scion- Stock relationship study
Study the effect of temperature and light duration on various crop plants of an area
Role of computer in Agriculture
GM crops and techniques
Study of Agrobiodiversity of Punjab

BOOKS

Plant Physiology, 2010 – K.N. Bhatia – Trumen Publishers, Jalandhar
Plant Physiology, 2010 – H.N. Srivasava – Pardeep Publishers, Jalandhar
Fruits – Ranjit Singh and S.K. Sexena – National Book Trust, New Delhi, 2008
Fruit Physiology – Amar Singh – Kalyani Publishers, Ludhiana, 1997

PAPER-B

Max. Marks : 75
Theory : 70 Marks
Int. Ass. : 05 Marks
Time : 03 Hours

Theory

Period per week:
1. Theory – Three of 45 minutes duration
2. Practical – One of three hours duration

INSTRUCTIONS FOR THE PAPER SETTER AND THE STUDENTS :

1. Question paper shall have five sections.
2. Section A shall comprise of 10 short answer type questions covering the whole syllabus and will be compulsory. Each question will carry 1 (one) mark.
3. Section B, C, D and E shall have two questions each from respective section, out of which one from each Unit is to be permitted. Each question will be of 15(fifteen) marks.
4. Total five questions are to be attempted.

Unit – I

Plant Protection
Crop diseases? How to control them?
Biological control, chemical control
Systemic fungicides
Weedicides
Compatibility of various fungicides and weedicides
Rodenticides

Unit – II

Fungal diseases of wheat
Rice
Maize
Citrus
Grapes
Cotton
Vegetables
Oil crops

Unit – III

What are insects and pests?
Parts of insect body
How to control insect diseases?
Classification of insects

Unit – IV

Insect and nematode diseases of Cereals crops; Oil crops; Fruit crops
Vegetable crops mentioned in package and practices for agricultural workers

PRACTICAL

Max. Marks : 25
Practical : 20 Marks
Int. Ass. : 05 Marks

Study of all the diseases mentioned in theory
Symptoms and their control measures

BOOKS

Plant Pathology –R.S. Singh, 2007- Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi
Plant Pathology – R.S. Malhotra and Aggarwal, 2009 – Tata MacGraw Hill Publishing, New Delhi
Economic and Applied Entomology – Kumar and Nigam, 2008 – Emkay Publishing, New Delhi
Fungicides – Y.L. Naine and Thaplial, 2008 – Oxford and IBH Publishing Co. Pvt. Ltd., Calcuta, Bombay
Elements of Entomology – Rajendra Singh, 2007 – Rastogi Publications, Meerut
A Text book of Applied Entomology – Vol I and Vol II – Srivastva, 2007 – Kalyani Publishers, Ludhiana
Indian Crop Pests – Kumar and Nigam, 2007 – Emkay Publishers, Allahabad
Integrated Pest Management – G.S. Dhaliwal and R. Arora, 2006 – Kalyani Publishers, Ludhiana

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MATHEMATICS

B.A./B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Note : 1. The syllabus has been split into two sections – Section A and Section B. Four questions will be set from each Section.

2. A student will attempt five questions in all selecting atleast two questions from each Section. Each question will be of 12 marks.

3. The teaching time shall be five periods (45 minutes each) per paper per week, including tutorial.

Paper- I : ANALYSIS

Max. Marks	:	67
Theory	:	60 marks
Internal Assessment	:	7 marks
Time	:	3 Hours

SECTION-A

Countable and uncountable sets. Riemann integral. Integrability of continuous and monotonic functions. The fundamental theorem of integral calculus. Mean value theorems of integral calculus.

Improper integrals and their convergence, Comparison tests, Abel's and Dirichlet's tests. Beta and Gamma functions. Frullani's integral. Integral as a function of a parameter. Continuity, derivability and integrability of an integral of a function of a parameter.

Double and triple integrals. Fubini's Theorem without proof, Change of order of integration in double integrals, volume of a region in space, Triple integrals in spherical and cylinderical coordinates, substitution in multiple integrals.

SECTION-B

Vector Integration, Gauss Divergence Theorem and Green Theorems.

Sequences and series of functions, pointwise and uniform convergence, Cauchy criterion for uniform convergence, Weierstrass M-test, Abel's and Dirichlet's tests for uniform convergence, uniform convergence and continuity, uniform convergence and Riemann integration, uniform convergence and differentiation, Weierstrass approximation theorem, (Statement only), Abel's and Taylor's theorems for power series.

Fourier series. Fourier expansion of piecewise monotonic functions.

References :

1. T. M. Apostol : *Mathematical Analysis,* Narosa Publishing House, New Delhi, 1985.
2. R. R. Goldberg : *Real Analysis,* Oxford & IBH Publishing Co., New Delhi, 1970.
3. S. Lang : *Undergraduate Analysis,* Springer-Verlag, New York, 1983.
4. D. Somasundaram : *A First Course in Mathematical Analysis,* Narosa Publishing House, New Delhi, 1997.
5. Shanti Narayan : *A Course of Mathematical Analysis,* S. Chand & Co., New Delhi.
6. P.K.Jain and S.K.Kaushik : *An Introduction to Real Analysis,* S. Chand & Co., New Delhi, 2000.
7. S.C.Malik and Savita Arora : *Mathematical Analysis,* 2nd edition, New Age International Publishers.
8. G.B.Thomas and R. L. Finney : *Calculus and Analytic Geometry* (Ninth edition), Pearson Publication.

Paper-II : ABSTRACT ALGEBRA

Max. Marks	:	67
Theory	:	60 marks
Internal Assessment	:	7 marks
Time	:	3 Hours

Note :
1. The Syllabus has been split into two sections – Section A and Section B. Four questions will be set from each Section.
2. A student will attempt five questions in all selecting atleast two questions from each Section. Each question will be of 12 marks.
3. The teaching time shall be five periods (45 minutes each) per paper per week, including tutorial.

SECTION- A

Groups, Subgroups, Lagrange's Theorem, Normal Subgroups and Quotient Groups. Homomorphisms, Isomorphism Theorems, Conjugate Elements, Class Equation, Permutation Groups, Alternating Groups,

Simplicity of A_n, $n \geq 5$(without proof).

Rings, Integral domains, Subrings and Ideals, Characteristic of a ring, Quotient Rings, Prime and Maximal Ideals, Homomorphisms, Isomorphism, Theorems, Polynomial Rings.

SECTION- B

Vector Spaces, Subspaces, Algebra of subspaces, Linear dependence and independence of vectors, Basis and Dimension of a vector space, Direct sums and complements.

Linear Transformations, Rank and Nullity of a linear transformation, Matrices and Linear transformations.

Characteristic roots and characteristic vectors of a linear transformation or of a matrix, Algebraic and Geometric multiplicity of a characteristic value, Cayley-Hamilton theorem, Diagonalizable operators and matrices. Minimal polynomial of a linear operator (matrix).

References :

1. S. Singh and Q. Zameerudin : *Modern Algebra*, 8th Edition, Vikas Publication, New Delhi.
2. I.N. Herstein : *Topics in Algebra*, Wiley Eastern Ltd., New Delhi.
3. J. A. Gallian : *Contemporary Abstract Algebra*, Narosa Publishing House, New Delhi.
4. M. Artin : *Algebra*, Prentice Hall of India, New Delhi, 1994.
4. K. Hoffman and R. Kunze : *Linear Algebra*, 2nd Edition, Prentice Hall, New Jersey, 1971.

Paper-III : OPTIONAL

Choose any one from the following three optional papers:

Option (i) : DISCRETE MATHEMATICS

Max. Marks	:	66
Theory	:	60 marks
Internal Assessment	:	6 marks
Time	:	3 Hours

Note:
1. The syllabus has been split into two Sections – Section-A and Section-B. Four questions will be set from each Section.
2. A student will attempt five questions in all selecting atleast two questions from each Section. Each question will be of 12 marks.
3. The teaching time shall be five periods (45 minutes each) per paper per week including tutorial.

SECTION- A

The pigeonhole principle : Simple form, strong form and its applications. A Theorem of Ramsey (without proof). Permutations and combinations of sets and multisets.

Identities involving binomial co-efficients. Pascal's formula. The multinomial theorem. Newton's binomial theorem. The inclusion exclusion principle and its applications. Derangements. Permutations with forbidden positions.

Recurrence relations – Linear Recurrence relations with constant co-efficients. Homogeneous solutions, particular solutions, total solutions. Non homogeneous recurrence relation.

Generating functions, Exponential generating functions, solution of recurrence relations using generating functions, Catalan numbers, Difference sequences and stirling numbers, Partition numbers.

[Scope as in Chapters 2, 3, 5-8 of Introductory Combinatorics, 3rd Edition by R. A. Brualdi.]

SECTION- B

Combinatorial designs, block designs, Steiner triple system, Latin squares.

Graphs : Basic properties, Eulerian trails, Hamilton Chains and Cycles, Bipartite multigraphs. Plane and Planar Graphs.

Trees, Spanning trees : Breadth first, Depth first, Dijkstra's, Kruskal and Prim's algorithms to generate spanning trees.

Directed graphs and networks. Chromatic number.

[Scope as in Chapters 10-13 (Sections 13.1, 13.2 only) of Introductory Combinatorics, 3rd Edition by R. A. Brualdi.]

References :

1.	R. A. Brualdi	:	*Introductory Combinatorics, 3rd Edition.*
2.	Mott, Kendal & Baker	:	*Discrete Mathematics for Computer Scientists and Mathematicians.*
3.	Schaum Series	:	*Discrete Mathematics.*
4.	C. L. Liu	:	*Elements of Discrete Mathematics,* Tata McGraw Hill, International Edition, Computer Science Series, 2000.
5.	C. L. Liu	:	*Introduction to Combinatorial Mathematics,* McGraw Hill Book Company, New York.

Option (ii) : PROBABILITY THEORY

Max. Marks	:	66
Theory	:	60 marks
Internal Assessment	:	6 marks
Time	:	3 Hours

Note:
1. The syllabus has been split into two Sections – Section-A and Section-B. Four questions will be set from each Section.
2. A student will attempt five questions in all selecting atleast two questions from each Section. Each question will be of 12 marks.
3. The teaching time shall be five periods (45 minutes each) per paper per week including tutorial.

SECTION-A

Notion of probability: Random experiment, sample space, axiom of probability, Combinatorial problems, elementary properties of probability, Conditional Probability and independence, Bayes' Theorem.

Random Variables: Concept, probability density function, cumulative distribution function, discrete and continuous random variables, expectations, mean, variance, moment generating function, skewness and kurtosis.

Discrete Random Variable: Bernoulli random variable, binomial random variable, negative binomial random variable, geometric random variable, Poisson random variable.

SECTION-B

Continuous random variables: Uniform random variable, exponential random variable, Beta random variables, Gamma random variable, Chi-square random variable, normal random variable.

Bivariate random variables: Joint distribution, joint and conditional distributions, Conditional Expectations, Independent random variables, the correlation coefficient, Bivariate normal distribution.

Distribution of Functions of Random Variable : Transformation of random variables of discrete and continuous type, Student's t–distribution, F-distribution.

References :

1. Ross,S.M. : *Introduction to Probability Models* (Sixth edition) Academic Press, 1997
2. Hogg., R.V and Craig, A.T : *Introduction to Mathematical Statistics,* MacMillan, 2002.
3. Blake, I. : *An Introduction to Applied Probability,* John Wiley & Sons,1979.
4. Pitman, J. : *Probability,* Narosa,1993.
5. Meye, P.L. : *Introductory Probability and Statistical Applications,* 2nd Edition, Oxford and IBH Publishing Co.

Option-(iii) NUMERICAL TECHNIQUES

Max. Marks : 66
Theory : 50 Marks
Practical : 10 Marks
Internal Assessment (Theory) : 6 Marks
Time (Theory) : 3 Hours
Time (Practical) : 1 Hour

***Note* :** 1. The syllabus has been divided into two Sections - Section A and Section B. Five questions will be set from Section A and three questions from Section B.
2. Candidate will attempt five questions in all selecting three questions from Section-A and two questions from Section- B. Each question will carry 10 marks.
3. The teaching time shall be five periods (45 minutes each) per paper per week including tutorial.
4. Only non-programmable scientific calculator is allowed.
5. In theory part emphasis should be given to discussions of methods.
6. There will be no internal assessment for practical.

SECTION A

Approximate numbers, Significant figures, Round off error and truncation error, Absolute, Relative and Percentage errors, General formula to estimate the error in a function.

Descarte's rule of signs, Location of roots, Bisection, Secant, Regula Falsi, Newton-Raphson Methods, Order of convergence of these methods.

Direct Methods-Gauss Elimination, Guass-Jordan, LU Decomposition, Cholesky Decomposition, Iterative Methods-Jacobi, Guass-Seidal.

Eigen values-Jacobi Method, Givens' Method, Householders' Method, Power Method, QR Method.

Interpolation-Finite differences, Forward. Backward and central differences, Symbolic Relations, Detection of error using difference table, Divided differences and their properties, Interpolation formulas – Newtons, Gauss, Stirling and Bessels, Lagranges.

Numerical Differentiation using interpolation formulas.

Numerical Integration, General formula, Trapezoidal, Simpson one-third and three-eight forulae, Newton-Cote's Formulae.

Euler Method, Euler modified, Picard methods, Runge Kutta second and forth order Methods, Milne-Thompson Method.

SECTION B

Introduction to FORTRAN, Character set, constants, variables, Arithmetic expressions, Library functions, Arithmetic statements, Structure of a FORTRAN program, Format specification, Read and write statement, Terminating of a Program, Working sheet. Unconditional and Computed Go To, Arithmetic and Logical IF, IF-THEN-ELSE, Nested IF-THEN-ELSE, ELSE-IF-THEN, DO-loops, nested DO-loops, DATA statement, double precision, WHILE structure, Arrays and Subscripted variables, Sub programmes and subroutines, data files.

References:

1. C. Xavier : *Fortran 77 and numerical methods,* New age international, 2012.
2. S. S. Sastry : *Introductory Methods of Numerical Analysis,* Fourth Edition, PHI Learning Private Limited, 2010.
3. M. K. Jain, S.R.K. Iyengar and R.K.Jain : *Numerical Methods for Scientific and Engineering Computation,* New Age International (P) Ltd., 1999.
4. R.L. Burden, J. D. Faires: *Numerical Analysis,* CENGAGE Learning, 2012.
5. K E Atkinson : *An Introduction to Numerical Analysis,* Hohn Wiley and Sons, NY, 1978.

Numerical Techniques (Practical)

Writing programs in FORTRAN for the simple problems and problems based on the methods studied in the theory paper and run it on computer.

Practical will be conducted by the Internal and external examiners. External examiner will be appointed by the university. The following criteria will evaluate the candidate:

List of Practical

1. Magnitude of a vector
2. Slope and mid-point of a line
3. Area of triangle
4. Simple and compound interest
5. Mean and standard deviation
6. Bisection method
7. Regula –Falsi method
8. Newton-Raphson
9. Gauss elimination
10. Gauss-Seidal
11. Trapezoidal rule
12. Simpson one-third rule
13. Euler method
14. Runge-Kutta method
15. Sum of Sin and Cosine series

STATISTICS

B.A./B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Note :
1. A candidate shall offer this subject in B.A./B.Sc. only if he/she takes up Mathematics as a subject in B.A./B.Sc.
2. A candidate shall offer this subject in B.A./B.Sc. Third year only if he/she had taken up the corresponding subject in B.A./B.Sc. First and Second years.
3. There are three papers code named papers 301, 302 and 303 in the subject of Statistics in B.A./B.Sc. Third year. These are to be taught simultaneously throughout the year.
4. 4 lectures (45 minutes each) per paper per week amounting in all to 12 lectures for three papers shall be allocated for the teaching.

Paper-301 : DEMOGRAPHY AND ECONOMIC STATISTICS

Max. Marks	:	75
Theory	:	65 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Notes :

1. There will be in all nine (9) questions, all of equal marks. The first question is **compulsory** and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each section. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each section.
2. Simple non-programmable calculator is allowed.
3. Statistical tables and log tables will be provided on request.

SECTION-I

Sources of demographic data - census, vital statistical registers, adhoc surveys and hospital records. Measurement of mortality - crude death rate, specific death rates, standardized death rate, infant mortality rate and cause of death rate, complete life Table and its description. Measurement of fertility - crude birth rate, general fertility rate, total fertility rate, gross reproduction rate and net reproduction rate.

Economic time series, its different components, illustrations, additive and multiplicative models, determination of trend, growth curves (exponential and modified exponential), analysis of seasonal fluctuations, construction of seasonal indices.

SECTION-II

Index numbers**,** definition, Methods to construct price, quantity and value index numbers. Problems involved in the construction of index numbers, use of averages, simple aggregative and weighted average methods. Laspeyre's, Paasche's, Edgeworth - Marshall and Fisher's index numbers. Time and factor reversal tests of index numbers. Chain Base index numbers, Cost of living index number, interpretation and applications of index numbers.

Static laws of demand and supply, price elasticity of demand, Pareto distribution, log normal distribution and their properties.

References :

1. Goon, A.M., Gupta, M.K., Das Gupta, B. (2005) : *Fundamentals of Statistics,* Vol. II, World Press, Calcutta.
2. Srivastava, O.S. (1983) : *A Textbook of Demography,* Vikas Publishing.

Additional References :

1. Croxton, F.E and Cowden, D.J. (1969) : *Applied General Statistics,* Prentice Hall of India.
2. Gupta and Mukhopadhyay, P.P. (1999) : *Applied Statistics,* Central Book Agency.
3. S.C. Gupta & V.K. Kapoor (2001) : Fundamentals of Applied Statistics, Sultan Chand & Sons, Educational Publishers, New Delhi.

Paper - 302 : STATISTICAL QUALITY CONTROL AND COMPUTATIONAL TECHNIQUES

Max. Marks	:	75
Theory	:	65 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Notes:

1. There will be in all nine (9) questions, all of equal marks. The first question is **compulsory** and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each section. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each section.
2. Simple non-programmable calculator is allowed.
3. Statistical tables and log tables will be provided on request.

SECTION-I

Importance of Statistical methods in industrial research and practice, meaning of quality, quality assurance and process control. General theory of control charts, causes of variations in quality, control limits, sub-grouping summary of out of control criteria, Charts for variables - $\overline{X}$ and R charts. Charts for attributes - np, p, c and u - charts.

Principle of acceptance sampling - problem of lot acceptance, stipulation of good and bad lots, concepts of producer's and consumer's risks, AQL, LTPD, AOQL, ATI, ASN and OC functions. Single and double sampling plans and their ATI, ASN and OC functions.

SECTION-II

Difference tables and methods of interpolation, Newton's and Lagrange's methods of interpolation, divided differences, numerical differentiation and integration. Trapezoidal rule, Simpson's one third formula, iterative solution of linear equations by Gauss–Seidel Method.

Linear Programming : Elementary theory of convex sets, definition of general linear programming problems (LPP), formulation of LPP, examples of LPP. Graphical and simplex methods of solving an LPP, artificial variables, duality of LPP, Transportation problem (non-degenerate and balanced cases), initial basic feasible solution through North-West Corner Rule, Matrix Minima & VAN method.

References :

1. Goon, A.M., Gupta, M.K. and Dasgupta, B. (2005) : *Fundamental of Statistics,* Vol. I., World Press, Calcutta.
2. Goon, A.M., Gupta, M.K. and Dasgupta, B. (2005) : *Fundamental of Statistics,* Vol. II., World Press, Calcutta.
3. Kantiswarrop, Gupta, P.K. and Manmohan (2008) : Operation Research, Sultan Chand & Sons, Educational Publishers, New Delhi.
4. Montgomery, D.C. (2001) : Introduction to Statistical Quality Control, Wiley.
5. Jain, M.K., Iyengar, S.R.K. & Jain, R.K. (2010) : Numerical Methods, 5th edition.

Additional References :

1. Grant, E.L . (1964) : *Statistical Quality Control,* McGraw Hill.
2. Duncan, A.J. (1974) : *Quality Control and Industrial Statistics,* Taraporewala and Sons.
3. Gass, S.I. (1975) : *Linear Programming Methods and Applications,* McGraw Hill.
4. Rajaraman, V. (1981) : *Computer Oriented Numerical Methods,* Prentice Hall.

Paper-303: PRACTICAL

Marks : 50

Time : 3 Hours

(Viva voce: 10 marks; record of the year: 10 marks; Annual Paper: 30 marks)

Note : The Practical Question Paper will contain five questions from the following topics. A student will be required to attempt three questions, each of 10 marks, in three hours duration.

1. Computing measures of mortality & fertility, construction of complete life table and examples involving users of life tables.

2. Construction of Index Numbers by Laspeyre's, Paasche's, Edgeworth-Marshall's and Fisher's methods.

3. Determination of trend, construction of seasonal indices in a time series.

4. Drawing of $\overline{X}$ - R, np, p,c and u – charts, Drawing of OC, AOQ and ATI curves for single and double sampling plans for attributes.

5. Construction of difference tables, use of Newton, Lagrange and divided difference interpolation formulae, numerical evaluation of integrals using Trapezoidal and Simpson one-third formulae, solution of system of linear equations by Gauss – Seidel iterative method.

6. Formulation of LPP's and their duals. Solving LPP's by graphical and Simplex methods, solution of transportation problem.

APPLIED STATISTICS

B.A./B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper- A : ESTIMATION AND TESTING OF HYPOTHESIS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Notes :
1. There will be in all nine (9) questions, all of equal marks. The first question is compulsory and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each section. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each section.
2. Simple non-programmable calculator is allowed.
3. Statistical tables and log tables will be provided on request.

SECTION-I

Estimators and estimates, unbiased, consistent, efficient estimators. Methods of moments, maximum likelihood estimators for the parameters of Binomial, Poisson and normal distributions, confidence intervals.

Tests of a statistical hypothesis, two types of errors, power of a test, Tests for the parameters of Binomial, Poisson and normal distributions, Chi-squared tests of goodness of fit. Wilcoxon, and sign test.

SECTION-II

Analysis of variance, one and two way classifications. Estimates of main effects, tests of significance for equality of effects.

Principles of design of experiments - Randomization, replication and local control. Completely randomized and randomized block designs.

Multivariate Techniques (upto 4 variables only). Estimators of mean vector and variance - covariance matrix of multivariate (upto 4) normal distribution, multiple regression, multiple correlation and partial correlation.

Books Recommended

1. Goon, A.M., Gupta, M.K. and Das Gupta, B. (2005) : Fundamentals of Statistics, Vol. I Chapters: 14 to18 (only the relevant portion from these chapters as suggested by the body of the Syllabus).

2. Goon, A.M., Gupta, M.K. and Das Gupta, B. (2005) : Fundamentals of Statistics, Vol. II, Chapters: 1, 2 (only the relevant portion from these chapters as suggested by the body of the syllabus).

Additional references:

1. Daniel, W.W. (7th edition) (1998) : Biostatistics. A Foundation for Analysis in the Health Sciences, Wiley India: New Delhi.

2. Gupta, S.C. & Kapoor, V.K. (2001) : Fundamentals of Applied Statistics, Sultan Chand & Sons, Educational Publishers, New Delhi.

Paper-B : ECONOMICS AND INDUSTRIAL STATISTICS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Notes : 1. There will be in all nine (9) questions, all of equal marks. The first question is compulsory and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each section. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each section.

2. Simple non-programmable calculator is allowed.

3. Statistical tables and log tables will be provided on request.

4. 4 to 5 lectures (40 minutes each) per paper per week amounting in all to 9 lectures for two papers shall be allocated for the teaching.

SECTION-I

Sampling : Simple random and stratified sampling, optimum allocation in stratified sampling. Ratio and regression estimates.

Index Numbers : Index Numbers-as weighted averages, Price Index numbers, Quantity Index numbers, Fisher's tests for index numbers.

Time Series : The four components of a time series, moving average, the Slutsky Yule effect, determination of trend by curve fitting and moving average methods.

SECTION-II

Quality Control : Construction, use and interpretation of control charts for mean, range, fraction defective and number of defects. Single sampling inspection plans. Concepts of producer's and consumer's risks, O.C. and A.O.Q. Curves.

Vital Statistics : Rates and ratios, crude death rate, age specific death rate, infant mortality rates, standardized death rates, direct and indirect methods. Measurement of fertility, crude birth rate, general, specific and total fertility rates. Standardized birth rates. Calendar Year rates.

Books Recommended

1. Goon, A.M., Gupta, M.K. and Das Gupta, B. (2005) : *Fundamentals of Statistics,* Vol. II , Ch. 3, 4, 6, 7, 9. (only the relevant portion from these chapters as suggested by the body of the syllabus).

2. Daniel, W.W. (7th edition) (1998) : Biostatistics. A Foundation for Analysis in the Health Sciences, Wiley India : New Delhi.

Additional references:

Gupta, S.C. & Kapoor, V.K. (2001) : Fundamentals of Applied Statistics, Sultan Chand & Sons, Educational Publishers, New Delhi.

PHYSICS

B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Note :

1. There will be three papers of theory and one laboratory (practical) course.
2. The number of lectures per week will be three for each theory paper and six for practicals.
3. The examination time for each theory paper will be three hours and four hours for practicals.
4. Each theory paper will consist of nine questions carrying equal marks and spread over five prescribed units.
5. Eight questions in each theory paper will be set from units I to IV with two questions from each unit.
6. Ninth question in each paper is compulsory and will comprise of seven small answer type questions covering the whole syllabus.
7. The numerical problems/exercises in the question paper should be 25-30%.
8. Student will attempt one question from unit (I-IV) and any six parts of question nine.
9. The use of Non-programmable calculators will be allowed (paper setters should explicitly mention this on the question paper) in the examination centre but these will not be provided by the University/College. Mobile phones and pagers are not allowed in the examination hall.

Papers, marks and teaching hours allocation :

Paper A :	Condensed Matter Physics	: (45+5*) marks	Total Teaching hours**	60
Paper B :	Electronics and Solid State Devices	: (45+5*) marks	Total Teaching hours**	60
Paper C :	Nuclear & Particle Physics	: (45+5*) marks	Total Teaching hours**	60
	Physics Practicals	: (45+5*) marks	Total Teaching hours	90

***** Marks allotted for internal assessment.

** within the prescribed teaching load, teachers must give two assignments per paper per academic term (July – September, October – December, January – March) with stress on problem solving to enhance the skill component of the students.

Paper A : CONDENSED MATTER PHYSICS (60 Hrs.)

UNIT-I

Crystal structure, Symmetry operations for a two dimensional crystal. Two dimensional Bravais lattices, Three dimensional Bravais lattices, Basic primitive cells, Crystal planes and Miller indices, Diamond and NaCl structure. Crystal diffraction : Bragg's Law, Experimental methods for crystal structure studies, Laue equations, Reciprocal lattices of SC, BCC and FCC, Bragg's law in reciprocal lattice, Brillouin zones and its derivation in two dimensions, structure factor and atomic form factor.
Determination of crystal structure.

UNIT-II

Lattice vibrations, concepts of phonons, Scattering of photons by phonons, vibrations of mono and diatomic, linear chains, Density of modes, Einstein and Debye models of specific heat, Free electron model of metals, Free electron, Fermi gas and Fermi energy.

UNIT-III

Band Theory : Kronig-Penney model, metals and insulators, conductivity and its variation with temperature in semi-conductors, Fermi levels in intrinsic and extrinsic semi-conductors, Qualitative discussion of band gap in semi-conductors. Dielectric constant & polarisability, frequency dependence, ferroelectrics and piezoelectrics.

UNIT-IV

Magnetic classification of solids (Dia, para, ferro, ferri, antiferro), Langevin theory, Quantum theory, Weiss theory, temperature dependence, hysteresis of ferromagnetic materials. Superconductivity, Meisner effect, penetration depth, critical field and temperature, BCS theory (formation of cooper pairs, ground state and energy gap.

UNIT-V

Spread over the entire syllabi of all the four units above.

Recommended Books :

Essential Readings :

1. Kittel, C. : *Introduction to Solid State Physics,* Wiley Eastern.
2. Patil, S.H. : *Elements of Modern Physics,* TMGH, 1985.

Further Readings :

1.	Pillai, S.O.	:	*Solid State Physics,* 6th Edition, New Age International Publishers.
2.	Srivastava, J.P.	:	*Elements of Solid State Physics,* 2nd Edition, Prentice Hall.
3.	M. Ali Omar	:	*Elementary Solid State Physics,* Pearson.
4.	Verma, A.R. Srivastava, O.N.	:	*Crystallography for Solid State Physics,* Wiley Eastern.

Paper-B : ELECTRONICS AND SOLID STATE DEVICES (60 Hrs.)

UNIT-I

Concepts of current and voltage sources, p-n junction, biasing of diode, V-A characteristics, zener diode, LED, LCD, rectification : half wave, full wave rectifiers and bridge rectifiers, filter circuits (RC, LC and π filters), efficiency, ripple factor, voltage regulation, voltage multiplier circuits.

UNIT-II

Junction Transistor : Structure and working, relation between different currents in transistor, sign conventions, amplifying action. Different configurations of a transistor and their comparison, CB and CE characteristics, structure of JEFT and MOSFET, Transistor biasing and stabilization of operating point, fixed bias, collector to base bias, bias circuit with emitter resistor, voltage divider biasing circuit.

UNIT-III

Working of CE amplifier, Amplifier analysis using h-parameters, equivalent circuits, determination of current gain, power gain, input impedance, FET amplifier and its voltage gain, operational amplifier, characteristics and applications, feed back in amplifiers, different types, voltage gain, advantages of negative feed back, emitter follower as negative feed back circuit.

UNIT-IV

Barkhausen criterion of sustained oscillations, LC oscillator (tuned collector, tuned grid, Hartley), RC oscillators, phase shift and wein bridge, Modulation and detection, AM and FM, Power in AM and generation of AM detector, Radio transmitter, Radio wave propagation, Ionosphere, radio receiver, TV receiver.

UNIT-V

Spread over the entire syllabi of all the four units above.

Recommended Books :

Essential Readings :

1. Bhargave, N.N., Kulshreshtha, D.C. and Gupta, S.C. : *Basic Electronics and Linear Circuits,* Tata Mc Graw Hill.
2. Chatopadhyay, D., Rakshit, P.C., Saha, B., and Purkit, N.N. : *Foundations of Electronics,* New Age International.

Further Readings :

1. Thareja, B.L. : *Basic Electronics,* 5th Edition, S. Chand.
2. Ben G. Streetman : *Solid State Electronic Devices,* 3rd Edition.
3. Malvino, A.P. : *Electronic Principles,* McGraw Hill.

Paper-C : NUCLEAR AND PARTICLE PHYSICS (60 Hrs.)

UNIT-I

Constituents of nucleus and their intrinsic properties, Qualitative facts about size, mass, density, energy, charge, binding energy, angular momentum, magnetic moment and electric multipole moments of the nucleus, Wave mechanical properties of nucleus, Average Binding energy and its variation with mass number, properties of nuclear forces and saturation, Non-existence of electrons in the nucleus and neutron-proton model, Assumptions of liquid drop model, semi-empirical mass formula, conditions of nuclear stability, Fermi gas model, Nuclear Shell Model, Experimental evidence of magic numbers and its explanation.

UNIT-II

Radioactivity, Modes of decay and successive radioactivity, Aplha emission, electron emission, positron emission, electron capture, gamma-ray emission, internal conversion, Qualitative discussion of alpha, beta and gamma-ray spectra, Geiger-Nuttal rule, Neutrino hypothesis of beta decay, Evidence of existence of Neutrino, Qualitative discussion of alpha and beta decay theories. Nuclear Reactions, Reaction cross section, conservation laws, Kinematics of nuclear reaction, Q-value and its physical significance, compound nucleus, possible reaction with high energy particles.

UNIT-III

Energy loss due to ionization (Bethe Block formula), Energy loss of electrons, Bremsstrahlung, Multiple Coulomb scattering, Gamma-ray through matter, pair production, radiation loss by fast electrons, radiation length, electron-positron annihilation, Cyclotron, Betatron, Qualitative discussion of Synchrotron, Collider machines and Linear accelerators.

UNIT-IV

Ionization chamber, proportional counter, G.M. counter, Scintillation counter, Solid State detectors. Subatomic particles and their masses, lifetimes, decay modes, classification of these particles, types of interactions, Conservation laws and quantum numbers, concepts of isospin, strangeness, charge conjugation, antiparticles, introduction to quarks and qualitative discussion of the quark model.

UNIT-V

Spread over the entire syllabi of all the four units above.

Recommended Books :

Essential Readings :

1. Kaplan, I. : *Nuclear Physics,* Addition-Wesley, Publishing Company Inc.
2. Bhiday, M.R. and Joshi, V.A. : *An Introduction to Nuclear Physics,* Orient Longman.
3. Beiser : *Concept of Modern Physics,* McGraw Hill.

Further Readings :

1. Cohen, B.L. : *Concepts of Nuclear Physics,* TMH Edition.
2. Segre, E. : *Nuclei & Particles.*
3. Verma, J. : *Fundamentals of Nuclear Physics,* CBS.

PHYSICS PRACTICALS (90 Hours)

Total Marks : 50
Practical : 45
Int. Asses. : 05

General Guidelines for Physics Practical Examinations :

1. *The distribution of Practical 45 marks is as follows :*

   I One full experiment out of Section–A requiring the student to take some data, analyse it and draw conclusions (candidates are expected to state their results with limits of error). : 20 marks

   II One exercise based on experiment or Computer Programming. : 10 marks

   (To be allotted by the external examiner at the time of examination).

III	Viva-Voce.	:	10 marks
IV	Record (Practical file).	:	05 marks

Note for Examiners :

The marks scored under each head must be clearly written on the answer sheet.

2. There will be one session of 4 hours duration. The paper will have two sections. Section–A will have 8 experiments out of which an examinee will mark 6 experiments and one of these is to be allotted by the external examiner.

3. Section–B will consist of exercises which will be set by the external examiner at the spot. The length of the exercises should be such that any of these could be completed in one hour.

4. The examiner should take care that the experiment allotted to an examinee from section–A and exercise allotted from section–B are not directly related to each other.

5. *Number of candidates in a group for practical examination should not exceed 12.*

6. In a single group, no experiment be allotted to more than three examinees in the group.

LIST OF EXPERIMENTS :

Note : Each student should perform ***atleast eighteen experiments*** in the laboratory.

I CONDENSED MATTER PHYSICS :

Activities :

(i) Measurement of reverse saturation current in p-n junction diode at various temperatures and to find the approximate value of energy gap.

(ii) To draw forward and reverse bias characteristics of a p-n junction diode and draw a load line.

(iii) Study of a diode as a clipping element.

(iv) To measure the magnetic susceptibility of $FeCl_2$ solution by Quincke's method.

(v) To trace the B-H curves for different materials using CRO and find the magnetic parameters from these.

(vi) To find the conductivity of a given semi-conductor crystal using four probe method.

(vii) To determine the Hall coefficient for a given semiconductor.

II ELECTRONICS AND SOLID STATE DEVICES :

Activities :

(i) To study the response of RC-circuit to various input voltages (square, sine and triangular).

(ii) To measure the efficiency and ripple factors for (a) Half-wave, (b) Full wave, and (c) Bridge rectifier circuits.

(iii) To study the reduction in the ripples in the rectified output with RC, LC and π-filters.

(iv) To draw the characteristics of a Zener diode.

(v) To study the stabilization of output voltage of a power supply with Zener diode.

(vi) To measure and plot Common Emitter Characteristics of a transistor (pnp or npn).

(vii) To plot Common Base Characteristics and determine h-parameters of a given transistor.

(viii) To draw output and mutual characteristics of an FET and determine its parameters.

(ix) To study the gain of an amplifier at different frequencies and to find band-width and gain-band-width product.

(x) To set up an oscillator and study its output on CRO for different V values.

(xi) To study the characteristics of a thermistor and find its parameters.

III NUCLEAR PHYSICS :

Activities :

(i) To draw the Plateau of a GM counter and find its dead time.

(ii) To study the statistical fluctuations using GM counter.

(iii) To study the absorption of beta-particles and determine the end point energy using GM counter. Also determine the absorption co-effficient (for aluminium) from it.

(iv) Verification of Rutherford Scattering experiment-mechanical analogue.

Exercises :

1. Any one exercise based on the above given experiments.

Computer based Activities :

2. To solve simultaneous equations by elimination method.

3. Fitting a straight line or a simple curve of a given data.

4. Convert a given integer into binary and octal systems and vice versa.

5. Inverse of a matrix.
6. Spiral array.

Text and Reference Books :

1. Khandelwal, D.P. : *"A Laboratory Manual of Physics for Undergraduate Classes"*.
2. Arora, C.L. : *"B.Sc. Practical Physics"*.
3. Dixon, C. : *"Numerical Analysis"*.
4. Lipsdutz, S., and Poe, A. : *"Schaum's Outline of Theory and Problems of Programming with Fortran"*.

CHEMISTRY

B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Scheme of Teaching and Examination

Paper	*Course*	*Teaching Hrs.*		*Max. Marks*
IX	Inorganic Chemistry	60	3 periods per week	45+5 internal assessment
X	Organic Chemistry	60	3 periods per week	45+5 internal assessment
XI	Physical Chemistry	60	3 periods per week	45+5 internal assessment
XII	Laboratory Practicals		6 periods per week	45+5 internal assessment
		Total	**15 periods/week**	**200**

Paper – IX : INORGANIC CHEMISTRY

Time : 3 Hrs.
Max. Marks : 45+5
60 Hours (2 Hrs. week)
3 Periods/Week

OBJECTIVE OF THE COURSE :

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (GENERAL) (3 Year course) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I **15 Hrs.**

Metal–Ligand Bonding in Transition Metal Complexes :

Limitations of valence bond theory, an elementary idea of crystal – field theory, crystal field splitting in octahedral, tetrahedral and square planar complexes, factors affecting the crystal – field parameters, Spectro chemical Series.

Thermodynamic and Kinetic Aspects of Metal Complexes:

A brief outline of thermodynamic and Kinetic stability of metal complexes and factors affecting the stability, substitution reactions of square planar complexes.

UNIT-II 15 Hrs.

Organometallic Chemistry :

Definition, nomenclature and classification of organometallic compounds. Preparation, properties, bonding and applications of alkyls and aryls of Li, Al, Hg, Sn and Ti, a brief account of metal – ethylenic complexes and homogeneous hydrogenation, mononuclear carbonyls and the nature of bonding in metal carbonyls.

Silicones and Phosphazenes :

Silicones and phosphazenes as examples of inorganic polymers, nature of bonding in triphosphazenes.

UNIT-III 15 Hrs.

Bioinorganic Chemistry :

Essential and trace elements in biological processes, metalloporphyrins with special reference to haemoglobin and myoglobin. Biological role of alkali and alkaline earth metal ions. Nitrogen fixation.

Hard and Soft Acids and Bases (HSAB) :

Classification of acids and bases as hard and soft Pearson's HSAB concept, acid-base strength and hardness and softness. Symbiosis, theoretical basis of hardness and softness, electronegativity and hardness and softness.

UNIT-IV 15 Hrs.

Electronic Spectra of Transition Metal Complexes :

Types of electronic transitions, L – S coupling, selection rules for *d-d* transitions, spectroscopic ground states, Orgel – energy level diagram for d^1 and d^0 states, discussion of the electronic spectrum of $[Ti(H_2O)_6]^{3+}$ complex ion.

Magnetic Properties of Transition Metal Complexes :

Types of magnetic behaviour, methods of determining magnetic susceptibility, spin-only formula. Correlation of μ_s and μ_{eff} values, orbital contribution to magnetic moments, application of magnetic moment data for 3*d*-metal complexes.

Instructions for paper setters and candidates :

(i) *Examiner will set total of **Nine** questions comprising **Two** questions from each unit and **One** compulsory question of short answer type covering whole syllabi.*

(ii) *The students are required to attempt **Five** questions in all, **One** question from each unit and the compulsory question.*

(iii) *All questions carry equal marks.*

Books Suggested :

1. Cotton, F.A., Wilkinson, G., Gaus, P.L., *Basic Inorganic Chemistry;* 2nd edition, Pubs: John Wiley and Sons, 1995.

2. Lee, J.D., *Concise Inorganic Chemistry;* 4th edition, Pubs: Chapman and Hall Ltd., 1991.

3. Shriver, D.E., Atkins, P.W., Langford, C.H., *Inorganic Chemistry;* 4th edition, Pubs: Oxford University Press, 2006.

4. Dauglas, B., McDaniel, D., Alexander, J., *Concepts and Models of Inorganic Chemistry*; 3rd edition, Pubs: John Wiley and Sons Inc., 1999.

5. Porterfeild, W.W., *Inorganic Chemistry;* Pubs: Addison-Wesley Publishing Company, 1984.

6. Miessur, G.L., Tarr, D.A., *Inorganic Chemistry;* 3rd edition, Pubs: Pearson Education Inc., 2004.

7. Jolly, W.L., *Modern Inorganic Chemistry;* 2nd edition, Pubs: Tata McGraw-Hill Publishing Company Ltd., 1991.

8. Purcell, K.F., Kotz, J.C., *Inorganic Chemistry;* Pubs: W.B. Saunders Company, 1977.

9. Puri, B.R., Sharma, L.P., Kalia, K.C., *Principles of Inorganic Chemistry;* 30th edition, Pubs: Milestones Publishers, 2006-07.

Paper–X : ORGANIC CHEMISTRY

Time : 3 Hrs.
Max. Marks : 45+5
60 Hours (2 Hrs. week)
3 Periods/Week

OBJECTIVE OF THE COURSE :

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (GENERAL) (3 Year course) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I **15 Hrs.**

Carbohydrates :

Classification and nomenclature. Monosaccharides, mechanism of osazone formation, interconversion of glucose and fructose, chain lengthening and chain shortening of aldoses. Configuration of monosaccharides. Erythro and threo diastereomers. Conversion of glucose into mannose. Formation of glycosides, ethers and esters. Determination of ring size of monosaccharides. Cyclic structure of D (+) – glucose. Mechanism of mutarotation.

Structure of ribose and deoxyribose.

An introduction to disaccharides (maltose, sucrose and lactose) and polysaccharides (starch and cellulose) without involving structure determination.

Amino Acids, Peptides, Proteins and Nucleic Acids :

Classification, structure and stereochemistry of amino acids. Acid-base behavior, isoelectric point and electrophoresis. Preparation and reactions of α- amino acids.

Structure and nomenclature of peptides and proteins. Classification of proteins. Peptide structure determination, end group analysis, selective hydrolysis of peptides. Classical peptide synthesis, solid – phase peptide synthesis. Structures of peptides and proteins. Levels of protein structure. Protein denaturation/renaturation.

Nucleic Acids : Introduction. Constituents of nucleic acids. Ribonucleosides and ribonucleotides. The double helical structure of DNA.

UNIT-II **15 Hrs.**

Spectroscopy :

Nuclear magnetic resonance (NMR) spectroscopy.

Proton magnetic resonance (^{1}H NMR) spectroscopy, nuclear shielding and deshielding, chemical shift and molecular structure, spin-spin splitting and coupling constants, area of signals, interpretation of PMR spectra of simple organic molecules such as ethyl bromide, ethanol, acetaldehyde, 1, 1, 2-tribromoethane, ethyl acetate, toluene and acetophenone.

Problems pertaining to the structure elucidation of simple organic compounds using UV, IR and PMR spectroscopic techniques.

Synthetic Polymers :

Addition or chain-growth polymerization. Free radical vinyl polymerization, ionic vinyl polymerization, Ziegler – Natta polymerization and vinyl polymers.

Condensation or step growth polymerization. Polyesters, polyamides, phenol formaldehyde resins, urea formaldehyde resins, epoxy resins and polyurethanes.

Natural and synthetic rubbers.

UNIT-III **15 Hrs.**

Electromagnetic Spectrum: Absorption Spectra : .

Ultraviolet (UV) absorption spectroscopy – Absorption laws (Beer – Lambert Law), molar absorptivity, presentation and analysis of UV spectra, types of electronic transitions, effect of conjugation. Concept of chromophore and auxochrome. Bathochromic, hypsochromic, hyperchromic and hypochromic shifts. UV spectra of conjugated enes and enones.

Woodward Fieser Rules and their applications in calculating maximum values of conjugated alkenes (cyclic as well as acyclic) and conjugated carbonyl compounds.

Infrared (IR) absorption spectroscopy – Molecular vibrations, Hooke's law, selection rules, intensity and position of IR bands, measurement of IR spectrum, fingerprint region, characteristic absorptions of various functional groups and interpretation of IR spectra of simple organic compounds.

Organic Synthesis via Enolates :

Acidity of ά-hydrogens, alkylation of diethyl malonate and ethyl acetoacetate. Synthesis of ethyl acetoacetate: the Claisen condensation. Keto-enol tautomerism of ethyl acetoacetate. Alkylation and acylation of enamines.

UNIT-IV **15 Hrs.**

Heterocyclic Compounds :

Introduction : Molecular orbital picture and aromatic character of pyrrole, furan, thiophene and pyridine. Methods of synthesis and chemical reactions with particular emphasis on the mechanism of electrophilic substitution. Mechanism of nucleophilic substitution reactions in pyridine derivatives. Comparison of basicity of pyridine, piperidine and pyrrole.

Introduction to condensed – five and six – membered heterocycles. Preparation and reactions of indole, quinoline and isoquinoline with special reference to Fisher indole synthesis. Skraup synthesis and Bischler–Napieralski synthesis. Mechanism of electrophilic substitution reactions of indole, quinoline and isoquinoline.

Organometallic Compounds :

Organomagnesium Compounds: The Grignard reagents – Formation, structure and chemical reactions.

Organozinc Compounds : Formation and Chemical reactions.

Organolithium Compounds: Formation and Chemical reactions.

Instructions for paper setters and candidates:

(i) *Examiner will set total of* ***Nine*** *questions comprising* ***Two*** *questions from each unit and* ***One*** *compulsory question of short answer type covering whole syllabi.*

(ii) *The students are required to attempt* ***Five*** *questions in all,* ***One*** *question from each unit and the Compulsory question.*

(iii) *All questions carry equal marks.*

Books Suggested :

1. Morrison, R.T., Boyd, R.N., *Organic Chemistry;* 6th edition, Pubs: Prentice-Hall, 1992.

2. Wade Jr., L.G., Singh, M.S., *Organic Chemistry;* 6th edition, Pubs: Pearson Education, 2008.

3. Mukherji, S.M., Singh, S.P., Kapoor, R.P., *Organic Chemistry;* Pubs: New Age International, 1985, Vols. I, II, III.

4. Carey, F.A., *Organic Chemistry;* 4th edition, Pubs: McGraw-Hill, 2000.

5. Solomons, T.W., *Fundamentals of Organic Chemistry;* 5th edition, Pubs: John Wiley & Sons, 1997.

6. Streitwieser, A., Clayton, Jr., Heathcock, H., *Introduction to Organic Chemistry;* 3rd edition, Pubs: Macmillan Publishing Company, 1989.

Paper–XI : PHYSICAL CHEMISTRY

Time : 3 Hrs.
Max. Marks : 45+5
60 Hours (2 Hrs. week)
3 Periods/Week

OBJECTIVE OF THE COURSE :

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (GENERAL) (3 Year course) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I 15 Hrs.

Elementary Quantum Mechanics :

Black-body radiation, Planck's radiation law, photoelectric effect, heat capacity of solids, Bohr's model of hydrogen atom (no derivation) and its defects, Compton effect.

De Broglie hypothesis, the Heisenberg's uncertainty principle, Sinusoidal wave equation, Hamiltonian operator, Schrodinger wave equation and its importance, physical interpretation of the wave function, postulates of quantum mechanics, particle in a one dimensional box.

Schrodinger wave equation for H-atom, separation into three equations (without derivation), quantum numbers and their importance, hydrogen like wave functions, radial wave functions, angular wave functions.

Molecular orbital theory, basic ideas – criteria for forming M.O. from A.O., construction of M.O.'s by LCAO – H_2^+ ion. Calculation of energy levels from wave functions, physical picture of bonding and antibonding wave functions, concept of σ, σ^*, π, π^* orbitals and their characteristics. Hybrid orbitals – sp, sp^2, sp^3; calculation of coefficients of A.O.'s used in these hybrid orbitals.

Introduction to valence bond model of H_2, comparison of M.O. and V.B. models.

UNIT-II 15 Hrs.

Spectroscopy :

Introduction : Electromagnetic radiation, regions of the spectrum, basic features of different spectrometers, statement of the Born-Oppenheimer approximation, degrees of freedom.

Rotational Spectrum :

Diatomic molecules. Energy levels of a rigid rotor (semi – classical principles), selection rules, spectral intensity, determination of bond length, qualitative description of non-rigid rotor, isotope effect.

Vibrational Spectrum:

Infrared Spectrum : Energy levels of simple harmonic oscillator, selection rules, pure vibrational spectrum intensity, determination of force constant and qualitative relation of force constant and bond energies, effect of anharmonic motion and isotope on the spectrum, idea of vibrational frequencies of different functional groups. Raman Spectrum : Concept of polarizability, pure rotational and pure vibrational, Raman spectra of diatomic molecules, selection rules.

Electronic Spectrum :

Concept of potential energy curves for bonding and antibonding molecular orbitals, qualitative description of selection rules and Franck- Condon principle.

Qualitative description of σ, π – and n M.O., their energy levels and the respective transitions.

UNIT-III **15 Hrs.**

Solid State :

Definition of space lattice, unit cell.

Laws of Crystallography – (i) Law of Constancy of Interfacial Angles, (ii) Law of Rationality of Indices, (iii) Law of Symmetry. Symmetry elements in crystals.

X-ray diffraction by crystals. Derivation of Bragg equation. Determination of crystal structure of NaCl, KCl and CsCl (Laue's method and powder method).

UNIT-IV **15 Hrs.**

Photochemistry :

Interaction of radiation with matter, difference between thermal and photochemical processes. Laws of Photochemistry : Grothus – Drapper law, Stark – Einstein law, Jablonski diagram depicting various processes occurring in the excited state, qualitative description of fluorescence, phosphorescence, non-radiative processes (internal conversion, intersystem crossing), quantum yield, photosensitized reactions – energy transfer processes (simple examples). Photochemistry of carbonyl compounds and alkenes.

Instructions for paper setters and candidates :

(i) *Examiner will set total of **Nine** questions comprising **Two** questions from each unit and **One** compulsory question of short answer type covering whole syllabi.*

(ii) *The students are required to attempt **Five** questions in all, **One** question from each unit and the Compulsory question.*

(iii) *All questions carry equal marks.*

Books Suggested :

1. Atkins, P., Paula, J.de, *Atkins, Physical Chemistry;* 8th edition, Pubs: Oxford University Press, 2008.

2. Puri, B.R., Sharma, L.R., Pathania, M.S., *Principles of Physical Chemistry;* 43rd edition, Pubs: Vishal Publishing Co., 2008.

3. Barrow, G.M., *Physical Chemistry;* 6th edition, Pubs: McGraw Hill Company Inc., 1996.

4. Rao, C.N.R., *University General Chemistry;* Pubs: Macmillan of India, 1985.

5. Berry, R.S., Rice, S.A., Ross, J., *Physical Chemistry;* 2nd edition, Pubs: Oxford University Press, 2000.

6. Albert, R.A., Silbey, R.J., *Physical Chemistry;* I edition, Pubs: John Wiley & Sons Inc., 1992.

7. Dogra, S.K., Dogra, S., *Physical Chemistry Through Problems,* Pubs: Wiley Eastern Ltd., 1991.

8. Levine, I.N., *Physical Chemistry;* 5th edition, Pubs : Tata McGraw Hill Publishing Co. Ltd., 2002.

9. Moore, W.J., *Basic Physical Chemistry;* Pubs : Prentice Hall of India Pvt. Ltd., 1983.

10. Metz, C.R., Theory and Problems of Physical Chemistry; Schaum's outline series, 2nd edition, Pubs: McGraw-Hall Book Company, 1989.

11. Banwell, C.N., McCash, E.M., *Fundamentals of Molecular Spectroscopy;* 4th edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 1999.

12. Atkins, P. Friedman, R., *Molecular Quantum Mechanics;* 4th edition Pubs: Oxford University Press, 2007.

13. Levine, I.N., *Quantum Chemistry;* 5th edition, Pubs: Prentice Hall International Inc., 2000.

Paper–XII : LABORATORY PRACTICALS **Max. Marks : 45+5**

INORGANIC CHEMISTRY

Synthesis and Analysis: **6 Periods/ week**

(a) Preparation of sodium trioxalatoferrate (III), Na_3 $[Fe(C_2O_4)_3]$ and determination of its composition by permaganometry.

(b) Preparation of Ni – DMG complex, $[Ni (DMG)_2]$.

(c) Preparation of copper tetraammine complex $[Cu (NH_3)_4] SO_4$.

(d) Preparation of cis-and trans-bisoxalatodiaqua chromate (III) ion.

Instrumentation

Solvent Extraction

Separation and estimation of Mg(II) and Fe(II).

ORGANIC CHEMISTRY

Laboratory Techniques

Column Chromatography

Separation of fluorescein and methylene blue.

Separation of leaf pigments from spinach leaves.

SYNTHESIS OF ORGANIC COMPOUNDS

(a) Aliphatic electrophilic substitution.

Preparation of iodoform from ethanol and acetone.

(b) Aromatic electrophilic substitution.

Nitration

Preparation of m-dinitrobenzene

Preparation of p-nitroacetanilide

Preparation of p-iodoaniline from aniline.

Preparation of methyl orange from N, N-dimethyl aniline and sulphanilic acid.

Halogenation

Preparation of p-bromoacetanilide

Preparation of 2,4,6 – tribromophenol

(c) Oxidation

Preparation of benzoic acid from toluene.

(d) Reduction

Preparation of aniline from nitrobenzene

Preparation of m-nitroaniline from m – dinitrobenzene

Stereochemical study of Organic Compounds via Models

R and S configuration of optical isomers.

E, Z configuration of geometrical isomers

Conformational analysis of cyclohexanes and substituted cyclohexanes.

PHYSICAL CHEMISTRY

Electrochemistry :

(a) To determine the strength of the given acid conductometrically using standard alkali solution.

(b) To determine the solubility and solubility product of a sparingly soluble electrolyte conductometrically.

(c) To study the saponification of ethyl acetate conductometrically.

(d) To determine the ionization constant of a weak acid conductometrically.

(e) To study the distribution of iodine between water and CCl_4.

(f) To study the distribution of benzoic acid between benzene and water.

Molecular Weight Determination :

(a) Determination of molecular weight of a non–volatile solute by Rast method.

(b) Determination of the apparent degree of dissociation of an electrolyte (e.g. NaCl) in aqueous solution of the substance.

General Instructions to the Examiners :

Note : Practical examination will be of four hours duration & shall consist of the following questions:

Q.No. 1*. Preparation of an inorganic complex : 10 marks

Q.No. 2*. Preparation of an organic compound : 10 marks

Q.No. 3. Physical Chemistry : 11 marks

Students shall be allowed the choice to opt for one experiment out of the three offered. The candidate will write theory, short procedure and calculations of that experiments in the next 10 minutes. Note–Book/Books is/are not allowed during writing.

Q.No. 4. Viva-Voce : 8 marks

Minimum of four questions (2 marks each) be asked on the background of practical course.

Q.No. 5. Note Book : 6 marks

*If a question on preparation is asked, then the students shall be required to give Equation, requirements & short procedure in the first 10 minutes. Note Books are not allowed during writing.

Books Suggested (Laboratory Courses) :

1. Denny, R.C., *Vogel's Quantitative Inorganic Analysis;* 4th edition, Pubs: English Language Book Society, 1985.

2. Harwoor, L.M., *Moody, J., Experimental Organic Chemistry;* 1st edition, Pubs: Blackwell Scientific Publications, 1989.

3. Palmer, W.G., Jamer, C., Swinehart, S., *Experimental Inorganic Chemistry;* 1st edition, Pubs: Perlkin-Elmer Corporation, 1969.

4. Forniss, B.S., Rogers, V., *Vogel's Text Book of Practical Organic Chemistry;* Pubs: Dorling Kindhsky Pvt. Ltd., 1989.

5. Garland, C.W., *Experiments in Physical Chemistry;* 1st edition, Pubs: McGraw Hill Book Company, 1989.

6. Bansal, R.K., *Laboratory Manual of Organic Chemistry;* 3rd edition, Pubs: Wiley Eastern Limited, 1994.

7. Furniss, B.S., Hannaford, A.J., Rogers, V., Smith, P.W.G., Tatchell, A.R., *Vogel's Text Book of Practical Organic Chemistry;* 4th edition, Pubs: Longman group, 1978.

8. Khosla, B.D., Garg, V.C., Gulati, A., *Senior Practical Physical Chemistry;* 11th edition, Pubs: R.Chand & Co., New Delhi, 2002.

9. Das, R.C., Behra, B., *Experimental Physical Chemistry;* Pubs: Tata McGraw Hill Publishing Co. Ltd., 1983.

10. Levitt, B.P., *Findlays Practical Physical Chemistry;* 8th edition, Pubs: Longman group Ltd., London & New York, 1978.

BOTANY

B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

		Time	*Theory*	*Int. Assess.*	*Max. Marks*
Theory Paper-A:	Plant Physiology, Biochemistry and Biotechnology	3 hrs.	68	7	75
Theory Paper-B:	Ecology and Utilization of Plants	3 hrs.	68	7	75
One practical pertaining to entire syllabus included in both theory papers		4 hrs.	45	5	50
				Total :	**200 marks**

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each papers A & B. Queston No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in papers A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

Paper-A : PLANT PHYSIOLOGY, BIOCHEMISTRY AND BIOTECHNOLOGY

Max. Marks : 75
Theory : 68
Int. Assessment : 07

Objective : The basic aim of this paper is to familiarize the students with various concepts of functions and metabolism of plants. The course material of this paper would enable the students to correlate structural diversity of various plant forms with functional differentiation. Study of this paper would help the students in understanding several other applied disciplines of Botany like Agriculture, Plant breeding, Agronomy, Tissue culture, Biotechnology, Horticulture etc.

Teaching Methodology : Teaching methodology includes series of lectures making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc.

UNIT-I

1. *Solutions and Colloids :* True solutions, electrolytes and non-electrolytes; Colloidal solutions and colloids, types of colloids, characteristics, gels and emulsions.

2. *Plant Water Relations :* Importance of water to plant life; physical properties of water; imbibition, diffusion, osmosis, plasmolysis and deplasmolysis, concept of osmotic potential, water potential and pressure potential; absorption of water, active and passive mechanism of water absorption; transport of water, mechanism and theories to explain ascent of sap; transpiration types, mechanism of opening and closing of stomata, mechanism of transpiration, factors affecting transpiration, antitranspirants.

3. *Mineral Nutrition :* Essential macro and micro elements and their role; mineral uptake; mechanism of mineral uptake.

UNIT-II

1. *Nitrogen and Lipid Metabolism :* Biological nitrogen fixation; Importance of nitrate reductase and its regulation; ammonia assimilation; structure and function of lipids; fatty acid biosynthesis; β–oxidation; saturated and unsaturated fatty acids; storage and mobilization of fatty acids.

2. *Proteins :* Classification, role and structure (primary, secondary and tertiary) synthesis of amino acids.

3. *Basics of Enzymology :* Discovery and nomenclature; classification, structure, properties, factors affecting enzyme activity, mechanism of enzyme action.

UNIT-III

1. *Photosynthesis :* Significance, historical aspect; photosynthetic pigments; action spectra and enhancement effects; concept of two photosystems, cyclic and non-cyclic photophosphorylation; Calvin cycle; C_4 pathway; CAM plants; photorespiration; factors affecting photosynthesis; Transport of organic substances : Mechanism of phloem transport, source-sink relationship, factors affecting translocation.

2. *Respiration :* ATP—The biological energy currency; aerobic and anaerobic respiration; Kreb's cycle; electron transport mechanism (Chemi-osmotic theory); redox potential; oxidative phosphorylation; pentose phosphate pathway; Respiratory quotient.

UNIT-IV

1. *Growth and Development :* Definitions; phases of growth and development; kinetics of growth, factors affecting growth; seed dormancy, seed germination and factors of their regulation; plant movements; the concept of photoperiodism; physiology of flowering; florigen concept; physiology of senescence, fruit ripening; plant hormones—auxins, gibberellins, cytokinins, abscisic acid and ethylene, history of their discovery, biosynthesis and mechanism of action; photomorphogenesis; phytochrome and cryptochrome, their discovery, physiological role and mechanism of action.

2. *Biotechnology :* Functional definition; basic aspects of plant tissue culture; cellular totipotency, differentiation and morphogenesis.

Suggested Readings :

1. Bhatia, K.N. : *Plant Physiology - A Modern Treatise,* Trueman Book Co., Jalandhar, 2009.
2. Bhojwani, S.S. : *Plant Tissue Culture : Applications and Limitations,* Elsevier Science Publishers, New York, USA, 1990.
3. Dennis, D.T., Turpin, D.H. Lefebvre, D.D. and Layzell (eds.) : *Plant Metabolism* (2nd Edition), Longman, Essex, England, 1997.
4. Galston, A.W. : *Life Processes in Plants,* Scientific American Library, Springer-Verlag, New York, U.S.A., 1989.
5. Hopkins, W.G. : *Introduction to Plant Physiology,* John Wiley & Sons, Inc., New York, U.S.A., 1995.
6. Lea, P.J. and Leegood, R.C. : *Plant Biochemistry and Molecular Biology*, John Wiley and Sons, Chichester, England, 1999.
7. Mohr, H. and Schopfer, P. : Plant Physiology, Springer-Verlag, Berlin, Germany, 1995.
8. Raghavan, V. : *Embryogenesis in Angiosperms : A Developmental and Experimental Study,* Cambridge University Press, New York, USA, 1986.
9. Salisbury, F.B. and Ross, C.W. : *Plant Physiology* (4th Edition), Wadsworth Publishing Co., California, USA, 1992.
10. Srivastava, H.N. : *Plant Physiology, Biochemistry & Bio-technology,* Pradeep Publication, Jalandhar, 2008.
11. Srivastava, H.S. : *Plant Physiology, Bio-chemistry & Bio-technology,* Rastogi Publications, Meerut, 2008.
12. Taiz, L. and Zeiger, E. : *Plant Physiology* (2nd Edition), Sinauer Associates, Inc., Publishers, Massachusetts, USA, 1988.
13. Vasil, I. K. and Thorpe, T. A. : *Plant Cell and Tissue Culture,* Kluwer Academic Publishers, The Netherlands, 1994.

Paper B : ECOLOGY AND UTILIZATION OF PLANTS

Max. Marks	:	75
Theory	:	68
Int. Assessment	:	07

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each papers A & B. Question No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in papers A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

Objective :

The basic objective of this paper is to make students aware about the role of environment in causing structural and functional variation in plants. Since the present day problems of varied nature like pollution, Global Warming etc. are directly or indirectly related to ecology, it is more than desired to provide the students with knowledge of basic concepts of ecology. The second part of this paper is aimed to give an insight into plant wealth such as medicinal plants; crop plants; beverages; spices; condiments; sugar, fiber pulp & oil yielding plants of commercial & economic importance. Both the aspects of this paper give a sound basis of ecology and economic botany so that students can venture into fields like Environmental Biology, Conservation Biology, Forestry, Agriculture, Horticulture, Crop production etc.

Teaching Methodology : Teaching methodology includes series of lectures making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc.

SECTION-I : ECOLOGY

UNIT-I

1. Definition, scope, relationship with other sciences.

2. *Plant Environment :* Climatic, edaphic, topographic and biotic factors affecting growth and distribution of plants.

3. *Ecosystem :* Concept, structure; abiotic and biotic components; trophic levels, food chain, food web, ecological pyramids, energy flow, biogeochemical cycles of carbon, nitrogen and water.

UNIT-II

1. *Community Ecology :* Community characteristics, frequency, density cover, life forms, biological spectrum; ecological succession – Hydrosere and Xerosere.

2. *Applied Ecology :*
   (a) Air, water and soil pollution and their control.
   (b) Conservation and management of natural resources (renewable and non-renewable)

SECTION-II : UTILIZATION OF PLANTS

UNIT-III

1. *Crop Production :*
   Area of cultivation, soil requirement, cultivation practices and high yielding varieties of :
   (i) Cereals (Wheat, Rice and Maize).
   (ii) Fibres (Cotton).
   (iii) Vegetables (Potato).
   (iv) Fruits (Mango, Grapes, Lemon).
   (v) Sugar-yielding plants (Sugarcane).
   (vi) Oil-yielding plants (Groundnut, Mustard).

2. Brief introduction on genetically modified crops.

UNIT-IV

1. *Elementary Knowledge of the following plants (Botanical names, families, part used and economic importance) :*
   (i) Wheat, Maize, Rice, Moong, Gram (Food).
   (ii) Teak, Shisham, Deodar, Sal (Timbers).
   (iii) Cotton, Jute, Coir, Flax (Fibres).
   (iv) Fennel, Coriander, Turmeric, Ginger, Mint, Clove (Spices and Condiments).
   (v) Bamboo, Eucalyptus (Pulp plants).
   (vi) Liquorice, Belladona, Aconite, Ashwagandha, Arjun, Poppy, Amla (Medicinal plants).
   (vii) Tea and Coffee (Beverages).

2. *Forestry :* Forest conservation, wood seasoning and its preservation.

Suggested Readings :

1. Kochhar, S.L. : *Economic Botany in Tropics,* 2nd Edition, Macmillan India Ltd., New Delhi, 1998.
2. Kormondy, E.J. : *Concepts of Ecology,* Prentice-Hall of India Pvt. Ltd., New Delhi, 1996.
3. Mackenzie, A. *et al.* : *Instant Notes in Ecology,* Viva Books Pvt. Ltd., New Delhi, 1999.
4. Odum, E.P. : *Basic Ecology,* Saunders, Philadelphia, 1983.
5. Sambamurthy, A.V.S.S. and Subramanyam, N.S. : *A Textbook of Economic Botany,* Wiley Eastern Ltd., New Delhi, 1989.
6. Sharma, O.P. : *Hill's Economic Botany* (Late Dr. A.F. Hill, Adapted by O.P. Sharma), Tata McGraw Hill Co. Ltd., New Delhi, 1996.
7. Simpson, B.B. and Conner-Oghorzaly, M. : *Economic Botany – Plants in Our World,* McGraw Hill, New York, 1986.

Suggested Laboratory Exercises :

Plant Physiology :

1. To determine osmotic pressure of cell sap by plasmolytic method.
2. To demonstrate imbibition pressure using :
   (i) Imbibition pressure apparatus.
   (ii) Plaster of Paris cone.
3. To demonstrate osmosis through animal membrane/potato osmoscope.
4. To demonstrate plasmolysis and deplasmolysis.
5. To demonstrate mechanical and electrical adsorption.
6. To demonstrate the measurement of transpiration using simple potometer.
7. To demonstrate transpiration pull.
8. To study the effect of light intensity and wind velocity on the rate of transpiration using Ganong's potometer.

9. To compare the rate of transpiration from the two sides of a leaf using :

   (i) Vaseline method.

   (ii) Cobalt chloride method.

10. To demonstrate the mechanism of opening and closing of stomata.

11. To demonstrate the path of ascent of sap.

12. To demonstrate that chlorophyll is necessary for photosynthesis.

13. To demonstrate that light is necessary for photosynthesis.

14. To demonstrate that CO_2 is essential for photosynthesis.

15. To demonstrate evolution of oxygen during photosynthesis in an aquatic plant.

16. To study the effect of light intensity and CO_2 concentration on the rate of photosynthesis using an aquatic plant.

17. To demonstrate aerobic respiration using flask method.

18. To demonstrate anaerobic respiration in germinating seeds or yeast.

19. To demonstrate the measurement of respiratory quotient using Ganong's respirometer.

20. To demonstrate the activity of amylase.

21. To demonstrate the activity of catalase in plant tissue (germinating seeds).

22. To demonstrate phototropism.

23. To demonstrate geotropism using clinostat.

24. To separate chlorophyll pigments by solvent method.

25. To perform chemical analysis of plant ash for K^{+1}, Ca^{2+}, Fe^{++}, Fe^{+++}, B^{3+}, Mn^{2+}, S^{2-} and Mg^{2+}.

26. To test for the presence of starch, proteins, amino acids and reducing sugars in plant material.

Ecology and Utilization of Plants :

1. Study of ecological adaptations in external characters of :
   Hydrilla, Potamogeton, Ceratophyllum, Vallisneria, Lemna, Eichhornia, Nelumbium, Calotropis, Nerium, Acacia, Zizyphus, Casuarina, Capparis, Asparagus, Ruscus, Opuntia, Euphorbia royleana.

2. To prepare permanent stained slide to show ecological adaptations in the internal structure of the following :

   (a) T.S. stem of *Hydrilla*.

   (b) T.S. leaf of *Potamogeton* and *Vallisneria*.

   (c) T.S. leaf and petiole of *Eichhornia*.

   (d) T.S. leaf and petiole of *Nelumbium*.

   (e) T.S. leaf of *Nerium.*

   (f) T.S. stem of *Casuarina* and *Capparis*.

3. Identification and morphology of economically important part/s of crop plants mentioned below : Cereals (wheat, rice); Fibres (cotton); Vegetables (potato); Fruits (mango, grapes, lemon); sugar-yielding plants (sugarcane) and oil-yielding plants (groundnut, mustard).

4. To determine soil pH using pH paper/solution/pH meter.

5. To determine water holding capacity of soil.

6. To assess soil texture through ball-making technique.

Guidelines for Botany Practical Examination :

Max. Marks	:	50
Practical	:	45 marks
Internal Assessment	:	05 marks
Time	:	4 Hours

1. Write material required, procedure and precautions for physiology experiment A (to be announced by the examiner). Perform the experiment, record observations, calculations if any, and results and show the experiment to the examiner. 11

2. Write procedure of physiology experiment B (minor experiment to be announced by the examiner). Perform the experiment and show it to the examiner after recording observations. 5

3.	(a)	Identify and write illustrated ecological note pertaining to external characters of specimen C.	4
	(b)	Identify and cut T.S. of stem, leaf or petiole (to be announced by the examiner) of specimen D. Make its permanent stained slide and show it to the examiner. Draw its labelled diagram and write its anatomical characters of ecological importance.	7
4.		Identify and write illustrated morphological notes on specimens E and F giving their economic importance.	8
5.		Practical Note-Book.	5
6.		*Viva-Voce.*	5

ZOOLOGY

B.Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Paper-A	**:**	Developmental Biology & Genetics	75 (67 + 8) marks
Paper-B	**:**	Applied Zoology	75 (67 + 8) marks
Practical	**:**	One paper (covering the entire syllabus of both the papers) of 4 hours duration	50 (45 + 5) marks

Note : The number of hours for Theory and Practical per week shall be 5 hours and 4 hours respectively.

OBJECTIVES OF THE COURSE :

The syllabus pertaining to B.Sc. (General) Part-III, in the subject of Zoology has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Zoology working in the Panjab University, Chandigarh and affiliated colleges.

The syllabus contents are duly arranged section wise as well as unit wise. The contents are included in such manner so that due importance may be given to skill oriented components.

The course contents are also given due stress for excursion/field trips to Zoological Parks, Sea-shores, Hill Stations, Museum, Fossil Park and Apiary/godowns for better academic outlook. The Department of Zoology, P.U., Chandigarh usually organizes workshop/seminars from time to time for updating the teachers.

PAPER-A : DEVELOPMENTAL BIOLOGY AND GENETICS (ZOO 301)

Max. Marks	:	75
Theory	:	67 marks
Internal Assessment	:	8 marks
Time	:	3 Hours

Note : Nine questions are to be set. Question No. 1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

UNIT-I

Gametogenesis with particular reference to differentiation of spermatozoa:
Vitellogenesis; role of follicle/subtesticular cells in gametogenesis.

Egg maturation; egg membranes : polarity of egg.

Fertilization; parthenogenesis.

Fate maps of chick and frog embryos.

UNIT-II

Cleavage patterns; Cleavage; determination and differentiation, development upto three germ layers in *Herdmania, Amphioxus*, frog, chick and rabbit.

Foetal membranes, their formation and role. Mammalian placenta – Its formation, types and functions.

Tissue interactions, basic concepts of organizers and inductors and their role.

Metamorphosis in *Herdmania* and *Rana* (frog).

UNIT-III

Modification of Mendelian ratios :

Non-allelic gene interaction, Modified F_2 ratios.

(9:7, 9:3:4, 9:3:3:1, 12:3:1, 13:3,15:1, 9:6:1).

Gene modifications due to incomplete dominance, lethal factors (2:1), Pleiotropic genes.

Multiple Alleles : Blood group inheritance, eye colour in *Drosophila*, pseudo-allelism.

Multiple Factors : Qualitative and quantitative characters, inheritance of quantitative traits (skin colour in man).

Linkage, crossing over and recombination : Linkage, sex-linked characters, crossing over, frequency of crossing over, cytological basis of crossing over, synaptonemal complex. Recombination in Fungi (tetrad analysis).

Gene and Genetic Code : Structure of nucleic acids (DNA & RNA). Replication of DNA and transcription. Expression of gene (protein synthesis in Prokaryotes and Eukaryotes). Properties of genetic code, codon assignment, wobble hypothesis.

Extranuclear inheritance : Kappa particles in *Paramoecium*.

UNIT-IV

Mutations : Spontaneous and induced mutations, physical and chemical mutagens.
Detection of mutations in Maize and *Drosophila*. Inborn errors of metabolism in man (Phenylketonuria, Alcaptonuria, Albinism). Somatic mutations and carcinogenesis.

Regulation of gene expression in prokaryotes (Operon model) and in eukaryotes.

Population genetics : Equilibrium of gene frequencies and Hardy Weinberg Law.

Genetic recombination in bacteria (conjugation, transduction and transformation) plasmids.

Applied Genetics : Recombinant DNA, genetic cloning and its applications in medicine and agriculture, DNA finger printing.

Note : Nine questions are to be set. Question No. 1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

PRACTICALS : Practical based on Theory Paper ZOO 301 (ZOO 351)

A. Genetics :

1. Demonstration of Law of segregation, Independent assortment and epistasis (use of coloured beads or capsules etc.). Numericals for segregation and Independent assortment.
2. Segregation demonstration in preserved material (Maize).
3. Cytoplasmic inheritance in *Mirabilis jalapa*.
4. Inheritance of other human characteristics, ability to taste PTC, thiourea.
5. Comparison of variance in respect of pod length and number of seeds/pods.
6. Calculation of gene frequencies and random mating (coloured beads or capsules).
7. Study of polytene chromosomes of *Chironomus*/*Drosophila* through permanent slide.
8. Dermatographics : Palm print taking and finger tip patterns.

B. Frog Embryology :

1. Collection of spawn.
2. Identification of stages and preservation.
3. Preparation of permanent/temporary slide of representative developmental stages of frog.
4. Study of the development of frog from permanent slides.
5. Window preparation and identification of stages of development in chick egg.
6. Study of the development of chick embryo from permanent slides upto 96 hours.
7. Study of the following prepared slides :
   (a) Stages of gametogenesis, structure of egg and sperm of a mammal.
   (b) Larva of *Herdmania*.

Books Recommended (Latest Edition)

1. De Robertis, E.D.P. and De Robertis, E.M.F. : *Cell and Molecular Biology,* W.B. Saunders Co., Philadelphia, 1995.
2. Powar, C.B. : *Cell Biology,* Himalaya Publishing House, Bombay, 1999.

3. Swanson, C.P., Merz, T. and Young, W.J. : *Cytogenetics – The Chromosome in Division, Inheritance and Evolution,* Prentice Hall, 1981.

4. Gupta, P.K. : *Cytology, Genetics and Molecular Biology,* Rastogi Publishers, Meerut, 1993.

5. Chaudhry, S. and Sharma, A. : *PV's Cytology & Ecology,* S. Vikas & Co., Jallandhar, 2003.

6. Ballinsky, B.I. : *An Introduction to Embryology*, 5th Edition, Saunders Co., 1981.

7. Gilbert, S.F. : *Developmental Biology,* Sinauer Ass. Inc. Publ., 2000.

8. Urspaung, H. : *Major Problems in Developmental Biology,* Academic Press, New York, 1966.

9. Gardner, E.J., Simmons, M.J. and Snustad, D.P. : *Principles of Genetics,* 8th ed., John Wiley and Sons, Inc., New York, 2004.

Paper-B : APPLIED ZOOLOGY

Max. Marks	:	75
Theory	:	67 marks
Internal Assessment	:	8 marks
Time	:	3 Hours

Note : *Students are required to opt **any** one of the following :*

1. Medical Zoology and Medical Laboratory Technology.
2. Economic Entomology and Pest Management.
3. Inland Fisheries & Aquaculture.

OPTION-I : MEDICAL ZOOLOGY & MEDICAL LABORATORY TECHNOLOGY (ZOO 302)

UNIT-I

Introduction to parasitology (pertaining to various terminologies in use).

Brief introduction to pathogenic microbes. Viruses, Rickettsiae, Spirochaetes and Bacteria.

Brief accounts of life history, mode of infection and pathogenicity of the following pathogens with reference to man; prophylaxis and treatment:

(a) Pathogenic protozoans : *Entamoeba, Trypanosoma, Leishmania, Giardia, Trichomonas;* and *Plasmodium.*

(b) Pathogenic helminthes: *Fasciolopsis, Schistosoma, Echinococcus, Ancylostoma, Trichinella, Wuchereria, Dracunculus* and *Oxyuris*.

Life cycle, disease caused and control measures of arthropod vectors :

Anopheles stephensi, Anopheles culicifaces, Aedes aegypti, Aedes albopictus, *Culex pipiens, Culex tritaeniorhynchus*, *Mansonia* sp., *Xenopsylla, Cheopis, Pediculus*.

UNIT-II

Epidemic diseases, such as typhoid, cholera, small pox; their occurrence and eradication programmes.

Brief introduction to human defence mechanisms.

Humoral and cell mediated immune - response, Antigens - physical & chemical properties, Antibodies - structure and function of immunoglobulin M, G, A, E and D.

Antigen and antibody interactions : Serodiagnostic assays.

Vaccines.

UNIT-III

Laboratory safety rules, hazards and precautions during sample collection and laboratory investigations.

Laboratory techniques : Colorimetry, Microscopy (student's and electron microscope), Autoclaving. Centrifugation, Spectrophotometry.

Haematology : Collection of blood (Venous and Capillary), Anticoagulants (merits and demerits), Romanowsky's stains, Total RBC count, Erythrocyte sedimentation rate, TLC, DLC, Eosinophil count, Platelet Count, Reticulocyte count.

UNIT-IV

Bacteriology : Sterilisation (dry heat, moist heat, autoclave, filtration), Disinfection, Staining techniques (gram's stain, AFB stain, etc.), Culture media (Defined & Synthetic media & routine laboratory media), Bacterial culture (aerobic and anaerobic), Antibiotic sensitivity.

Biochemistry : Protein estimation, Estimation of blood urea, blood sugar and urine analysis.

Histopathology : Common fixatives and staining techniques, Histochemistry : Principle and method : Staining of carbohydrates, proteins and fats with bromo phenol blue, Periodic acid Schiff, Sudan Black blue and Feulgen reaction.

Note : Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each Unit. One question is to be attempted from each Unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

Practical based on Theory Paper ZOO 302 (ZOO 352)

1. Demonstration of safety rules in laboratory like proper handling of patients specimens and disposal of syringes, needles etc.

2. Demonstration of the use of autoclave, centrifuge and spectrophotometer.

3. Cleaning and sterilization of glassware using hot air oven, autoclave etc.

4. Demonstration of parts of microscope, its functioning and care.

5. Processing of clinical samples for culture and identification of pathogens : blood, urine and stool.

6. Estimation of haemoglobin using Sahli's haemometer.

7. Preparation of thick and thin film for malarial parasite.

8. Counting of WBC, RBC & DLC.

9. Examination of stool for demonstration of intestinal parasites.

10. Study of permanent slides and specimens of parasitic protozoans, helminthes and arthropods mentioned in theory syllabus.

11. Analysis of blood group, A, B, AB, O and Rh.

12. ESR, haematocrit, bleeding time, coagulation time, prothrombin time.

13. Estimation of blood sugar and protein.

14. Fixation, embedding, cutting of tissue sections and their staining (Routine Haematoxytin and Eosin and special staining with BPB, PAS, SBB and Feulgen reaction).

Suggested Readings :

1. Baker, F.J., and Silverton, R.E. : *Introduction to Medical Laboratory Technology,* 6th edition, Butterworth and Co. Ltd., 1985.

2. Cheesborough, M. : *Medical Laboratory Technology for Tropical Countries,* 2nd edition, Butterworth and Co. Ltd., 1987.

3. Talib, V.H. : *Essential Laboratory Manual,* Mehta Publishers, New Delhi, 1999.

4.	Goldsby, R.A., Kindt, T.J., Osborne, B.A. and Kuby, J.	:	*Immunology,* 6th edition, Freeman, W.H. & Co., New York, U.S.A., 2007.
5.	Chatterjee, K.D.	:	*Parasitology, Protozoology and Helminthology,* 13th ed., CSB Publishers and Distributors Pvt. Ltd., 2009.
6.	Garcia, L.S.	:	*Diagnostic Medical Parasitology,* 4th ed., ASM Press Washington, 2001.
7.	Parija, S.C.	:	*Text Book of Medical Parasitology, Protozoology* and *Helminthology,* 1st edition, 2001.

OPTION-II : ECONOMIC ENTOMOLOGY & PEST MANAGEMENT (ZOO 303)

UNIT-I

Systematic position, habits and nature of damage of the following pests of crops and vegetables :

I. Sugarcane :

1. Sugarcane leaf hopper (*Pyrila perpusila*) alongwith life cycle and control measures.
2. Sugarcane top borer (*Scirpophaga nivella*)
3. Sugarcane stem borer (*Chilotrea infuscatellus*)

II. Cotton :

1. Pink bollworm (*Pectinophora gossypiella*) alongwith life cycle and control measures.
2. Red cotton bug (*Dysdercus cinglulatus*)
3. Cotton grey weevil (*Myllocerus maculosus*)
4. Surface grasshopper (*Chrotogonus trachypterus*)
5. Cotton jassid (*Empoasca devastans*)

III. Paddy :

1. Rice Gundhy Bug (*Leptocorisa varicornis*) alongwith life cycle and control measures.
2. Rice grasshopper (*Hieroglyphyus banian*)
3. Rice Hispa (*Dicladispa armigera*)

IV. Wheat :

1. Wheat stem borer (*Sesamia inferens*) alongwith life cycle and control measures.
2. Termites *(Microtermes obesi)*
3. Aphids, *(Macrosiphum miscanthi)* Jassids *(Amrasca sp.)*

V. Vegetables :

1. Red pumpkin beetle (*Aulacophora foveicollis*)
2. Pumpkin fruit fly (*Dacus cucurbitae*) alongwith life cycle and control measures.
3. Hadda beetle (*Epilachna vigintioctopunctata*)

UNIT-II

VI. Pests of Stored Grains : Systematic position, habits and nature of damage of the following pests of stored grains :

1. Pulse Beetle (*Callosobruchus maculates*) along with life cycle and control.
2. Rice weevil (*Sitophilus oryzae*)
3. Khapra beetle (*Trogoderma granarium*)
4. Rust red flour beetle (*Tribolium castaneum*)
5. Lesser grain borer (*Rhizopertha dominica*)
6. Rice moth (*Corcyra cephalonica*)

Systematic position, disease caused and control of the following insects of Medical and Veterinary importance :

1. Mosquitoes *(Aedes, Anopheles, Culex)*
2. Sand fly (*Phlebotomus minutus*)
3. House fly (*Musca domestica*) along with life cycle of house fly.
4. Horse fly (*Tabanus striatus*)
5. Blow fly (*Calliphora erythrocephala*)
6. Warble fly (*Hypoderma lineatum*)
7. Poultry louse (*Menopon gallinae*)
8. Sucking louse/(*Haematopinus eurysternus*)
9. Fleas *(Xenopsylla cheopis)*

UNIT-III

Development of Insects : Different types of metamorphosis along with a study of different kinds of larvae and pupae.

Comparative studies of mouth parts in Grasshopper, Honeybee, Butterfly, Red-Cotton bug and Mosquito.

Major modifications in the antennae and legs of insects.

1. Sericulture

1. Species of silkworm
2. Requirements of Sericulture Industry
3. Grainage Management
4. Pre and Post-cocoon processing (Stifling & Reeling)
5. Diseases of silkworm.

2. Apiculture

(i) Species of Honeybees

(ii) Flora for Apiculture

(iii) Methods & Appliances of Bee Keeping

(iv) Products - (a) Honey (b) Bee wax (c) Propolis (d) Pollen (e) Royal Jelly (f) Bee Venom

(v) Diseases of Honey bee

3. Lac Culture :

(i) Species and Varieties of Lac insect

(ii) Host Plants

(iii) Cultivation of Lac

(iv) Processing of Lac

(v) Lac Industry

(vi) Enemies of Lac insect.

UNIT-IV

I. Chemical Control :

Types and Classification of Insecticides (a) Insecticides of plant origin with special reference to Nicotine; Pyrethrum; Rotenone and Azadirachtin (b) Chlorinated Hydrocarbons insecticides with special reference to DDT; Toxaphene; BNC; Chlordane; Aldrin; Endrin and Endosulfan (c) Organophosphorus Insecticides with special reference to Malathion; TEPP; Parathion and DDVP (d) Carbamate Insecticides with reference to Carbaryl and Carbofuran (e) Fumigants with reference to Hydrogen cyanide; Methyl bromide; Ethylene dichloride; Carbon tetrachloride and Aluminium phosphide. Hazards of chemical control. List of banned pesticides.

II. Recent Methods of Pest Control :

Biological Control : History; Techniques in biological control, Agents of biological control (a) Vertebrates (b) Nemathelminthes (c) Arthropods (d) Protozoan; Microbial control with the help of Bacteria, Virus and Fungi.

Integrated Pest Control : Introduction of IPM : Pre-requisites; Implementation Strategy; Framework of IPM programme and perspectives in IPM.

Note : Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each unit. One question is to be attempted from each Unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

Practical based on Theory Paper ZOO 303 (ZOO 353)

1. Feeding apparatus : Mouth parts of honey bee, butterfly and red cotton bug by preparing permanent mounts.
2. Study of different types of antennae, legs and wings through specimens and slides.
3. A study of different types of larvae and pupae of insects.
4. External morphology and identification marks of the crops and vegetables pests : *Pyrilla perpusilla* (Sugarcane leaf hopper), *Pectinophora gossypiella* (Pink bollworm), *Leptocorisa varicornis* (Gundhi bug) *Hieroglyphus banian* (Paddy grass hopper), *Dacus cucurbitae* (Pumpkin fruit fly).
5. External morphology and identification marks of the following stored grain pests : *Sitophilus oryzae* (Rice weevil), *Tribolium castaneum* (Rust red flour beetle), *Rhizopertha dominica* (Lesser grain borer/susri), *Trogoderma granarium* (Khapra beetle), *Callosobruchus maculatus* (Pulse beetle/Dhora).
6. External morphology and identification marks of the following insects of Medical/Veterinary importance - Mosquitoes *(Culex, Anopheles* and *Aedes),* house fly (*Musca domestica),* blow fly *(Calliphora erythrocephala),* warble fly, *(Hypoderma lineatum)* and horse fly *(Tabanus striatus)*.
7. A study of life stages of silk worm and honey bees.
8. Collection of insects representing different orders; storage and preservation of insect material.
9. Structure and working of common sprayers : Hand Compression sprayer, Knap sack sprayer.
10. Visit to apiary and godowns for study of infestation and project report.

Books Recommended

1. Attwal, A.S. : *Agricultural Pests of India and South East Asia,* Kalyani Publishers, New Delhi, 1991.
2. Nair, M.R.G.K. : *Insects and Mites of Crops in India,* ICAR, New Delhi, 1975.
3. Kumar, A. & Nigam, P.M. : *Economic and Applied Entomology,* Emkay Publications, 1991.
4. Matheson, R. : *Medical Entomology,* Comstock Publishing Company, Inc., 1950.
5. Metcalf , R.L. & Metcalf, R.A. : *Destructive and Useful Insects,* McGraw Hill Book Company, Inc. New York, Toronto, London, 1951.
6. Dent, D. : *Integrated Pest Management,* Chapman & Hall, London, NewYork, Tokyo, Madras, 1995.
7. Howse, P., Stevens, I. and Jones, O. : *Insect Pheromones and their use in Pest Management,* Chapman & Hall, London, New York, Tokyo, Madras, 1998.
8. Mishra, R.C. : *Honey Bees and their Management in India,* ICAR Publication, New Delhi, 1995.

Option-III : AQUACULTURE (ZOO 304)

UNIT-I

Components of Aquaculture: Capture, Culture, Intensive, Extensive, Semi-Intensive, Pen Culture, Cage Culture.

Morphology of a typical fish (Carp, Catfish, Eel, Perch)

Culturable fishes: Characters of culturable fishes, Indian major carps, saltwater fishes, exotic species and air breathing fishes.

Structure of mouth of different fishes in relation to feeding habits.

Bionomics of *Labeo rohita, Catla catla, Cirrhinus mrigala, Wallago attu.*

Food value of Fish (Vitamins, Amino acids, Minerals etc.)

UNIT-II

Exotic Fishes : History, their introduction, morphology, their role in fish culture and impact on native fish fauna.

Induced Breeding : History, technique, chemicals involved in induced breeding and impact on fish culture.

Pond Culture : Construction of pond, types of pond, hydrobiological factors of water and soil of a fish pond, fertilization of pond, maintenance of pond.

Aquatic weeds and their control both biological and chemical.

UNIT-III

Fishing gears: Classification of the fishing gears according to the habitats in which they are being used, Electrical fishing.

Culture Systems : Monoculture and polyculture.

Integrated fish farming : Duck-cum-fish farming, Dairy-cum-fish farming and Poultry-cum-fish farming.

Pearl culture in India: Species used, implantation procedures, water quality requirements and economics.

Fish seed resources and their transport.

UNIT-IV

Prawn culture: Culture of fresh water prawn, culture technology.

Cold water fisheries : Mahseer fisheries and trout fisheries

Fish Diseases and their control : Viral, bacterial, fungal, protozoan, helminth, crustacean, diseases due to unhygienic conditions, diseases during transportation.

Fish by-products

Fish Marketing

Fish preservation : Principles of fish preservation, preservation by curing (drying, salting and smoking), chilling and freezing of fish, canning of fish and fish, products, spoilage of fish, rigor mortis.

Note : Nine questions are to be set. Question No.1 is compulsory consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each unit. One question is to be attempted from each Unit. In all, Five questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

PRACTICALS : Practical based on Theory Paper ZOO 304 (ZOO 354)

1. Morphology of a Carp, Cat fish and Perch.

2. Morphometric and meristic characters of a typical fish.

3. Identification of the following fishes using keys to the species :

   Notopterus spp. *Labeo rohita, L. colbasu, L. bata, Cirrhinus mrigala, Catla catla, Puntius sarana, Tor putitora, Schizothorax, Aorichthys seenghala, Wallago attu, Callichrous padda, Bagarius bagarius, Heterpneustes fossilis, Channa maruluis, C. striatus, Xenentondon cancila, Cyprinus carpio, Hypophthalmichthys molitrix,Ctenopharyngodon idella, Colisa fasciatus, Mastacembelus armatus.*

   For the identification of the fishes, the students can use already prepared keys or can prepare their own keys.

4. Determination of food and feeding habits of locally available fishes on the basis of stomach analysis adopting the following methods :

   a. Frequency occurrence method

   b. Feeding intensity

   c. Point method.

5. Determination of maturity stages (both male and female) of any commercial fish (preserved specimens).

6. Preparation of permanent slides of phytoplanktons and zooplanktons which constitute the food of commercial fishes. Their identification and study of important characters.

7. Identification of aquatic weeds of a fish pond.

8. Estimation of following chemical parameters of the water of a fish pond:

   a. Temperature

   b. pH

   c. Dissolved oxygen

   d. Phosphates

   e. Total dissolved solids

   f. Nitrates

   g. Hardness

   h. Chlorides

9. Study of exotic and endemic culturable fish species.

10. Visits to various fish ponds and fish market.

Suggested Readings

1. Jhingran, V.G. : *Fish and Fisheries of India,* Hindustan Publishing Corporation of India, Delhi, 1991.

2. Day, F. : *Fishes of India,* Vol. I & II, Reprinted Edition Jagmander Book Agency, New Delhi, 1994.

3. Johal, M.S. & Tandon, K.K. : *Monograph on the Fishes of Reorganised Punjab,* Punjab Fisheries Bulletin, Vols. I & II, 1979 - 1980.

4. Agarwal, S.C. & Johal, M.S. : *Fishery Development,* Narendra Publishing House, Delhi, 1997.

5. Johal, M.S. & Tandon, K.K. : *Fishes of Punjab,* Res. Bull., Panjab University, Vol. 32, pp. 143-154, 1981.

6. Karl, F.L. : *Freshwater Fishery Biology,* Wm. C-Brown Company Publ., Dubuque, IOWA, USA, 1969.

7. Brian R. M. & David W. W. (Ed.) : *Fisheries Techniques : American Fisheries Society*, Bethesda Maryland, USA, 1996.

Guidelines for the conduct of Practical Examinations, (ZOO 351)

Max. Marks	:	50
Practical Exam.	:	45 marks
Internal Assessment	:	5 marks
Time	:	4 Hours

1. Demonstrate the law of independent assortment/segregation/epistasis from the material provided. Identify the characters involved showing the dominance/recessiveness of characters. 5

2. Calculate the gene frequency from a known sample of characteristics using Hardy-Weinberg Law.

OR

Make a dermatographic print of your finger tips or palm pattern and classify the various visible pattern with the help of diagrams and demonstrate it to the examiner. 4

3. Make a permanent/temporary preparation and identify the material provided. Write a brief note on it. 4

4. Identify the slides A & B giving two reasons for each identification. 4

5. Viva-Voce 3

6. Practical Note Book and Project Report (3+2) =5

25

Option–I (ZOO 352)

7. To make a permanent stained preparation of blood smear showing different stages of *Plasmodium*/bacteria/sputum. Write briefly about your observation under the microscope. Draw a labeled sketch. (2+1+1) = 4

OR

To test the given sample of urine/stool under the microscope for its pathology. Write the procedure adopted. (2+2) = 4

8. To identify the specimens C & D. Write the disease caused by each & two reasons for their identification. (1+1+1) × 2 = 6

9. To find out the blood groups/erythrocyte sedimentation rate/bleeding time/ coagulation time/prothrombin time of the given sample and write the procedure adopted. 4

OR

To identify the slide F-G and write diagnostic features. 4

10. Determine RBC/WBC counts, DLC of the given blood sample. Write the procedure adopted. 6

OR

Demonstrate Feulgen/PAS reaction in given tissue section. 6

20

Option-II (ZOO 353)

7. To make a permanent preparation of the mouth parts of the given specimen. Make a labelled sketch of the same. 8

8. To identify specimens C, D, & E, belonging to crop pests, stored grain pests and insects of medical importance respectively. Give one outstanding morphological character and one identification mark of each. Mention their scientific names and economic importance also. (2.5 each) = 7.5

9. To mention the type of larva/pupa/stages of life history of silk worm and honeybee. Write a note on its external morphology. 2

10. To name the apparatus provided and explain its structure and working. 2.5

20

Option-III (ZOO 354)

7.	To identify the given sample C & D and write their morphometric and meristic characters. Make labeled sketch of given samples.	8
8.	To determine the food contents of the specimens E and F.	5
9.	To identify slides G & H and write two identifying characters of each.	5
10.	To determine the pH of a given sample of pond water by pH paper.	2
		20

ELECTRONICS

B. Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Note :

1. There will be two theory and one laboratory (practical) courses.
2. The number of lectures per week will be three for each theory paper.
3. The number of lectures per week will be six for practicals.
4. The time duration for each theory paper will be three hours.
5. The time duration for practical examination will be 4 hours.
6. The use of Non-programmable calculators will be allowed in the examination centre, but these will not be provided by the University/College.

Distribution of marks and total teaching hours will be as under :

Paper A :	Data Acquisition Systems	:	(45 + 5*) = 50 marks	Total Teaching Hours	75
Paper B :	Microcontrollers and Applications	:	(45 + 5*) = 50 marks	Total Teaching Hours	75

* Internal assessment

Practicals :	:	50 marks	Total Teaching Hours	125
Project Work :	:	50 marks		

Each paper will consist of five units :

UNIT-I : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry ten marks.

UNIT-II : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry ten marks.

UNIT-III : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry ten marks.

UNIT-IV : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry ten marks.

UNIT-V : There will be eight questions of small answer type covering the syllabi of all the four units (I-IV). Five questions are to be attempted. Each question will carry two marks.

Paper–A : DATA ACQUISITION SYSTEMS :

Objectives :

The objective of Data Acquisition Systems course is to familiarize the students with various kinds of Transducers, instrumentation amplifiers, data acquisition system. Understanding of Power conditioning equipment and Programmable Logic controllers.

UNIT-I

Transducers :

Classification, Electrical Transducer, Selecting a transducer, Resistive transducer, Inductive transducer, Capacitive transducer, Piezo-electric transducer, Photoelectric, Phototransistor and thermoelectric transducer.

UNIT-II

Signal Conditioning :

OP-AMP, Basic Instrumentation Amplifier and its application. Concept of Filtering.

UNIT-III

Data Acquisition Systems :

Objective, Block diagram and functioning of various blocks, types of DAS, Case study (possibly through field visit).

UNIT-IV

Power Conditioning Equipment :

Operation, Block diagram of CVT, UPS, SMPS and Inverter with their applications.

Programmable Logic Controllers :

Introduction, PLC operation, Architecture and applications, Relays Timers, Counters, Sequencers, Ladder diagram programming, Case Study.

References :

1.	Kalsi, H.S.	:	*Electronic Instrumentation.*
2.	Halfrick and Cooper .	:	*Modern Electronic Instrumentation and Measurement Techniques.*
3.	Gottlieb, I.M.	:	*Power Supplies etc.*

Paper-B : MICROCONTROLLERS AND APPLICATIONS

Objectives :

The objective of Microcontrollers and Applications course is to cover various aspects of 8051 hardware and instruction architecture, basic assembly language programming, and applications.

UNIT-I

Microcontroller 8051 Architecture :

Microcontrollers—Evolution and comparison with microprocessor.

Microcontroller 8051 Hardware-Oscillator and Clock, Registers, RAM, ROM, I/O ports; External Memory-Connecting, COUNTERS and TIMERS—Timer Counter Interrupts, Timing, Timer Modes of Operation, Counting.

Transmission Modes :

Serial Data Input/Output : Serial Data Interrupts, Data Transmission, Data Reception, Serial Data.

UNIT-II

Interrupts :

Timer Flag Interrupt, Serial Port Interrupt, External Interrupts, Reset, Interrupt Control, Interrupt Priority, Interrupt destination, Software Generated Interrupts.

Basic Assembly Language Programming :

Why assembly language, Flow Charts, Writing and Testing the Program, Programming 8051—lines of code, 8051 instruction set.

Addressing Modes :

Immediate, Register, Direct, Indirect, External data moves, Code memory—read only data moves; Opcodes—Push, Pop.

UNIT-III

Logical Operations :

Byte-level, Bit-level : Internal RAM, SFR, Boolean operations.

Arithmetic Operations :

Flags, Instructions Affecting, Flags, Incrementing and Decrementing.

Addition—Unsigned and signed, Unsigned, Signed, Multiple-Byte Signed Arithmetic.
Subtraction : Unsigned and signed, Unsigned, Signed; Multiplication and Division, Decimal Arithmetic.

UNIT-IV

Jump and Call Instructions :

JUMP and CALL program range, Relative, Short absolute, Long absolute; Jumps-Bit, Byte, Unconditional; Calls and Subroutines-subroutines, Calls & the stack, Calls and returns.

Applications of Microcontroller 8051 : 7-segment display, Traffic Lights.

References :

1. Ayala, A.J. : *Microcontroller 8051.*
2. Scott MacKenzie : *Microcontroller 8051.*
3. Stewart, J.W. : *Microcontroller 8051.*
4. Barnett, R.H. : *8051 Family of Microcontroller.*

PRACTICALS

Objective :

The objective of PRACTICALS course is to provide practical training to students using Microcontrollers, their interfacing with I/O devices and transducers, Use of PLC's for basic industrial applications. Make a project based on Microcontroller/PLC**.**

Note : Atleast 15 practicals as per availability of apparatus.

1. To interface a toggle switch and an LED with Microcontroller.
2. To generate 1kHz square wave at P 1.1 using timer.
3. To generate I kHz square wave at P 1.1 using timer and 10 kHz at P 1.7 using interrupts.
4. To transmit and receive a character using serial I/O.
5. To copy the contents of register R7 to external RAM location.
6. To interface 4 DIP switches and to drive 7-segment display.
7. To interface a loudspeaker to generate a tone of 440 Hz.
8. To interface a Hex/calculator keypad to microcontroller.

9. To interface a DAC and DAC to microcontroller.
10. To study various active and passive transducers (4 turns).
11. To design an LPF/HPF.
12. To design an LPF using OPAMP.
13. To study load/voltage regulation of a UPS.
14. To study load/voltage regulation of a CVT.
15. To realize AND, OR, NOT, NAND and NOR gates using PLCs.
16. To use single push button to switch ON/OFF of a motor using PLCs.
17. To use a single push button switch ON/OFF of a conveyor and switch off after 20s with PLCs.
18. To realize an astable multivibrator using PLCs.
19. To demonstrate traffic light control using PLCs.
20. To realize a control for an elevator.
21. To realize a paid car parking.
22. To realize the working of a bottling plant.

Proposed Projects (Using Microcontroller/PLCs) :

Note : Any one project for one student.

1. Intrusion Alarm System.
2. Furnace Temperature Controller.
3. Traffic Light Controller.
4. Washing Machine Controller.
5. Micro wave Oven Controller.
6. SMPS
7. CVT.
8. Inverter.
9. UPS.

BIO-CHEMISTRY

B. Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Scheme of Examination	*Duration*	*Marks + Internal assessment*
Theory Paper-A: Molecular Biology	3 hrs.	67 + 8
Theory Paper-B: Applied Biochemistry	3 hrs.	67 + 8
One Practical examination pertaining to the entire syllabus included in Theory Papers A & B	3 hrs.	45 + 5
	Total marks :	**200**

Paper-A: MOLECULAR BIOLOGY

Three periods per week **Max. Marks : 75**

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS :

1. Total No. of questions will be nine. Q.No. 1 will be of 15 marks while other questions will be of 13 marks each.
2. Q. No. 1 will be compulsory. It will consist of 10 short questions covering the entire syllabus.
3. Besides question Number 1, there will be 4 sections of 2 questions each.
4. All other questions may contain 2-3 parts.
5. Questions should be uniformly spread over the entire syllabus.
6. Students will be required to attempt 5 questions in all including Q. No. 1 and at least one question from each of the 4 sections.

Objectives :
Aspects of storage and expression of genetic information. Membrane structure and function.

SECTION-I

(Lectures: 10)

***DNA Organization*:** Structure of chromatin – Histones and Nucleosomes. Active and inactive chromatin. Compaction of Chromatin. Chromosomes, Structure of Genome in eukaryotes. Rearrangements in Genetic Material. Integration of Chromosomes with viruses. Transposition, Experimental proofs for DNA as genetic material.

SECTION-II

(Lectures: 14)

DNA Replication : Semiconservative replication-proof. Molecular events and enzymes involved in DNA replication. DNA repair mechanisms. Mutations.

RNA Synthesis : Initiation, elongation and termination during RNA synthesis. Transcription signals. Processing of RNA. Introns and Exons. Nucleases.

Genetic Code and Protein Biosynthesis : Characteristics of Genetic code, Deciphering of Genetic Code. Initiation, elongation and termination of protein chains. Post translational modifications in proteins. Inhibitors of protein biosynthesis.

SECTION-III

(Lectures : 14)

Regulation of Gene Expression : Temporal Responses, Lac operon-Jacob Monod Hypothesis. Eukaryotic Gene Expression.

Recombinant DNA Technology : Restriction endonucleases. Chimeric DNA. Gene Library, Basic Principles in Gene Cloning. Applications of Biotechnology.

SECTION-IV

(Lectures: 12)

Membranes **:** Structure and functions of biological membranes, various models of membrane structure. Transport of solutes across membranes, Sodium pump.

Elementary aspects of the Molecular Biology of cancer and introduction to stem cells.

Molecular basis of the Origin and Evolution of Life.

Books Suggested :

1. *Harper's Illustrated Biochemistry,* 27th Edition, 2006.
2. Biochemistry, 5th Ed., 2002., Berg, J. M. & L. Stryer, W.H. Freeman, San Francisco.
3. *Molecular Biology* by David Freifelder, 1987.
4. *Genes VIII,* by Benjamin Lewin, 2003.
5. *Molecular Cell Biology* by James Darnell, Harvey Lodish and Davit Baltimore, 5th Edition, 2003.

Paper-B : APPLIED BIOCHEMISTRY

Three periods per week **Max. Marks: 75**

INSTRUCTIONS FOR PAPER SETTER AND STUDENTS :

1. Total No. of questions will be nine. Q.No. 1 will be of 15 marks while other questions will be of 13 marks each.
2. Q. No. 1 will be compulsory. It will consist of 10 short questions covering the entire syllabus.
3. Besides question Number 1, there will be 4 sections of 2 questions each.
4. All other questions may contain 2-3 parts.
5. Questions should be uniformly spread over the entire syllabus.
6. Students will be required to attempt 5 questions in all including Q. No. 1 and at least one question from each of the 4 sections.

Objectives :

General aspects of vitamins, hormones, nutrition, immunology, blood coagulation, muscle contraction and nerve impulse transmission.

SECTION-I

(Lectures : 14)

Nutritional functions of water soluble Vitamins: B_1, B_2, Niacin, Pantothenic acid, B_6, Biotin, Folic Acid, B_{12} and Vitamin C. The role of water soluble vitamins as co-enzymes.

General characteristics and classifications of hormones .

Mechanism of Blood coagulation. Blood buffers and role of the kidneys in Acid Base regulation, Role of liver in detoxification of endogenous and exogenous substances.

Structure of voluntary muscle and mechanism of muscle contraction. Mechanism of nerve impulse transmission.

SECTION-II

(Lectures : 12)

Essential nutrients. Protein energy malnutrition, starvation and obesity. Respiratory Quotient (R.Q.) of carbohydrates, proteins and lipids. Basic metabolic rates and factors influencing it. Specific dynamic action of foods.

Protein quality and Nitrogen balance studies. Role of essential amino acids and fatty acids in human diet. Role of dietary fiber in Nutrition. Requirement of minerals (Macro & Micro) and their major Physiological functions.

SECTION-III

(Lectures : 12)

Definition of immune system and antigens. Cells involved in immune response. T-cell and B-cells, Immunoglobulins, chemical structure of the Antibody molecule. Haptens and carrier molecules, cell mediated immune response. Complement system, activation and its role in defense.

Brief discussion of various immunological techniques; Precipitation reactions in gels Haemagglutination, Immuno-fluorescence, radio-immunoassay (RIA), enzyme liked immunosorption assay (ELISA) and immunoblotting.

SECTION-IV

(Lectures : 12)

Biochemical Principles of Toxicology. Phase I reactions and Cytochrome P 450 enzyme systems. Phase II reactions and various conjugation systems. Effects of nutritional status and metabolic induction on xenobiotic toxicity; Importance of Physico–chemical properties of toxic chemicals. Biochemical basis of organophosphate and carbamate pesticides toxicity.

Books Recommended

1. Berg, J. M., Lubert Stryer *: Bio-chemistry,* W.F. Freeman and Co., New York, Edition 5th, 2002.

2. *Harper's Illustrated Biochemistry* by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 27th Ed. 2006, Prentice Hall International Ltd. (U.K.).

3. *Essential Immunology* by Ivan M. Roitt, English Language Book Society (ELBS) 8th Ed.,1994.

4. *Biochemical Toxicology of Environmental Agents* by A. De Bruin, Elsevier Pub.

5. *Human Nutrition & Dietetics* by R. Passmore & M.A. Eastwood.

PRACTICALS

Marks : 50

One Practical of three hours per week.

1. Estimation of DNA by diphenylamine method.
2. Effect of temperature on the Viscosity of DNA using Oswald's viscometer.
3. Assays of SGPT and SGOT in serum.
4. Extraction of RNA from yeast and its estimation by Orcinol method.
5. Determination of total protein and A/G ratio in serum.
6. Separation of serum proteins using paper electrophoresis.
7. Estimation of creatinine in urine.
8. Estimation of haemoglobin in blood.
9. Separation of proteins by SDS-Polyacrylamide Gel Electrophoresis.
10. Identification of Sugars in fruit juice using paper chromatography.
11. Determination of nature of inhibition of alkaline phosphatase by cysteine.
12. Determination of proteins by dye binding assay.

MICROBIOLOGY

B. Sc. (GENERAL) THIRD YEAR EXAMINATION, 2015

Scheme of Examination			*Duration*	*Marks*
Theory MIC-301	:	Pathogenic Microbiology	3 hrs.	75 (67 + 8*)
Theory MIC-302	:	Food & Industrial Microbiology	3 hrs.	75 (67 + 8*)
One practical pertaining to the entire syllabus included in theory papers A & B.			6 hrs.	50 (45 + 5*)
			Total :	**200 marks**

Note : *Denotes marks for the Internal Assessment.

Outlines of Tests and Syllabi

MIC-301 : PATHOGENIC MICROBIOLOGY

Max. Marks	:	75
Theory	:	67 marks
Internal Assessment	:	8 marks
Time	:	3 Hours

(Three periods per week)

Note : The question paper will consist of nine questions in total. The first question with sub-parts representing the entire syllabus will be compulsory. There will be no choice in the compulsory question. Apart from the compulsory question, students will attempt four other questions, one each from sections A, B, C and D. Each of the sections A, B, C and D will consist of two questions. All questions will carry equal marks.

Objectives :

The paper provides in depth knowledge about the important diseases caused by bacteria, fungi, viruses, protozoans, worms etc, their operative mechanisms, laboratory diagnosis, prevention and contol.

SECTION-A

Introduction to important diseases caused by ***Streptococcus, Pneumococcus, Neisseria, Corynebacterium, Bacillus, Clostridium, Enterobacteriaceae (Proteus, Shigella, Salmonella), Vibrio, Yersinia, Hemophilus and Mycobacterium.*** The operative pathogenic mechanisms, laboratory diagnosis, prevention and control of these diseases.

SECTION-B

Morphology, pathogenesis, life cycle, laboratory diagnosis, prevention and control of viral diseases such as Rabies, Polio, Small pox, Herpes, Measles, Influenza and AIDS.

SECTION-C

Introduction to human mycotic infections : Cryptococcosis, Dermatophytosis, Blastomycosis, Opportunisitc Mycosis; Candidiasis and Aspergillosis.

SECTION-D

Life cycle, pathogenetic, mechanisms and control of parasitic infections viz. Amoebiasis, Kala-azar, Toxoplasmosis, Ascariasis, Filarasis, Hook worm infections.

Reference Books :

1. Ananthanarayan, R. Panikar, C.K.J. (2002), *Textbook of Microbiology,* Orient Longman Ltd., 160, Anna Salai, Chennai.

2. Brooks, G.F, Butel J.S, Morse, S.A (2002), *Jawetz, Malnick and Adelberg's Medical Microbiology* Mc Graw Hill.

3. Tortora, G.J., Funke, B.R., Case, C.L (2009), *Microbiology: An Introduction,* Benjamin/Cummings Publishing Company Inc.

MIC-302 : FOOD & INDUSTRIAL MICROBIOLOGY

Note : The question paper will consist of nine questions in total. The first question with sub-parts representing the entire syllabus will be compulsory. There will be no choice in the compulsory question. Apart from the compulsory question, students will attempt four other questions, one each from Sections A, B, C and D. Each of the sections A, B, C and D will consist of two questions. All questions will carry equal marks.

Objectives :

The paper provides knowledge about contamination, preservation, spoilage of various food products and important food borne microbial diseases. It also provides awareness about fermentation technology, production media and various fermentation products.

SECTION-A

Food as a substrate for microorganisms, nutritive value of food stuffs, effect of hydrogen ion concentration (pH), moisture requirement on food, important food borne diseases viz, Staphyococcal intoxication, Botulism, Salmonellosis, Shigellosis, qualitative and quantitative analysis of food components (proteins, fats, lipids, carbohydrates), microbiological examination of food products including dairy products, food poisoning caused by bacteria and fungi.

SECTION-B

Contamination, preservation and spoilage in various foods viz cereals & cereal products (cereal grains, flour, bread, pasta, macroni), sugars & sugar products (Maple, Syrup, Honey, Candy), vegetables & fruits, meat (Fresh meat, fresh beef, hamburger, fish), milk and milk products (cheese, butter).

SECTION-C

Industrial strains : isolation & screening techniques, preservation and genetic modification of industrial microorganisms, fermentation media, characteristics of ideal production media, common substrates used in ideal fermentations, batch and continuous fermentations.

SECTION-D

Yeasts (Baker's) and its uses, fermentation of beer, wine and alcohol, production of organic acids viz acetic acid, lactic acid, propionic and butyric acid and mixed acids. Mass transfer in aerobic fermentation.

Reference Books :

1. Frazier, W.C., Westhoff, D.C. (1988), *Food Microbiology,* Mc Graw Hill.

2. Jay, J.M., Loessher, M.J., Golden, D.A. (2005), *Modern Food Microbiology* (Edition 7), Illustrated Publishers Springer.

3. Potter, N.N., Hotchkiss, J.H. (1997), *Food Science,* CBS Publishers.

4. Admas, M.R., Moss, M.O. (2005), *Food Microbiology,* Edition 3, Illustrated Publishers Royal Society of Chemistry.

5. Montville, T.J., Metthews, K.R. (2005), *Food Microbiology, An Introduction,* ASM Press Washington, D.C.

6. Prescott, S., Dunn, C.G. (1994), *Industrial Microbiology,* Mc Graw Hill, New York, CBS Publishers and Distributors.

7. Casida, (1995), *Industrial Microbiology,* N.Y., Wiley.

PRACTICALS

Max. Marks : 50
Theory : 45 marks
Internal Assessment : 5 marks
(One practical of 3 hrs. per week)

1. Identification of both gram positive and gram negative microorganisms on the basis of :
   (i) Morphology.
   (ii) Bio-chemical characteristics.
   (iii) Serological reactions.
2. Demonstration of pathogens (Viruses, fungi, parasites) in permanent mounted slides.
3. Demonstration of cysts/ovas of protozoa/Helminthes.
4. Demonstration of Laboratory grown fungi on Sabauraud's agar.
5. Germ tube test for *Candida albicans.*
6. Demonstration of fungi through normal saline/KOH preparation.
7. Quantitative examination of microbial types in raw/processed/preserved food stuffs.
8. Direct microscopic determination of bacteria in raw and pasteurized milk and reductase test.
9. Various biochemical tests and their importance in Food Microbiology.

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